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CspStat.Dll

CSP Printer Status API

Command Reference

Ver. 0.7.1.2_D

Dai Nippon Printing Co., Ltd.

October 17, 2024

Revision History Chart		First Edition: May 31, 2007			Page 1/10	
Revised Item	Type of Revision	Revision No.	Document No.	Revision Date	Approved	Designer
	First Edition for CX, CXW(DS40,DS80), CXM	0.4.0.0	0.40	05/31/2007		
P42-	Minilab API Function added. (Average Current, Power Voltage, Sorter Sensor etc.)	0.5.0.0	0.50	07/23/2007		
P22 P35	Get status code changed. Group No initial setting changed.	0.5.1.0	0.51	07/27/2007		
P22 P23	Get status code changed (CXM1,2,3,etc.) Status code added.(RS422 non connect,soter max)	0.5.2.0	0.52	08/10/2007		
P34	Get minilab tower status added.	0.5.3.0	0.53	08/14/2007		
P22 P34	GET status code changed (shooting). Get minilab tower status changed.	0.5.4.0	0.54	08/22/2007		
P24 P24	GET Printer Counter Value changed. Set Printer Counter Value added,	0.5.5.0	0.55	08/23/2007		
P13	It is possible to get "Version"and "Serial No." from Tower controller. GetPrinterPortNum changed.(for Tower controller)	0.5.6.0	0.56	08/27/2007		
P18	GetFirmwareVersion()function corrected.	0.5.7.0	0.57	08/29/2007		
P7 P48	The procedure to shop "Windows driver request wizard" when direct DLL is in use, is added. GetMiniLabPrinterAdd() function added	0.5.8.0	0.58	08/31/2007		
P7 P22	it does NOT return the device name GetPrinterStatus Technical reference "RS422 Non Connected" status added	0.5.8.0	0.58A	09/25/2007		

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Revised Item	Type of Revision	Revision No.	Document No.	Revision Date	Approved	Designer
	Modify to the processing during non-connection of USB printer cable regarding memory of GetStatus() value against V0.5.8.0. Delete P24,25 Technical reference "RS422 Non Connected status" from V0.58A depend on the version of F/W (0.133)	0.5.9.0	0.59	10/03/2007 (Official Release)		
P14 P18 P19 P35 P39, 40 P44	SetMiniLabPowerVolt, SetMiniLabECurrentMax GetMiniLabPowerVolt, GetMiniLabECurrentMax Function Deleted. About the API function movement of the KIOSK model Image pixel size added. Setting the Resolution function added. Send Color Control Data to Temporary buffer added. Get minilab tower status changed. Set Group,Serial No document changed.	0.6.0.0	0.60	06/02/2008		
	~~Self diagnosis function API added.~~	0.6.0.0	~~0.61~~	~~06/11/2008~~		
~~P38~~	~~Overcoat finishing control command added.~~	0.6.0.0	~~0.62~~	~~06/26/2008~~		
P28-29	Counter L/A/B/P Count-up Action added.	0.6.0.0	0.621	09/18/2008		
P17	For Ver 0.5.9.0 <Linx machine & high-speed PC Countermeasures> *Note) Improved image data transmition method	0.5.9.1	0.621_1	12/05/2008		
P17	For Ver 0.6.0.0 <Linx machine & high-speed PC Countermeasures> *Note) Improved image data transmition method	0.6.0.1	0.621_1	12/05/2008		
P40	For Ver 0.6.0.0 Overcoat finishing control command added (Page number correction in revision history chart)	0.6.2.2	0.622	11/06/2008		
P6 P15 P18	For Ver 0.6.2.2 Windwos Vista was added. Procedure to firmware update when using API. <Linx machine & high-speed PC Countermeasures> *Note) Improved image data transmition method	0.6.2.3	0.623	01/09/2009		
P13-15 P19 P20 P31 P33 P36 P47	For Ver 0.6.2.3 Status API function list was changed Printer filter function added USB transmission time out was added TIMEOUT error of printer status was added Counter Matte was added Clear counter M was added Print retry control command was added	0.6.3.0	0.630	10/31/2009		
P24	Send image data Change in output size limitations according to printer location added	0.6.3.0	0.630_1	11/20/2009		
	[Dll correction] The printer error doesn't occur when SetMediaSize Function is no specified and the image transmission is done.	0.6.3.1	0.631	02/10/2010		

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Revised Item	Type of Revision	Revision No.	Document No.	Revision Date	Approved	Designer
P7 P9 P22 P23 P25 P41,42 43,45	(Change of only a document) The description of Windows Vista(32bit)and Windows 7(32bit) is added. 2.Windows driver requirements・・Description revisions Addition of SetMediaSize() function Addition of SetImageData() function Deletion of redundant firmware compatibility descriptions Addition of descriptions relating to firmware and control data rewrite commands	0.6.3.1	0.631_1	05/20/2010		
P14,15, P18	< New model addition > It corresponded to CY and CW02 printer.	0.6.3.2	0.632	11/19/2010		
P15, P48-50	・Addition of a function which gets region code. ・Addition of a function which clears printer region code. ・Addition of a function which gets media class. ・Addition of a function which gets RF-ID reserve data.	0.6.3.3	0.633	11/30/2010		
P14,15, P18 P50	< New model addition > It corresponded to DS-RX1 printer. Get Initial Media count was added.	0.6.4.0	0.640	01/28/2011		
P15 P49	・Addition of a function which sets printer region code.	0.6.4.1	0.641	02/28/2011		
P15 P49,50	・The function of the region code was deleted.	0.6.4.2	0.642	05/12/2011		
P35 P45	・Change of Copyright information ・The case of a 2inch cut is added to a count-up operation table. ・The 2inch cut mode is added to a cutter control command.	0.6.4.3	0.643	09/14/2011		
P7 P21 P35	・Change of description of Copyright ・The L 2-image layout is added to page definition contents of media size setting. ・The multi cut mode 5x3.5x2 is added to a count-up operation table.	0.6.4.4	0.644	04/22/2013		
P14,15, P18,21, P23,30, P35,38	< New model addition > It corresponded to DP-TC10 printer.	0.6.4.5	0.642	06/14/2013		
P14-18, 21,25,27- 29, 32,35,364 1,42,44,53 ,54	＜Additional function＞ ・Compatibility with DP-DS80D（Duplex-capable 8-inch printer）	0.6.5.0	0.6.5.0	06/28/2013		
P37 P39 P40 iv-v	・Duplex Error code was changed ・Turning unit Status was added. ・CutPaper Infomatin was added. ・Duplex Printer Combination Chart was added.	0.6.5.1	0.6.5.1	08/05/2013		

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Revised Item	Type of Revision	Revision No.	Document No.	Revision Date	Approved	Designer
P35 P37 P48 P68	(Change of only a document) ・Revised the count-up operation explanation ・Added the count value for the DP-TC10 L card size 2-image layout continuous printing to the count-up operation ・Added the DP-TC10 L card size 2-image layout continuous printing designation to the Cutter Control commands ・Added the DP-TC10 L card size 2-image layout continuous printing explanation to the appendix	0.6.5.1	0.6.5.1_1	10/24/2013		
P9,10, 20,23, 36,37 41, v-vi	・Multi-cut paper size of cut paper was added. ・Cunter Duplex was added ・Turning unit, Cancelling the 2nd-side printing and ejecting the media was added. ・Duplex Printer Combination Chart was changed. ・List of function compatibility was changed.	0.6.5.3	0.6.5.3	11/14/2013		
P56	・Get turning unit firmware version added.	0.6.5.4	0.6.5.4	12/28/2013		
P53 P73	(Change of only a document) ・ Added DP-TC10 partial matte operation to the overcoat finish function. ・ Added about DP-TC10 Partial Matte Printing to the appendix	0.6.5.4	0.6.5.4_1	01/17/2014		
v	(Change of only a document) ・Duplex Printer Combination Chart was changed.	0.6.5.4	0.6.5.4_2	03/28/2014		
P1 P25,46 P12 P12 P50 P56 P58 P65 P66 P67 P68 P69 P11	・64-bit version is supported. ・DS80D cut paper 8x6x2 size was added. ・DP-DS620 was added. The following commands were added. ・Ribbon Rewind Function is added. ・The example of the ribbon rewind operation process is added. ・Get L/PC Size Conversion Media Counter of Remaining Sheets ・Get Color Data Version <Type Designation> ・Get Color Data Checksum <Type Designation> ・Full Cutter Set-up ・Set Standby mode transition time ・Get Standby mode transition time ・Set Media End Keep Mode ・Get Media End Keep Mode ・Set USB iSerialNumber availability setting ・Get USB iSerialNumber availability setting ・ Set Rewind Mode setting ・ Get Rewind Mode setting ・ The operation in standby mode was added.	0.6.6.1	0.6.6.1_1	08/22/2014		
v P17 P19 P54,55 P21	・Duplex Printer Combination Chart was changed. ・ 6. Status API function list was changed ・ 11-2 Procedure to update for turning unit was added. The following commands were added. ・ Set turning unit rewrite mode ・ Write turning unit firmware data ・ Get turning unit firmware rewriting status ・ DS-621 description deleted.	0.6.7.0	0.6.7.0	09/08/2014		
P69 P12 P48, 90 P14,16,	・ It was Changed the factory preset value of the Media End Keep Mode.(0:Not Keep Media End -> 1:Keep Media End) ・ Rewind (5 x 3.5) was supported. ・ It was clarified that DP-DS620 (A) is printed without rewinding the ribbon. ・ Appendix (4) About DP-DS620 Ribbon End Check Operation was added. ・ Get printer version information was added in the API that can be used in Standby Mode	0.6.7.0	0.6.7.0_1	10/31/2014		

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Revised Item	Type of Revision	Revision No.	Document No.	Revision Date	Approved	Designer
P1 P19 P12,,24 ,28,45 P50 P50-51 ,91 P38 P40	・Remove “Winodws 2000” from supported OS. ・Fixed the order of procedure to update of “11. Procedure to firmware update when using API” ・The following media sizes were added. (for DP-DS620) 6x9, 6x4.5, (6x4.5)x2 ・Ribbon media (6x9) was supported. (for DP-DS620) ・Change wording to “Turning unit” from “Turning unit”. ・Change wording to “cut sheet” from “cut paper”. ・Delete some wording from Get Printer Status description. ・Delete some wording from Getting Turning unit description.	0.6.7.2	0.6.7.2	02/27/2015		
	・64-bit version is supported.	0.6.7.2	0.6.7.2_D_2	03/31/2015		
P68 P17,69 P64 P62 P17,75 P92-93 P95-96	・6x4, 5x7, and 6x9 media sizes are now supported with Full Cutter Set-up. ・Full Cutter Set-up Extended Settings command added to support intermediate scrap cut setting. (SetFullCutterSetUpEX) ・Fine Matte, Partial Matte (Matte), and Partial Matte (Fine matte) added to the Overcoat Finishing Control command. ・Panorama printing operation added to the Cutter Control Command ・Panorama Printing Start Check command was added. (GetPanoramaPrintable) ・Information for DP-DS620 Partial Matte function added to Appendix (3) About Partial Matte Printing. ・Appendix (5) Panorama Printing was added.	0.6.7.3	0.6.7.3	04/28/2015		
P24, 28,45 P17, 68-70	・Paper size 6x6 was supported.(for DS40) ・Full cutter set-up function was supported.(for DS40) (paper size 6x8 only)	0.6.7.5	0.6.7.5	06/30/2015		
P55 P17 P57-59	・DP-DS80D Set Turning unit Rewrite Mode was changed. ・DP-DS80D Get Turning Unit Error Log function was added. Request Creation of Turning Unit Error Log Get Turning Unit Error Log Data Length Receive the Turning Unit Error Log Data Get the Turning Unit Error Log Creation Status	0.6.7.6	0.6.7.6	07/22/2015		
P17, P71, P79, P77 (vi-vii)	・DS40 the following functions was added. Get Initial Media Count Get Media Offset Count Set USB iSerialNumber availability setting Get USB iSerialNumber availability setting ・DP-DS80D Combination Chart and function compatibility list has been deleted.	0.6.7.7	0.6.7.7	08/31/2015		
P17, P79, P77 P68	・DS-RX1 the following functions was added. Get Media Offset Count Set USB iSerialNumber availability setting Get USB iSerialNumber availability setting ・Luster added to the Overcoat Finishing Control command.	0.6.7.8	0.6.7.8	12/25/2015		
P17,80	・Continuous Panoramic Prints settings function was added.	0.6.7.9	0.6.7.9	02/29/2016		

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P12,14 15-17 21-22 25,29,35, 37,44, 46-47 52-53 64, 66-83 102	・DP-DS820 was added. The following commands were added. ・Set Print Speed function was added. ・Set System Time function was added. ・Get Color Data Data Version <Type,Media designation> was added. ・Get Color Data Checksum <Type,Media designation> was added. ・L/PC Size Conversion Media Counter of Remaining Sheets was changed ⇒ Half Size Conversion Media Counter of Remaining Sheets	0.6.8.0	0.6.8.0	02/29/2016		
P17,18	・Get media class was added in the API that can be used in Standby Mode. (DP-DS820)	0.6.8.0	0.6.8.0_1	03/18/2016		
P1	・Support for Windows 10	0.6.8.1	0.6.8.1	03/31/2016		
ix P17 P88,89	・Add the following API functions. - Set Gamma Table - Get Gamma Table Checksum - Clear Data Table	0.6.8.2	0.6.8.2	05/27/2016		
P72 P14 P47	(Change of only a document) ・Get Media Class Function was append a description of the media type. DP-DS820 related portion of the table in the following section have been updated. - 8.Standby Mode API that can be used in Standby Mode - Get Printer Counter Value Counter L/A/B/P count-up action	0.6.8.2	0.6.8.2_2	06/24/2016		
P71, P81,82, P83, P106,107	The following section has been changed. - Print Retry Control command. - Panorama Print Start Check function. - Continuous Panoramic Prints Settings function. - Appendix: (5) Panorama Printing with White Border	0.6.8.3	0.6.8.3_D	09/13/2016		
P70,103	・Fine Matte was supported to the Overcoat Finishing Control command (for DP-DS620)	0.6.8.3	0.6.8.3_D2	02/28/2017		
P25,29, 46 P37 P46 P52	・Paper size 5x5 and 6x6 was supported. (for DS-RX1) ・Part of contents of explanation of media code was changed ・Appends the contents on the DS-RX1 to the annotation of multi-cut (L2 image layout) ・Update media list - 350 sheets is added to the 6x4 (PC) size of DS-RX1 - 400 sheets is added to the 6x4 (2L) size of DS-RX1	0.6.8.5	0.6.8.5_3_D	07/28/2017		
P52	・Get media counter command DP-DS620 5x3.5 (L) media support.	0.6.8.7	0.6.8.7_D	02/28/2019		

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P14,17, 19-23, 25-27, 30,33,37, 40,42-43, 53,57-58, 70-71 73-82,84, 87-88,93 P96 P96 P55, P40	< New model addition > - DP-QW410 was added. The following section has been added. - Set Decurl Control - Current time notification Added description when negative value is specified. The mistyping was fixed.	0.6.9.1	0.6.9.1_1_D	12/24/2019		
P30,33,42, 53,57,75,82 P70,71,73, 74,80,96	4.5x6 media was corresponded. (for DP-QW410) The errors was corrected.	0.6.9.3	0.6.9.3_D	04/24/2020		
P21,23 P14, 21-23,25, 71-72, 74-75 P76 P81 P55 P55 P98 P16-19, 22-23,26, 27,30, 93 P116 ・Changed the figure of in house confirmation pattern.	・Status API function list was changed ・Merged firmware for DP-DS620 was supported. (DP-DS620 firmware version 3.00 or later) ・Added 2image layout and non-scrap cutter operation parameter Cutter Control Command ・Added media type to Media Class Command ・The following function were added. - Get Life Counter Value Extended - Get Matte Counter Value Extended - Get supported media information command ・Revised typos.	0.6.9.7	0.6.9.7_D	09/09/2020		
vii P23,27,58, 60,98	(Change of only a document) ・Merged firmware for DP-DS620 was supported.	0.6.9.7	0.6.9.7_1_D	05/20/2020		
P2	・Changed the diagram of the Depiction of printing and processing using DLL	0.6.9.8	0.6.9.8_D	07/02/2021		

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P30,33,54, 77,85 P27-30,32, 37-43, 46,48-50, 57-68, 70-79, 81-93,95, 97-103, 105-114 111-120 P22,92-93	・The following paper size have been added to the DP-QW410. 4x3, 4x4.5, 4.5x3, 4.5x4, 4x3x2, 4.5x3x2, 4.5x4x2 ・Revised the description of each API function ・Supported the Rewind mode for DP-DS820.	0.6.9.9	0.6.9.9_D	08/31/2021		
	[DLL modification] Fixed the processing related to multi-cut printing in DP-QW410 inside the DLL.	0.7.0.0	0.7.0.0_D	06/01/2022		
P1	・Support for Windows 11	0.7.0.0	0.7.0.0_1_2	03/10/2023		
	[DLL modification] Fixed bugs in DLL internal processing latent since v.0.6.9.6.	0.7.0.5	0.7.0.5_D	06/27/2023		
P1,12-16, 19,29-31, 39,44,47-48,58-59, 63-64,76, 78-79,83, 85-86,101, 128 P20-27 P35,38-39 P106 P107 P108 P108 P109 P109 P48	Remove “Winodws XP” and “Windows Vista” from supported OS. ・Added restrictions. < New model addition > - DP-DS820DX was added. ・Status API function list was changed ・Added paper size (8x6.5) for cut paper The following section has been added. ・Setting crease line control ・Setting slitter control ・Creaser unit disabling settings ・Get the value of creaser unit disable setting ・Slitter unit disabling settings ・Get the value of slitter unit disable setting The following section has been changed. ・Get Printer Status	0.7.0.8	0.7.0.8_4_D	05/31/2024		
P21,23,27 P112 P107 P14, 34-35, 38-39, 58-59,107	・Status API function list was changed The following section has been added. ・Band unevenness prevention operation setting The following section has been changed. ・Setting crease line control ・The following paper size have been added to the DP-DS820DX. Roll paper : (8x7.1), (8x11) Cut paper : (8x7.1), (8x11), (8x5.9), (8x5.9)x2, (8x10.35), (8xA4)	0.7.1.2	0.7.1.2	10/17/2024		

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Introduction

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Application Scope

This document has been written about the handling of the CSP status api function.

Supported OS, operating environment

This API runs on Windows 7(32/64bit), Windows 8(32/64bit), Windows 10(32/64bit), and Windows 11(64bit).

Note : This API have a DLL of two types of 32-bit and 64-bit versions.

Restrictions

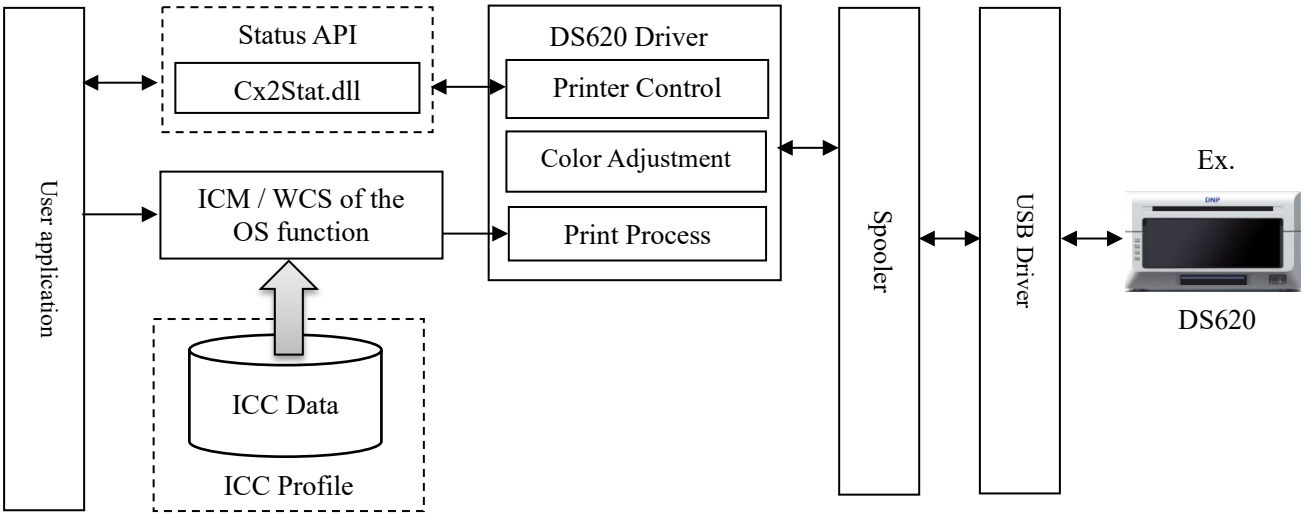
CspStat does not support multi-threaded operation. It must run single-threaded.
When printing with multiple printers (let's say printers A and B), make sure to send printer B's print data to the DLL after printer A's print data is sent from the DLL to the printer.
If the GetPrinterPortNum() function is executed while image data that has not been transferred to the printer remains in the DLL, the image data in the DLL will be lost.

1.Depiction of printing and processing using DLL

Printing and control data sent directly to the USB differs from that sent through the printer driver in the following ways.

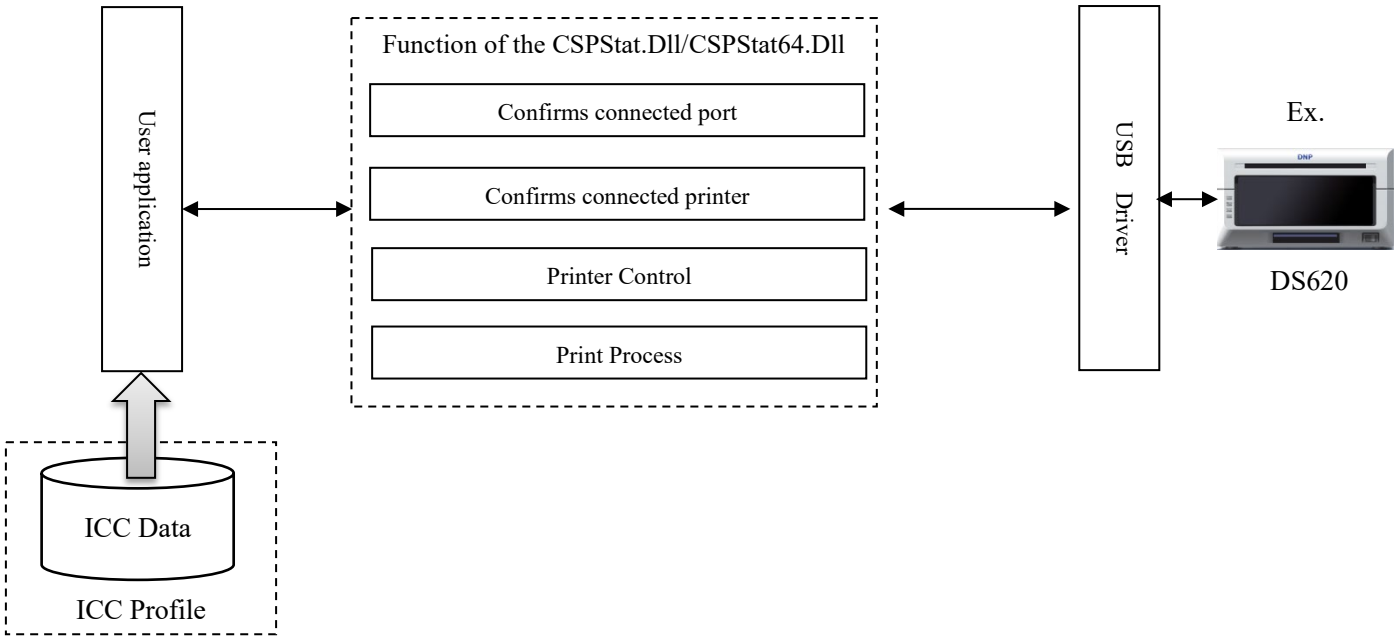
① **Printing and control where the data is sent through the printer driver**

The print and control data are processed through the respective drivers for each device.



② **Printing and control where the data is sent directly to the USB**

The print and control data are sent directly to the USB.



2.On the firmware stopping Window's driver requesting Wizard

In case the user uses direct DLL, and tower's firmware is used, it is possible to stop the driver requesting Wizard when connected to the USB.

Usually Windows, when printer is connected, starts USB connection (port formation)->Recognition of device name printer returns->Driver requesting Wizard, but tower's firmware at the time of connection, it does not return the device name from the printer, it would not start the driver request Wizard.
Windows dynamically allocates the ports, it makes the driver request every time the printer is connected to the new port.
This firmware can stop the driver request Wizard even at the time the printer is connected to a new USB port.

(Note) In case the user wants to print from printer driver on this firmware, he need manually install the printer driver.

3.Port Number Designation

① With control via the printer driver, Port Initialize is used to designate the port connected to the driver.

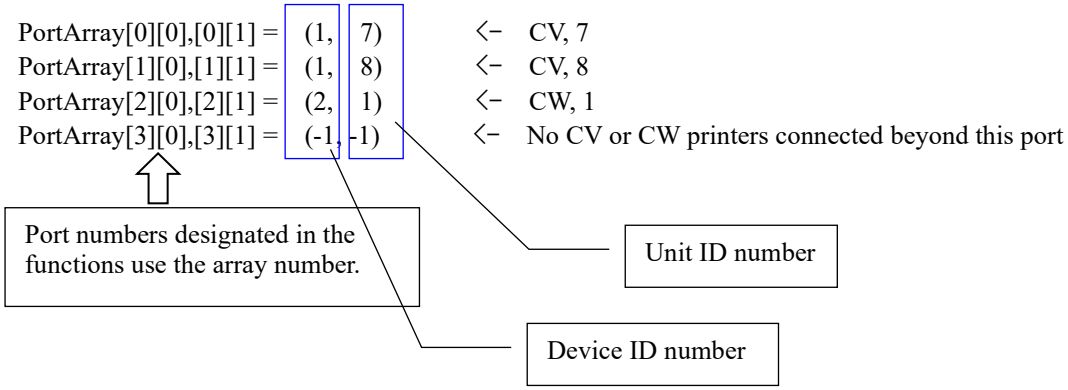
Ex.)PortNo = PortInitialize(StrPrt("USB001"));

For the various DLL functions, the Port Number gotten with this API function was used. "USB00n" is the name of the port connected to the printer driver, so in order to bypass the printer driver, the following function ② is used.

② In sending the data directly to the USB, the application must check the connection status and designate the port numbers.
Using the function GetPrinterPortNum, the printer connection status is gotten as follows.
The port number used in the DLL functions is arranged as shown below.

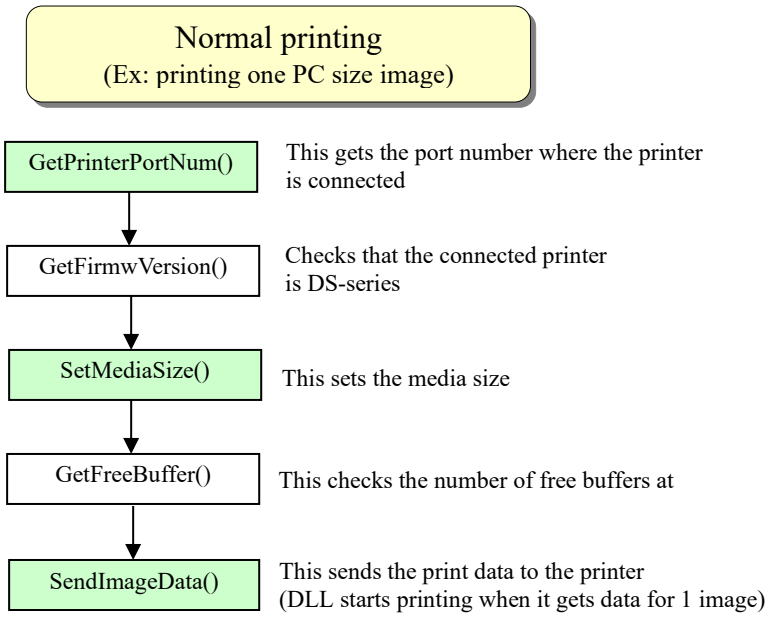
PortNum = GetPrinterPortNum(PortArray[ ][ ]);

Ex)In the case with three CV and CW printers connected to the ports, the PortNum is 3.

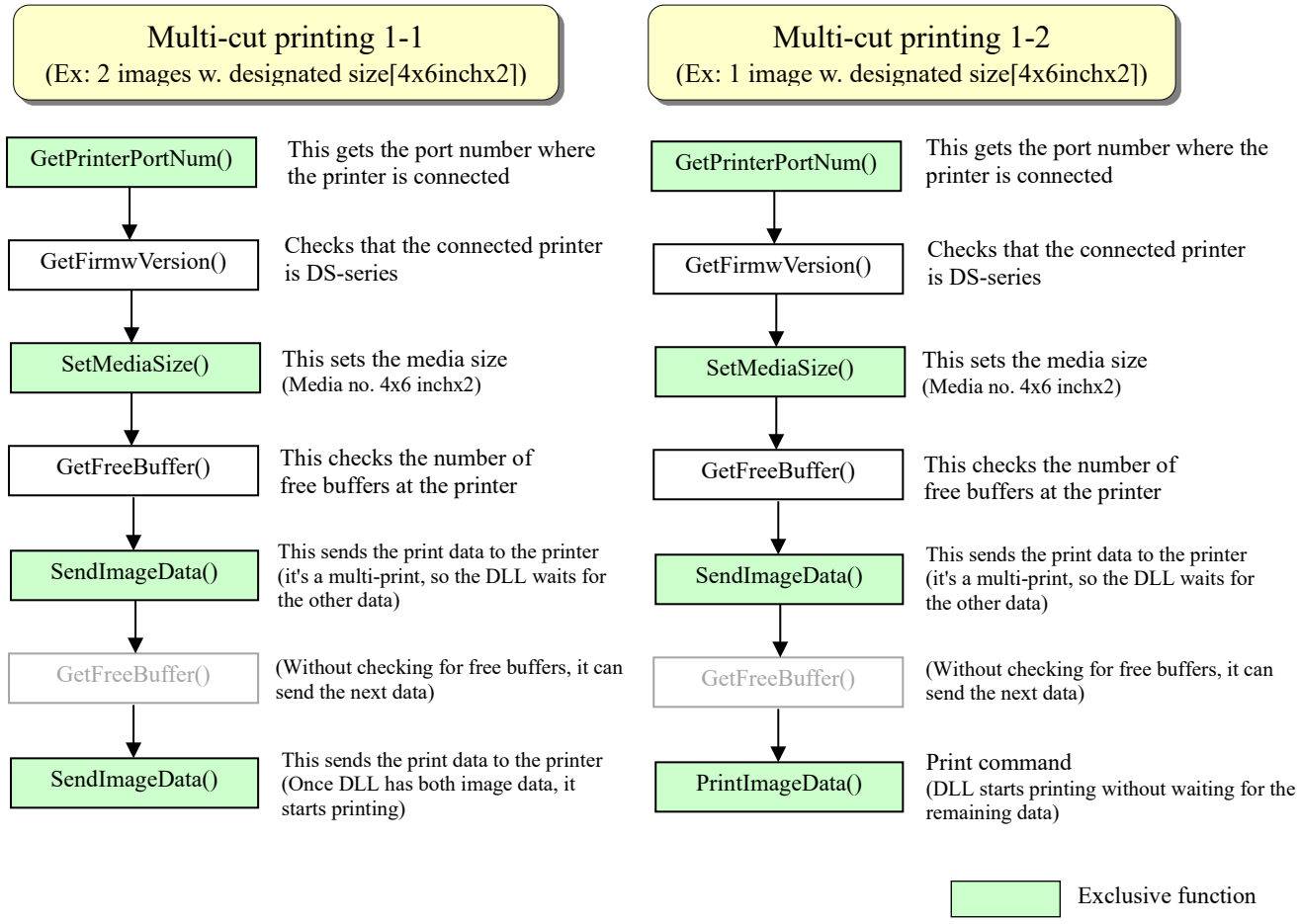


4.Print Process Flow

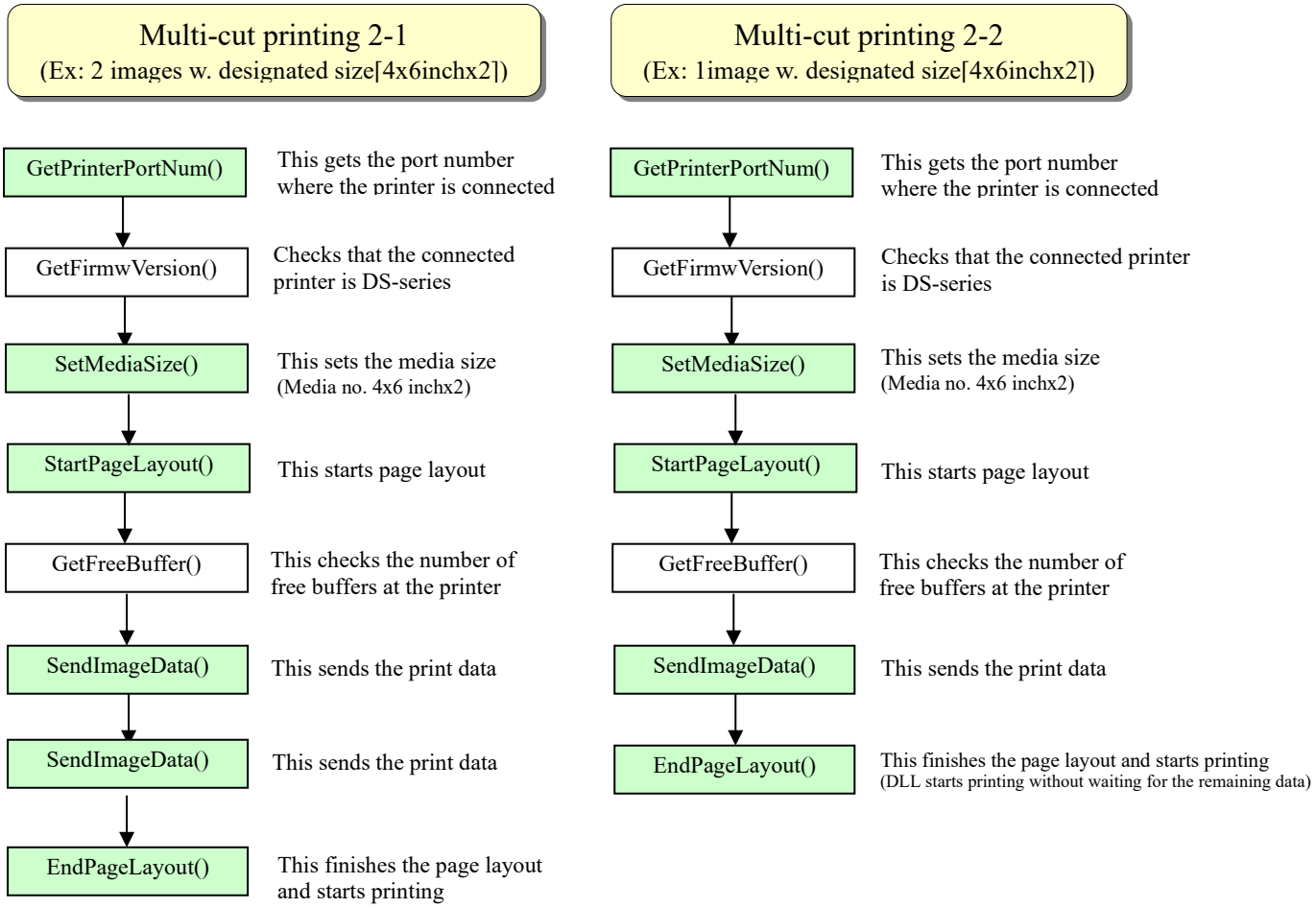
The print process order for the newly added API is as follows.



① Multi-cut print process not using the Page Layout function



② Multi-cut print process using the Page Layout function



For multi-cut printing, either method (① or ②) can be designated. The CV and CW send the data for printing one page in one installment, so the multi-print data are combined in the DLL and printer driver and sent. Whichever command is used, the data is sent when the necessary number of images is assembled.

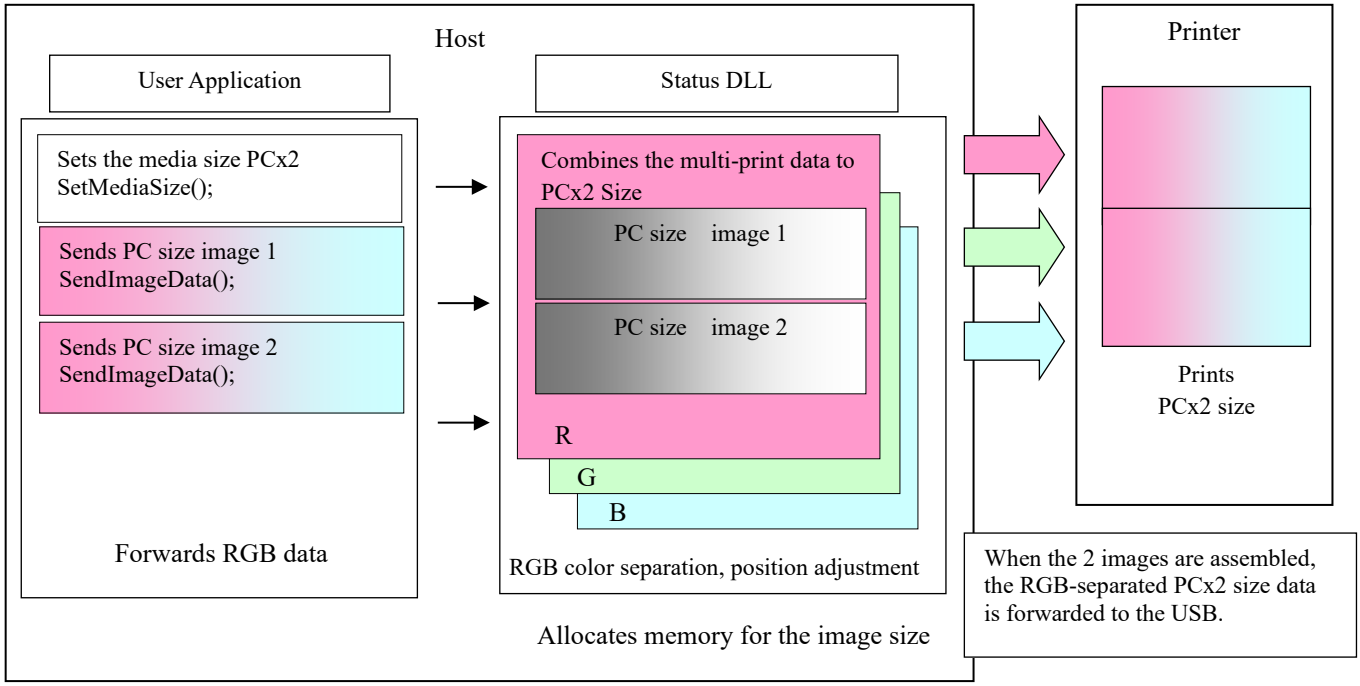
5.Image Flow Supplement Multi-cut printing

Multi-cut printing can be done in the two methods ① and ② shown below. The printer send all the page data for one printing in one installment, so the DLL or printer driver combines the multiple print data in the host memory, and sends it to the printer.

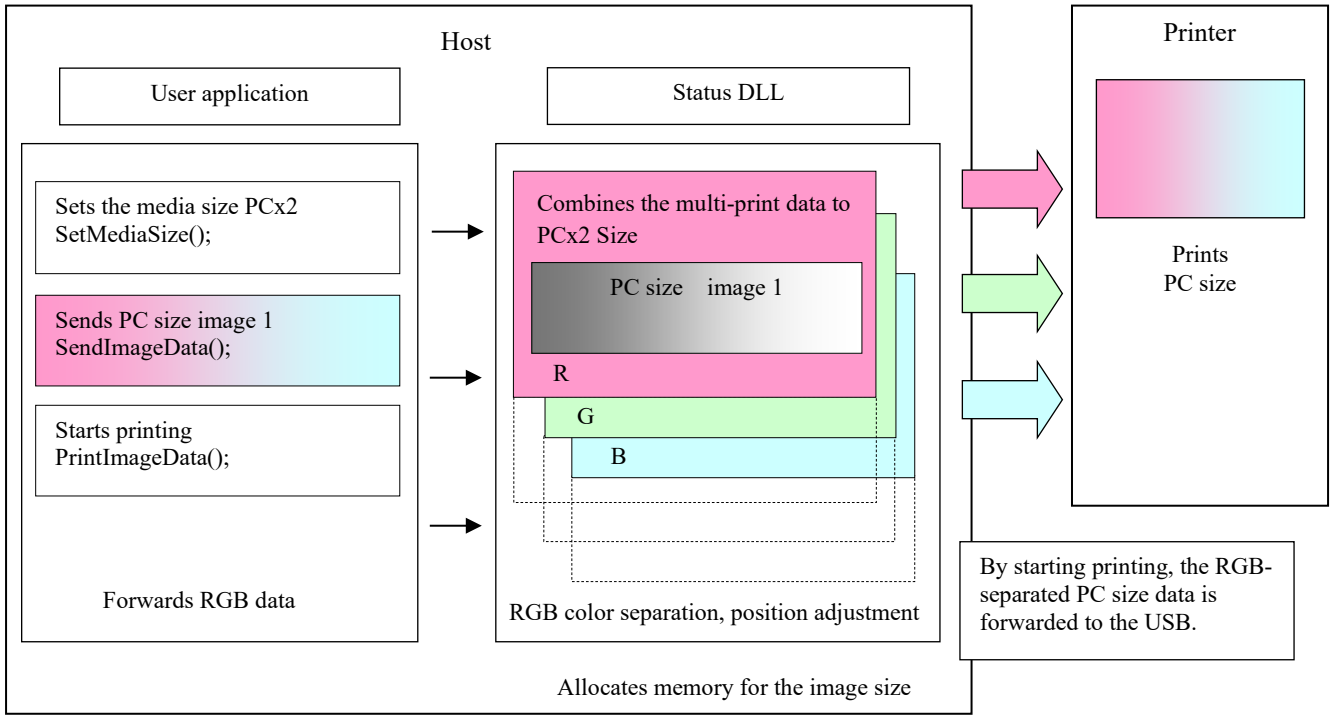
① Multi-cut print processing without using the Page Layout function

In printing without using the Page Layout function, the data is sent to the printer after the necessary images have been assembled in the memory allocated by the status DLL.

1-1 Image of the host process for dual-image layout



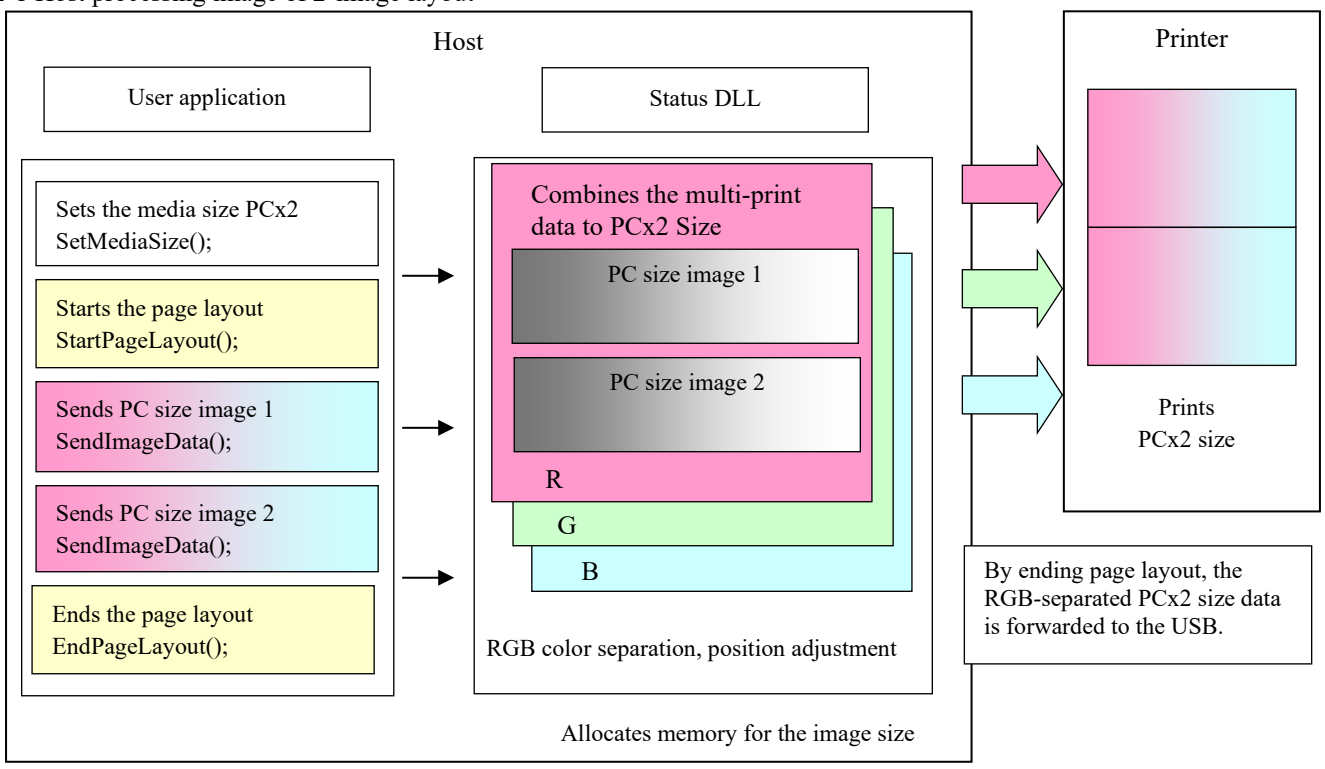
1-2 Host processing image for starting printing when only one image has been designated.



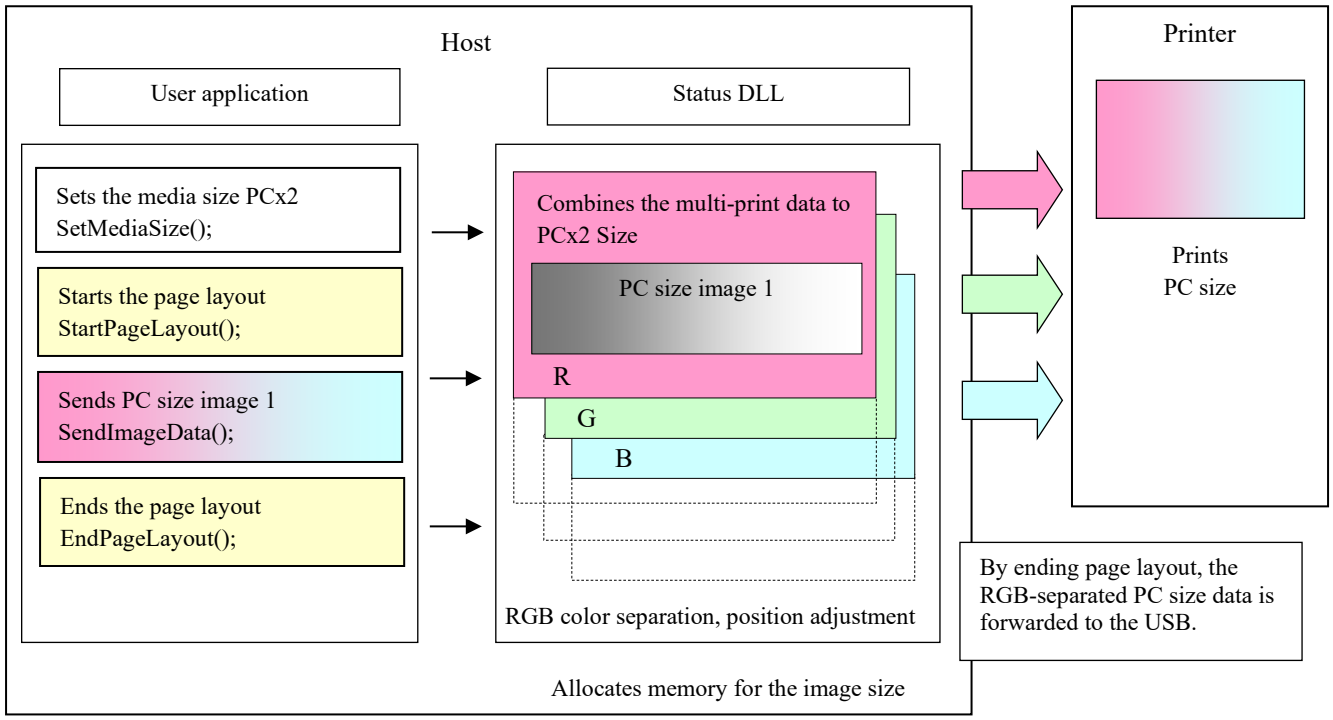
② **Multi-cut print processing using the Page Layout function**

When using the Page Layout function, once the host begins the page layout, the data is not forwarded to the printer until page layout is finished. In this way, the application forwards the data The application clearly defines when to send the data.

2-1 Host processing image of 2-image layout



2-2 Host processing image for starting printing when only one image has been designated.

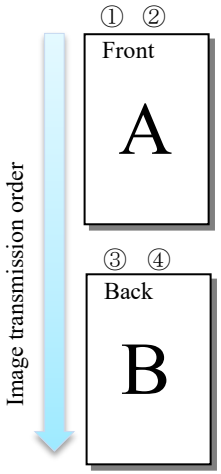
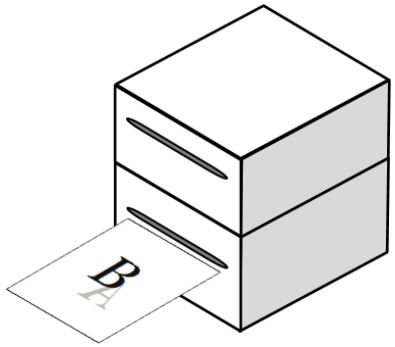
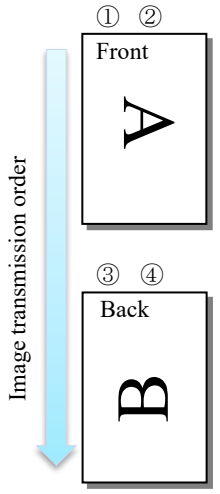
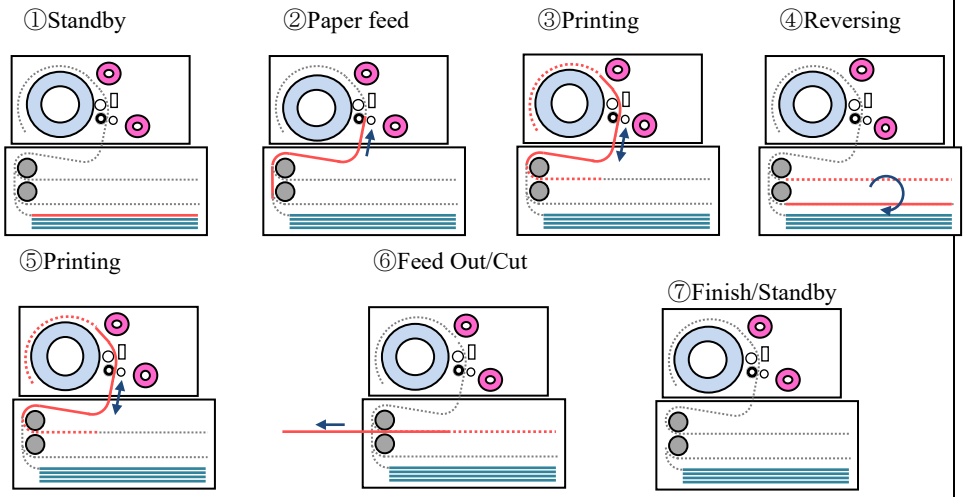
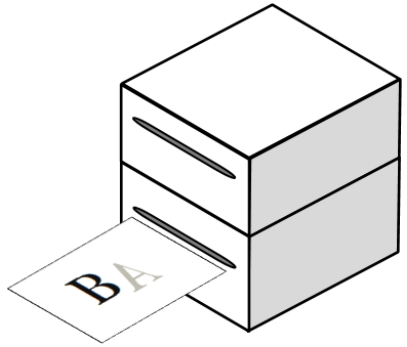


6.Duplex printer: Relationship between the image created by the application and the printer operation/output image

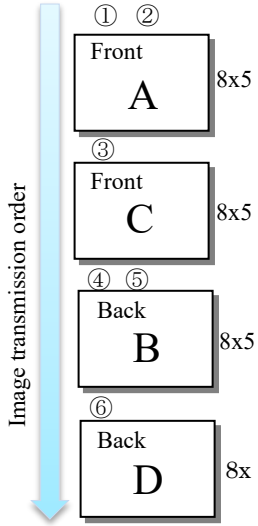
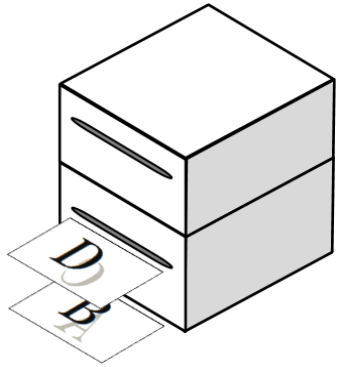
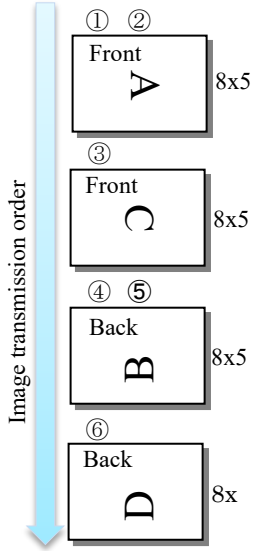
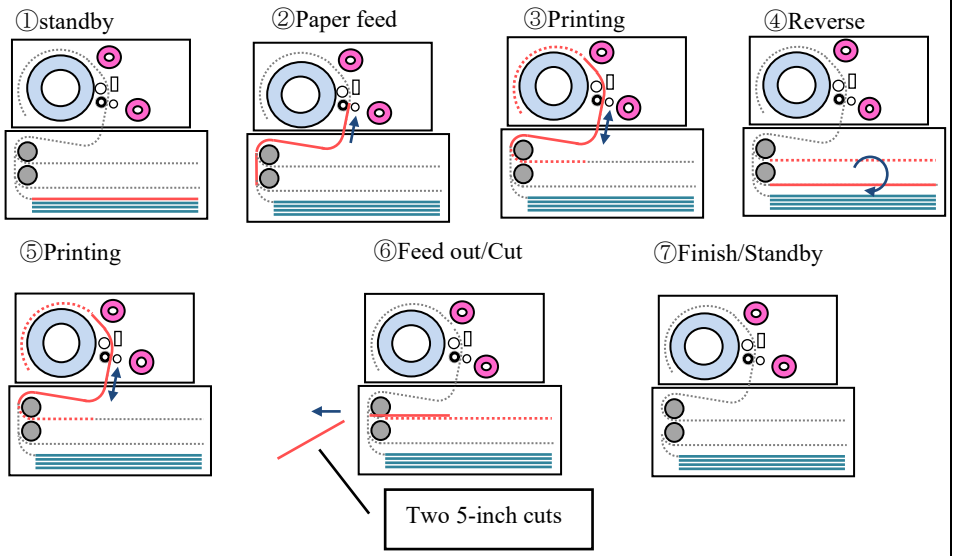
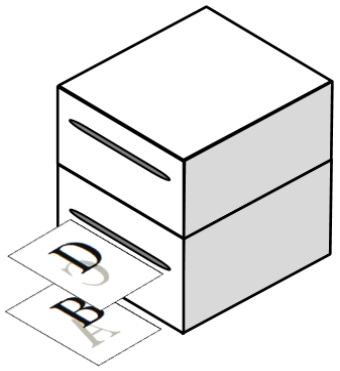
The application determined the order that images are sent to the printer for the desired result, and rotates and alters the image to be sent size depending on the print orientation, etc.
The relationship between the orientation of the image sent to the printer and the completed print is as shown in the chart below.

< For DP-DS80D >

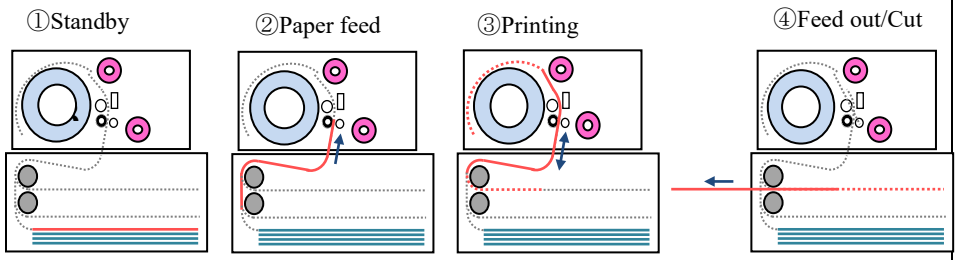
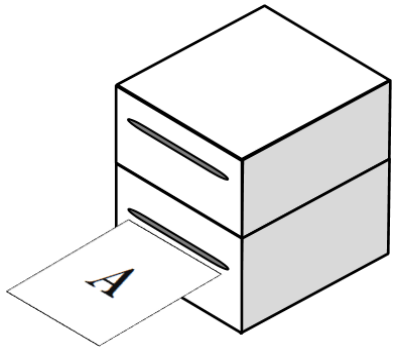
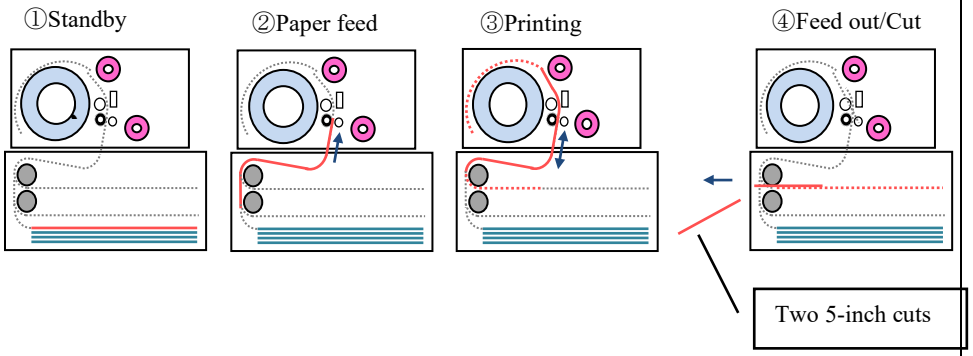
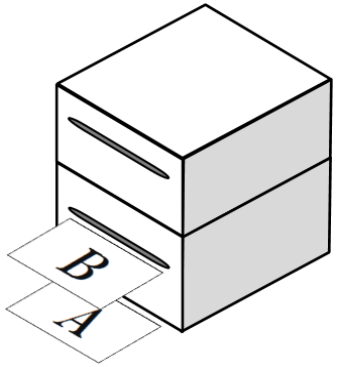
◆ For duplex printing (Duplex cut sheet, no layout. For 1-side printing with overcoating the other side, the process is the same.)

Paper type Output size	Duplex designation	Image creation、 Layout & transmit order		Image Transmit Process and Printer Operation	Media Delivery Image
Duplex Cut sheet Single-Layout	Long edge binding		⇒	Image transmission procedure ① Paper size setting SetMediaSize() Ex:for 8x10（front） set the parameter 231 ② Send BMP image A SendImageData() ③ Paper size setting SetMediaSize() Ex:for 8x10（back） set the parameter 331 ④ Send BMP image B SendImageData() (Example for printing 1 side, and putting an overcoat on the other side) ※If you only want to print on one side of the paper, and overcoat the other side, the BMP image data sent for the overcoat side should be blank. ①Paper size setting SetMediaSize() Ex：for 8x10(front) set the parameter 231 ②Send BMP image A SendImageData() ③Paper size setting SetMediaSize() Ex：for 8x10(back) set the parameter 331 ④Send blank BMP image B SendImageData()	
	Short edge binding		⇒		

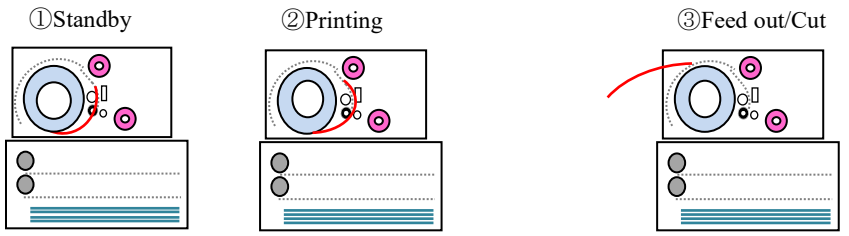
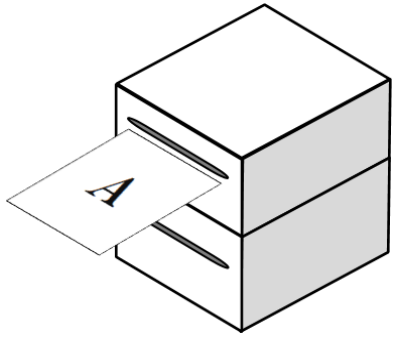
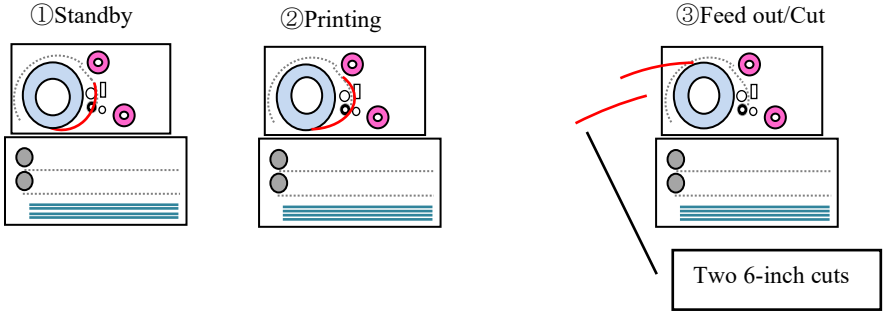
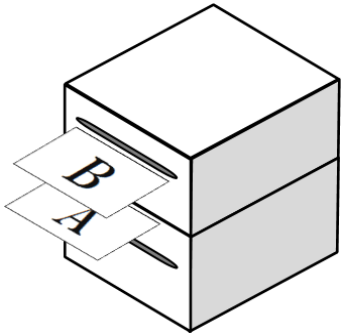
◆ For duplex printing (Duplex cut sheet, 2-image layout. When printing only 1 side and overcoating the other side, the process is the same.)

Paper type Output size	Duplex designation	Image creation、 Layout & transmit order		Printer Operation and Image Transmit Process	Media Delivery Image
Duplex Cut sheet Multi-Layout	Long edge		⇒	Image transmission procedure ①Paper size setting SetMediaSize() Ex:for 8x5x2(front) set the parameter 238 ②Send BMP image A SendImageData() ③Send BMP image C SendImageData() ④Paper size setting SetMediaSize() Ex:for 8x5x2(back) set the parameter 338 ⑤Send BMP image B SendImageData() ⑥Send BMP image D SendImageData() (Example for printing 1 side, and putting an overcoat on the other side) ※If you only want to print on one side of the paper, and overcoat the other side, the BMP image data sent for the overcoat side should be blank. ①Paper size setting SetMediaSize() Ex：for 8x5x2(front) set parameter 238 ②Send BMP image A SendImageData() ③Send BMP image C SendImageData() ④Paper size setting SetMediaSize() Ex：for 8x5x2(back) set parameter 338 ⑤Send blank BMP image B SendImageData() ⑥Send blank BMP image D SendImageData()	
	Short edge		⇒		

◆ For single-side printing (Cut media, using the Turning unit)

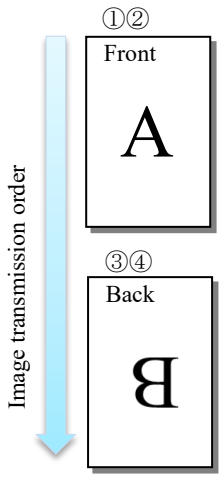
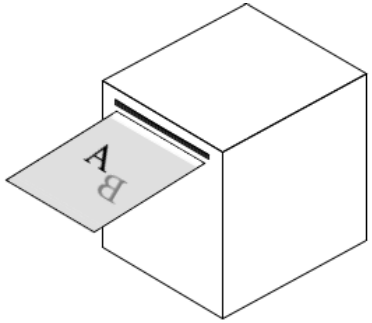
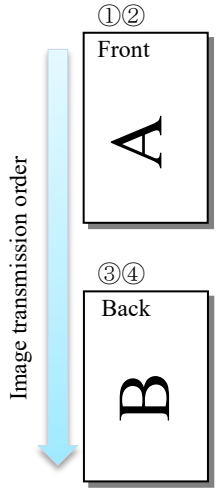
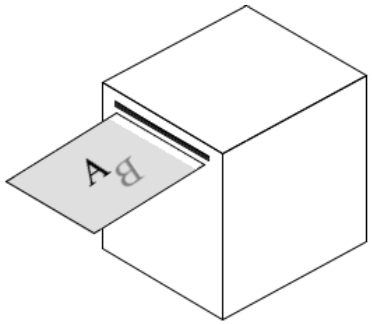
Paper type Output size	Duplex designation	Image creation、 Layout & transmit order		Printer Operation and Image Transmit Process	Media Delivery Image
Cut sheet Single-Layout	No	① ② Front A	⇒	Image transmission procedure ①Paper size setting SetMediaSize() Ex:for 8x10 set the parameter 131 ②Send BMP image A SendImageData() 	
Cut sheet Multi-layout	No	Image transmission order ① ② Front A 8x6 ③ Front B 8x6	⇒	Image transmission procedure ①Paper size setting SetMediaSize() Ex:for 8x5x2 set the parameter 138 ②Send BMP image A SendImageData() ③Send BMP image B SendImageData() 	

◆ For single-side printing (Roll media, not using the Turning unit)

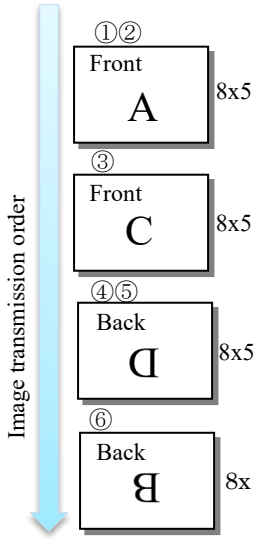
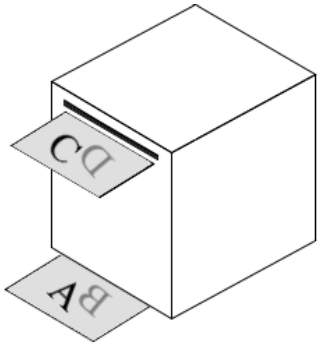
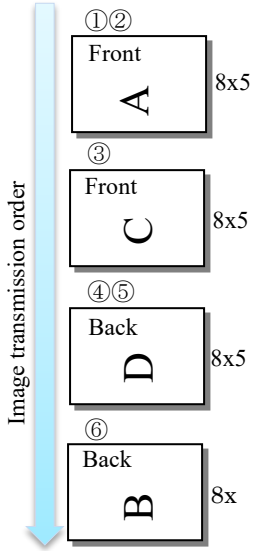
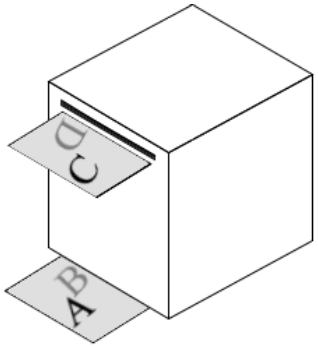
Paper type Output size	Duplex designation	Image creation、 Layout & transmit order		Printer Operation and Image Transmit Process	Media Delivery Image
Roll paper Single-Layout	No	① ② Front A	⇒	Image transmission procedure ①Paper size setting SetMediaSize() Ex:for 8x10 set the parameter 31 ②Send BMP image A SendImageData() 	
Roll paper Multi-layout	No	Image transmission order ① ② Front A 8x6 ③ Front B 8x6	⇒	Image transmission procedure ①Paper size setting SetMediaSize() Ex:for 8x6x2 set the parameter 39 ②Send BMP image A SendImageData() ③Send BMP image B SendImageData() 	

< For DP-DS820DX>

◆ For duplex printing (Duplex cut sheet, no layout.)

Paper type Output size	Duplex designation	Image creation、 Layout & transmit order		Image Transmit Process and Printer Operation	Media Delivery Image
Duplex Cut sheet Single-Layout	Long edge binding		⇒	Image transmission procedure ①Paper size setting SetMediaSize() Ex:for 8x10.75（front） set the parameter 252 ②Send BMP image A SendImageData() ③Paper size setting SetMediaSize() Ex:for 8x10.75（back） set the parameter 352 ④Send BMP image B SendImageData()	
	Short edge binding		⇒		

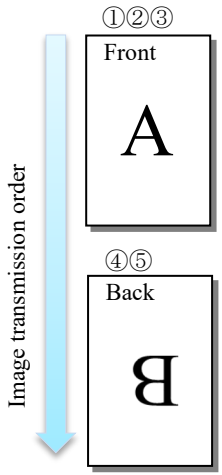
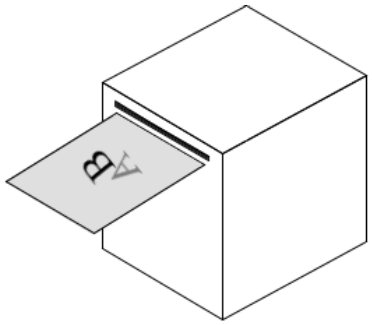
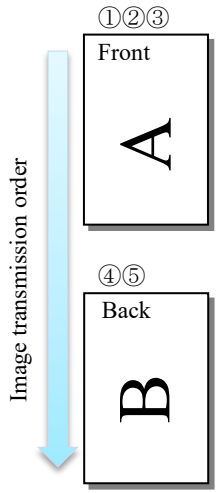
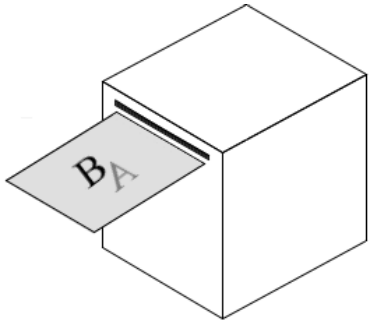
◆ For duplex printing (Duplex cut sheet, 2-image layout.)

<table>
<tr><th>Paper type
Output size</th><th>Duplex
designation</th><th>Image creation、
Layout & transmit order</th><th></th><th>Printer Operation and Image Transmit Process</th><th>Media Delivery Image</th></tr>
<tr><td rowspan="2">Duplex
Cut sheet
Multi-Layout</td><td>Long edge</td><td> 
</td><td>⇒</td><td rowspan="2">Image transmission procedure
①Paper size setting SetMediaSize() Ex:for 8x5x2(front) set the parameter 238
②Send BMP image A SendImageData()
③Send BMP image C SendImageData()
④Paper size setting SetMediaSize() Ex:for 8x5x2(back) set the parameter 338
⑤Send BMP image D SendImageData()
⑥Send BMP image B SendImageData()</td><td>  </td></tr>
<tr><td>Short edge</td><td> 
</td><td>⇒</td><td>  </td></tr>
</table>

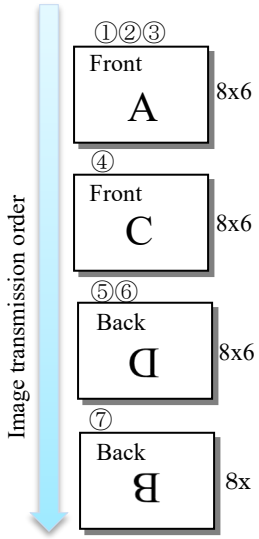
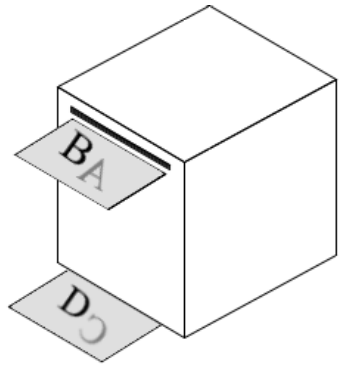
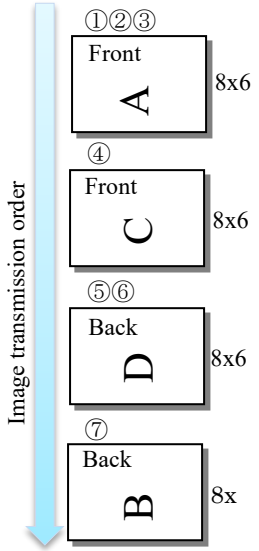
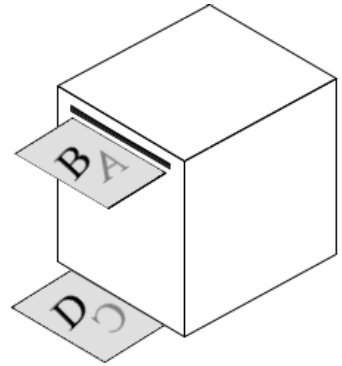
With DP-DS820DX, in the following cases, printed matter is output in a different direction than normal duplex printing.

- When the print size is 8x11, 8xA4, 8x12 and the crease line is set at a position exceeding 155.5mm from the reference point.
- When the print size is 8x6x2, 8x4x3 and the crease line is set.
- When the print size is 8x5.9x2

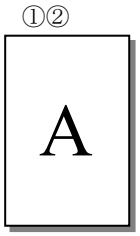
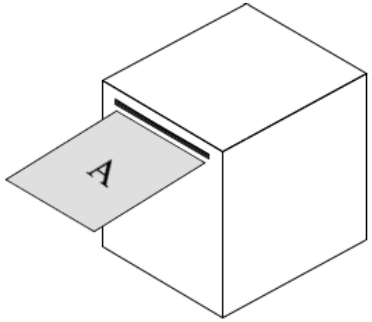
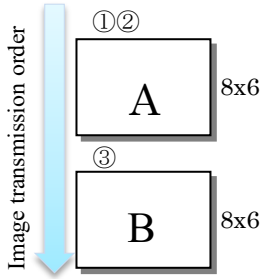
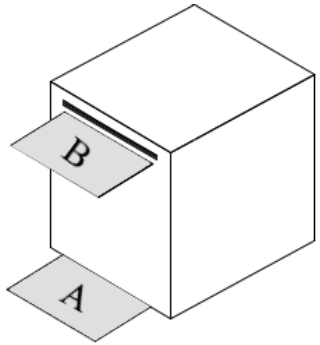
◆ When the print size is 8x12 and the crease line is set at a position exceeding 155.5mm from the reference point, and double-sided printing is performed.

Paper type Output size	Duplex designation	Image creation、 Layout & transmit order		Image Transmit Process and Printer Operation	Media Delivery Image
Duplex Cut sheet Single-Layout	Long edge binding		⇒	Image transmission procedure ①Setting crease line control SetCreaseLine() Ex: Set the crease line placement location to 1556 (155.6mm) ②Paper size setting SetMediaSize() Ex:for 8x12（front） set the parameter 232 ③Send BMP image A SendImageData() ④Paper size setting SetMediaSize() Ex:for 8x12（back） set the parameter 332 ⑤Send BMP image B SendImageData()	
	Short edge binding		⇒		

◆ When the print size is 8x6x2 and double-sided printing is performed with crease lines set.

<table>
<tr><th>Paper type
Output size</th><th>Duplex
designation</th><th>Image creation、
Layout & transmit order</th><th></th><th>Printer Operation and Image Transmit Process</th><th>Media Delivery Image</th></tr>
<tr><td rowspan="2">Duplex
Cut sheet
Multi-Layout</td><td>Long edge</td><td> 
</td><td>⇒</td><td rowspan="2">Image transmission procedure
①Setting crease line control SetCreaseLine() Ex: Set the crease line placement location to 762 (76.2mm)
②Paper size setting SetMediaSize() Ex:for 8x6x2(front) set the parameter 239
③Send BMP image A SendImageData()
④Send BMP image C SendImageData()
⑤Paper size setting SetMediaSize() Ex:for 8x6x2(back) set the parameter 339
⑥Send BMP image D SendImageData()
⑦Send BMP image B SendImageData()</td><td>  </td></tr>
<tr><td>Short edge</td><td> 
</td><td>⇒</td><td>  </td></tr>
</table>

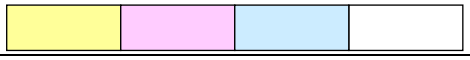
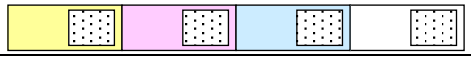
◆ For single-side printing (Roll media)

Paper type Output size	Duplex designation	Image creation、 Layout & transmit order		Printer Operation and Image Transmit Process	Media Delivery Image
Roll paper Single-Layout	No		⇒	Image transmission procedure ①Paper size setting SetMediaSize() Ex:for 8x10 set the parameter 31 ②Send BMP image A SendImageData()	
Roll paper Multi-layout	No		⇒	Image transmission procedure ①Paper size setting SetMediaSize() Ex:for 8x6x2 set the parameter 39 ②Send BMP image A SendImageData() ③Send BMP image B SendImageData()	

7.Ribbon Rewind Function (DP-DS620, DP-DS820)

・The host uses the Media Counter of Remaining Sheets information from the “Get Ribbon panel HALF Size Conversion Media Counter of Remaining Sheets” command to check the ribbon panel status. (DP-DS620 5x7*1, 6x8, 6x9*2, DP-DS820 8x10, 8x12, A4 media)

・If the ribbon panel Half Size Conversion Media Counter of Remaining Sheets returned is an even number, the ribbon panels are unused, and if it is an odd number, the panels are half-used.

Half Size Conversion Media Counter of Remaining Sheets	Ribbon Panel Status	
Even number	Unused condition	
Odd number	Half-used condition	

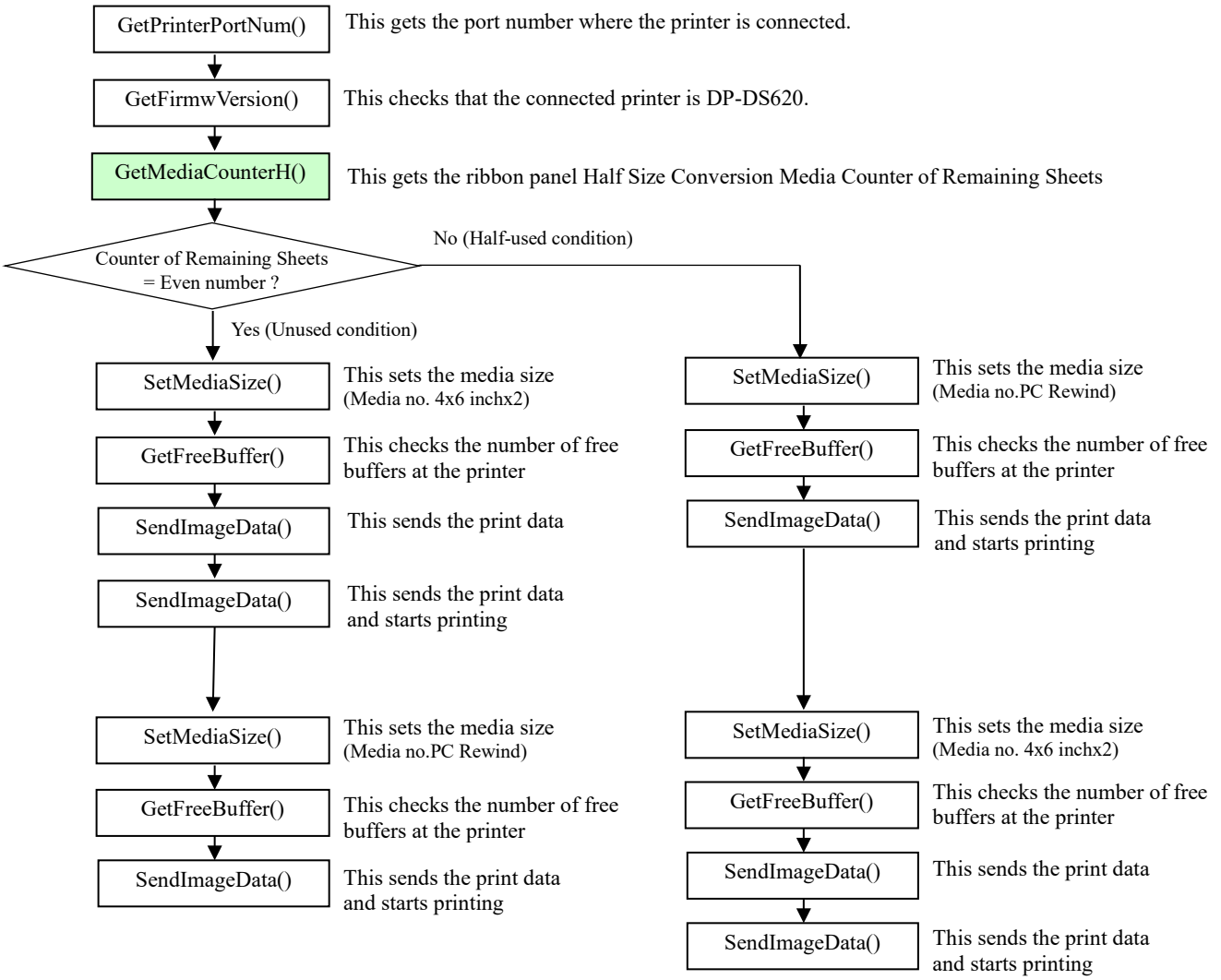
* When PC Rewind (CSP_PC_REWIND) is selected with the DP-DS620 (A), printing is performed without rewinding the ribbon. The same applies when L Rewind (CSP_L_REWIND *1) and 6x4.5 Rewind (CSP_6x4P5_REWIND *2) is selected.
Also if the DP-DS820 is used, printing is performed without rewinding the ribbon.

Note
*1 Firmware version 0.31 or later
*2 Firmware version 1.10 or later

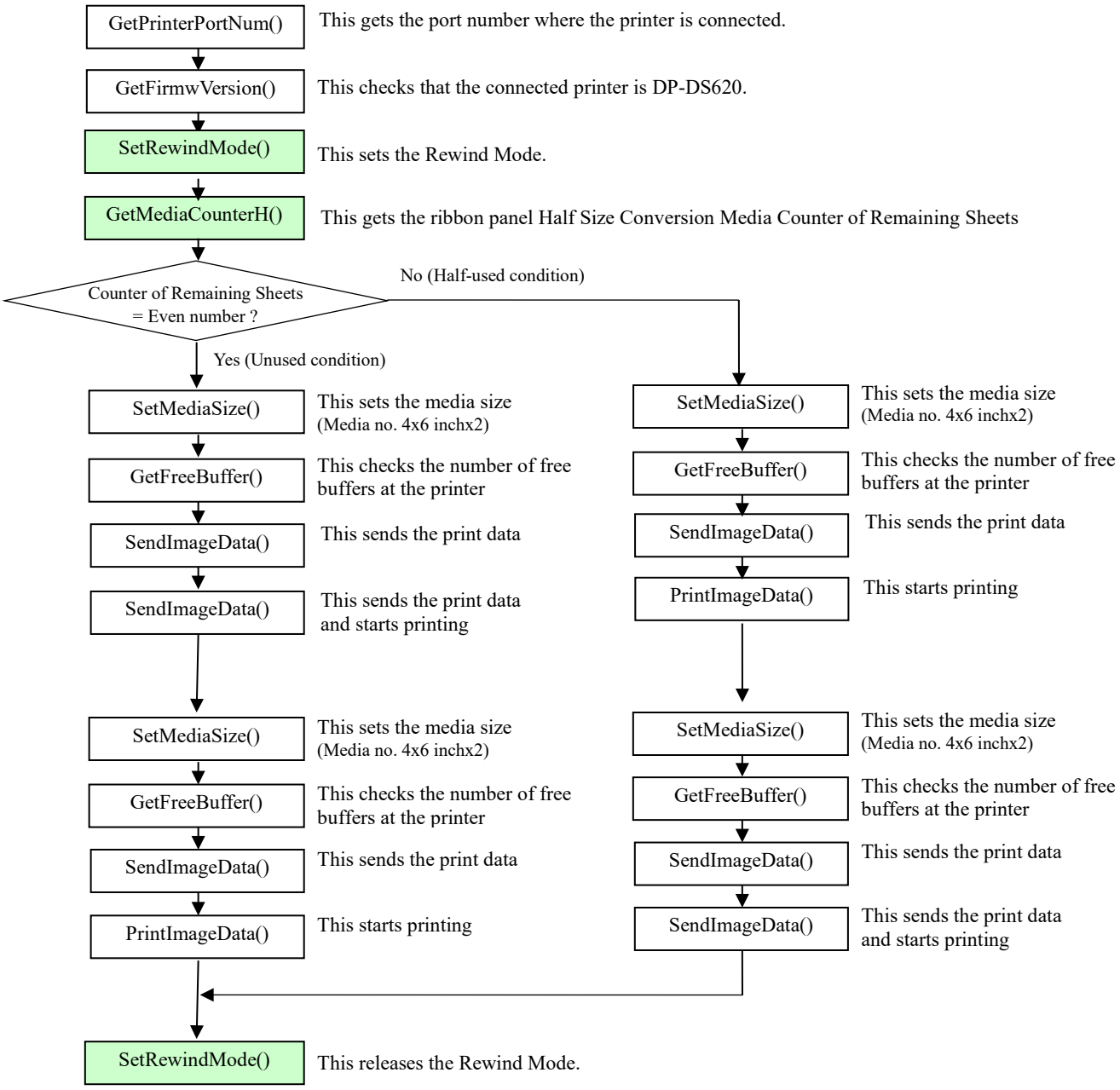
・It shows the example of printing to use the ribbon rewind function below.

Example of the Ribbon Rewind Function Process

1) This shows the process for the printing of three 6x4-size prints that specify the media size "PC Rewind (CSP_PC_REWIND)" or "PC 2-image layout (CSP_PCx2)".



2) This shows the process for the printing of three 6x4-size prints that specify the media size "PC 2-image layout (CSP_PCx2)" and use the Rewind Mode setting.



8.Standby Mode (DP-DS620, DP-DS820, DP-DS820DX)

- If the printer is Idle for 10 minutes (or the set time if the standby transition time command has been used to change it), it will enter Standby Mode.
- While in the Standby Mode, the printer status will be “STATUS_USUALLY_STANDBY_MODE”, and the green LED will change from lit to flashing.
- The API listed above can be used during Standby Mode.

< API that can be used in Standby Mode >

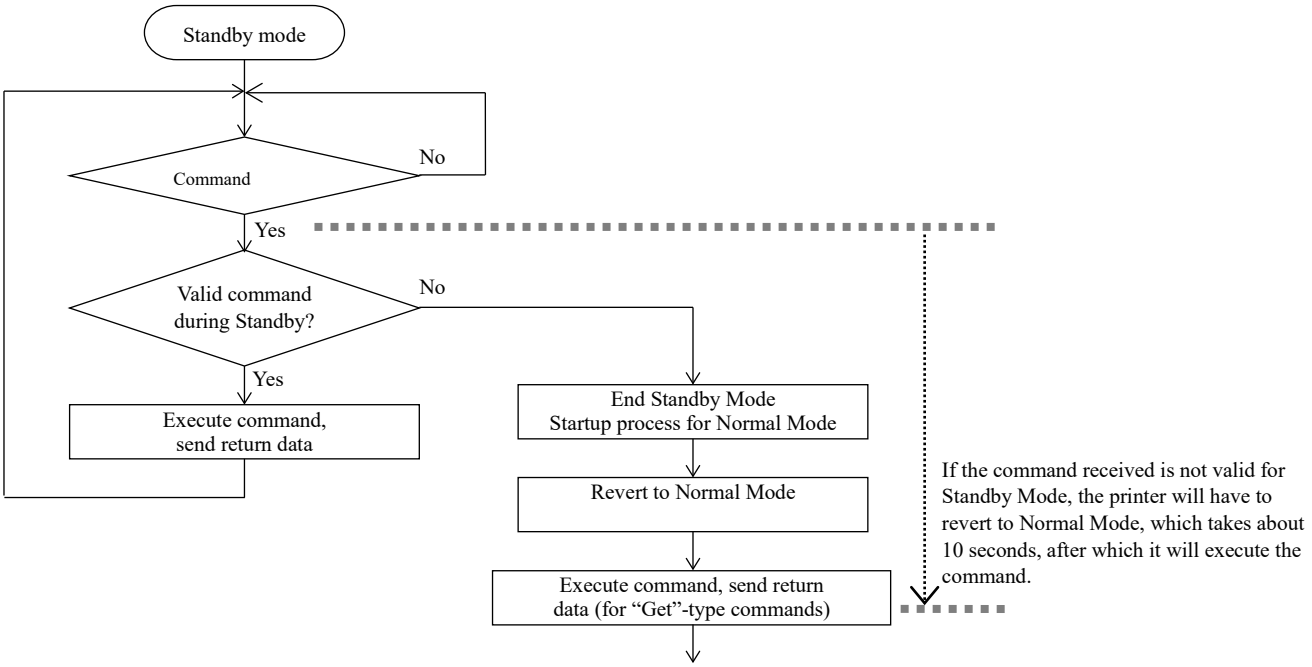
Function	API
Get Printer Status	CvGetStatus() / GetStatus()
Get Printer Serial Number	CvGetSerialNo() / GetSerialNo()
Get media counter default	GetInitialMediaCount()
Get Remaining Media Quantity	CvGetMediaCounter() / GetMediaCounter()
Get Half Size Conversion Ribbon Quantity	GetMediaCounterH()
Get Media User Info (Lot No.)	CvGetMediaLotNo() / GetMediaLotNo()
Get Media Code	CvGetMedia() / GetMedia()
Get Printer Version Information	CvGetVersion() /GetFirmwVersion() *1
Get Media Class	GetRfidMediaClass() *2

*1:DP-DS620 Firmware Version 0.31 or later

*2:Excluding DP-DS620 firmware version less than 3.00

- If any API other than those listed above while in Standby Mode is executed, or the cover is opened, the printer will come out of Standby Mode.
- If any API other than those listed above while in Standby Mode is executed, the printer will revert to Normal Mode before executing the process. In this case, the printer will need to undergo the startup process to revert to Normal Mode, and it will take approximately 10 seconds for the process to be executed. (See the chart below)

Printer operation when receiving commands during Standby



For “Get”-type commands that require return data from the printer, it will take approximately 10 seconds from the time of the command until the return data is sent from the printer. Therefore, after the Host issues the command, it will be necessary to wait while the return data is sent back from the printer.

9.Status API (CSPStat.Dll) function list

Operation of the API functions depends on the model connected and the firmware version compatibility. CSP Status API Ver.0.6.3.0 and later have the DLL filter function that checks the model and version. By using the filter function, the API function returns FALSE or (-1) if the printer is incompatible, and the command is not sent to the printer. By using the SetPrinterFilter() function on P32, you can turn the filter ON.

If the filter is turned OFF, operation may not be possible if the application uses printer-incompatible API functions. If the filter function is turned OFF, care must be taken that the application doesn't use printer-incompatible API functions.

9-1. Status API function list for CV, CW, CW02

○: Firmware compatible ×: Not firmware compatible(False or (-1))

Value: Compatible firmware version (Also compatible with subsequent versions)

API function	Direct Dll(CSP Stat.Dll) function name	Note	Standalone		
			CV	CW	CW02
Printer ID Number	GetPrinterPortNum();	-	○	○	○
Media size setting	SetMediaSize();	-	○	○	○
Send image data	SendImageData();	-	○	○	○
Printer resolution setting	SetResolution();	-	×	×	○
Designating print quantity	SetPQTY();	-	○	○	○
Start printing	PrintImageData();	-	○	○	○
Start page layout	StartPageLayout();	-	○	○	○
End page layout	EndPageLayout();	-	○	○	○
USB timeout setting	SetUSBTimeout();	-	○	○	○
CXM(for Tower) Function					
Get minilab tower status	GetMiniLabTowerStatis()	-	×	×	×
Minilab print data group no. Serial no. initial setting	SetMiniLabGroupNoIni()	-	×	×	×
Setting the miniLab print data group number	SetMiniLabGroupNo()	-	×	×	×
Setting the miniLab print data serial number	SetMiniLabSerialNo()	-	×	×	×
Setting miniLab back print data	SetMiniLabBackprintData()	-	×	×	×
Stop printing command	StopPrint()	-	×	×	×
Designation of paper feed destination	SetMiniLabOutputDest()	-	×	×	×
Set sorter paper max position detection sensor ON/OFF	SetMiniLabSortSensor()	-	×	×	×
Get sorter paper max position detection sensor ON/OFF	GetMiniLabSortSensor()	-	×	×	×
Get minilab group No.	GetMiniLabGroupNo()	-	×	×	×
Get minilab serial No.	GetMiniLabSerialNo()	-	×	×	×
Get printer address	GetMiniLabPrinterAdd()	-	×	×	×

API function	Direct Dll(CSP Stat.Dll) function name		Note	Standalone		
				CV	CW	CW02
Get printer version information	CvGetVersion();	GetFirmwVersion();	−	○	○	○
Get printer sensor information	CvGetSensorInfo();	GetSensorInfo();	−	○	○	○
Get printer resolution	CvGetResolutionH(); CvGetResolutionV();	GetResolutionH(); GetResolutionV();	−	○	○	○
Get printer media code	CvGetMedia();	GetMedia();	−	○	○	○
Get printer status	CvGetStatus();	GetStatus();	−	○	○	○
Get printer counter value	CvGetCounterL(); CvGetCounterA() CvGetCounterB();	GetCounterL(); GetCounterA(); GetCounterB();	−	○	○	○
		GetCounterP();	−	×	×	○
		GetCounterMatte() GetCounterM()	−	×	×	○
		GetCounterDuplex();	-	×	×	×
Set printer counter value (P)	SetCounterP();		−	×	×	○
Clear printer counter value	CvSetClearCounterA(); CvSetClearCounterB();	SetClearCounterA(); SetClearCounterB();	−	○	○	○
Clear printer counter value(Matte)	SetClearCounterM()		−	×	×	○
Get the number of free image buffers	CvGetFreeBuffer();	GetFreeBuffer();	−	○	○	○
Get remaining print quantity	CvGetPQTY();	GetPQTY();	−	○	○	○
Get the media counter of remaining sheets	CvGetMediaCounter();	GetMediaCounter();	−	○	○	○
Half Size Conversion Media Counter of Remaining Sheets	GetMediaCounterH();		−	×	×	×
Get media color offset value of the lot	CvGetMediaColorOffset();	GetMediaColorOffset();	−	○	○	○
Get media lot information	CvGetMediaLotNo();	GetMediaLotNo();	−	○	○	○
Get printer serial number	CvGetSerialNo();	GetSerialNo();	−	○	○	○
Set printer update mode	CvSetFirmwUpdateMode();	SetFirmwUpdateMode();	−	○	○	○
Write firmware data	CvSetFirmwDataWrite();	SetFirmwDataWrite();	−	○	○	○
Clear color data	CvSetColorDataClear();	SetColorDataClear();	−	×	○	×

API function	Direct Dll(CSP Stat.Dll) function name		Note	Standalone		
				CV	CW	CW02
Write color data	CvSetColorDataWrite();	SetColorDataWrite();	–	×	○	×
Set color data version	CvSetColorDataVersion();	SetColorDataVersion();	–	×	○	×
Write color data (temporary)	SetColorDataWriteTmp();		–	×	○	×
Set color data version(temporary)	SetColorDataVersionTmp();		–	×	○	×
Get color data version	CvGetColorDataVersion();	GetColorDataVersion();	–	○	○	○
Get Color Data Version <Type designation>	GetColorDataVersionRes();		-	×	×	×
Get Color Data Version <Type Media designation>	GetColorDataVersionResEX();		–	×	×	×
Get color data checksum	CvGetColorDataChecksum();	GetColorDataChecksum();	–	○	○	○
Get Color Data Checksum <Type designation>	GetColorDataChecksumRes();		-	×	×	×
Get Color Data Checksum Ex<Type、Media designation>	GetColorDataChecksumResEX();		–	×	×	×
Cutter control command	SetCutterMode();		–	○	○	○
Get media ID setting value	GetMediaIdSetInfo();		–	×	○	○
Set 1-image retention control command	SetRetentionMode();		–	×	×	×
Overcoat finish control command	SetOvercoatFinish();		–	×	×	○
Print retry control command	SetRetryControl();		–	×	×	○
Get media class	GetRfidMediaClass();		–	×	×	×
Get RF-ID reserve data	GetRfidReserveData();		–	×	×	×
Get media counter default	GetInitialMediaCount();		–	×	×	×
Get media offset count	GetMediaCountOffset();		–	×	×	×
Get Turning unit firmware version	GetVersionDuplex();		–	×	×	×
Full Cutter Set-up	SetFullCutterSetUp();		–	×	×	×
Full Cutter Set-up Extended Settings	SetFullCutterSetUpEX();		–	×	×	×
Set Standby Mode Transition Time	SetStandbyTime();		–	×	×	×
Get Standby Mode Transition Time	GetStandbyTime();		–	×	×	×
Set Media End Keep Mode	SetEndKeepMode();		–	×	×	×
Get Media End Keep Mode	GetEndKeepMode();		–	×	×	×
Set USB iSerialNumber availability setting	SetUSBSerialEnable();		–	×	×	×
Get USB iSerialNumber availability setting	GetUSBSerialEnable();		–	×	×	×
Set Rewind Mode setting	SetRewindMode();		–	×	×	×
Get Rewind Mode setting	GetRewindMode();		–	×	×	×
Panorama Print Start Check	GetPanoramaPrintable();		–	×	×	×
Continuous Panoramic Prints settings	SetContPanorama();		–	×	×	×
Set Print Speed	SetPrintSpeed();		–	×	×	×
Set Gamma Table	SetGammaTable();		–	×	×	×
Get Gamma Table Checksum	GetGammaTableChecksum();		–	×	×	×
Clear Data Table	SetDataTableClear();		–	×	×	×
Set Turning unit rewrite mode	SetDunitFirmwUpdateMode();		–	×	×	×
Write Turning unit firmware data	SetDunitFirmwDataWrite();		–	×	×	×
Get Turning unit firmware rewriting status	GetDunitUpdateStatus();		–	×	×	×
Request Creation of Turning Unit Error Log	DunitErrorLogCreateReq();		–	×	×	×
Get Turning Unit Error Log Data Length	GetDunitErrorLogDataLength();		–	×	×	×
Receive the Turning Unit Error Log Data	GetDunitErrorLogData();		–	×	×	×
Get the Turning Unit Error Log Creation Status	GetDunitErrorLogStatus();		–	×	×	×
Set decurl control	SetDecurlCtrl();		–	×	×	×
Current time notification	SetSysTime();		–	×	×	×
Get Supported Media Information	GetSupportedMediaInfo();		–	×	×	×
Setting crease line control	SetCreaseLine();		–	×	×	×
Setting slitter control	SetSlitterCtrl();		–	×	×	×
Creaser unit disabling settings	SetCreaserUnitDisable();		–	×	×	×
Get the value of creaser unit disable setting	GetCreaserUnitDisable();		–	×	×	×
Slitter unit disabling settings	SetSlitterUnitDisable();		–	×	×	×
Get the value of slitter unit disable setting	GetSlitterUnitDisable();		–	×	×	×
Band unevenness prevention operation setting	SetPreventBandUnevenness();		–	×	×	×
Common set command	CvSetCommand();	SetCommand();	–	○	○	○
Common get command	CvGetCommandEX();	GetCommandEX();	–	○	○	○

9-2. Status API function list for DS40, DS80, DS40T, DS80T

○: Firmware compatible　×: Not firmware compatible(False or (-1))

Value: Compatible firmware version (Also compatible with subsequent versions)

API function	Direct Dll(CSP Stat.Dll) function name	Note	Standalone		For Tower	
			DS40	DS80	DS40T	DS80T
Printer ID Number	GetPrinterPortNum();	-	○	○	○	○
Media size setting	SetMediaSize();	-	○	○	○	○
Send image data	SendImageData();	-	○	○	○	○
Printer resolution setting	SetResolution();	-	○	○	○	○
Designating print quantity	SetPQTY();	-	○	○	○	○
Start printing	PrintImageData();	-	○	○	○	○
Start page layout	StartPageLayout();	-	○	○	○	○
End page layout	EndPageLayout();	-	○	○	○	○
USB timeout setting	SetUSBTimeout();	-	○	○	○	○
CXM(for Tower) Function						
Get minilab tower status	GetMiniLabTowerStatis();	-	×	×	○	○
Minilab print data group no. Serial no. initial setting	SetMiniLabGroupNoIni();	-	×	×	○	○
Setting the miniLab print data group number	SetMiniLabGroupNo();	-	×	×	○	○
Setting the miniLab print data serial number	SetMiniLabSerialNo();	-	×	×	○	○
Setting miniLab back print data	SetMiniLabBackprintData();	-	×	×	○	○
Stop printing command	StopPrint();	-	×	×	○	○
Designation of paper feed destination	SetMiniLabOutputDest();	-	×	×	○	○
Set sorter paper max position detection sensor ON/OFF	SetMiniLabSortSensor();	-	×	×	○	○
Get sorter paper max position detection sensor ON/OFF	GetMiniLabSortSensor();	-	×	×	○	○
Get minilab group No.	GetMiniLabGroupNo();	-	×	×	○	○
Get minilab serial No.	GetMiniLabSerialNo();	-	×	×	○	○
Get printer address	GetMiniLabPrinterAdd();	-	×	×	○	○

API function	Direct Dll(CSP Stat.Dll) function name		Note	Standalone		For Tower	
				DS40	DS80	DS40T	DS80T
Get printer version information	CvGetVersion();	GetFirmwVersion();	−	○	○	○	○
Get printer sensor information	CvGetSensorInfo();	GetSensorInfo();	−	○	○	○	○
Get printer resolution	CvGetResolutionH(); CvGetResolutionV();	GetResolutionH(); GetResolutionV();	−	○	○	○	○
Get printer media code	CvGetMedia();	GetMedia();	−	○	○	○	○
Get printer status	CvGetStatus();	GetStatus();	−	○	○	○	○
Get printer counter value	CvGetCounterL(); CvGetCounterA(); CvGetCounterB();	GetCounterL(); GetCounterA(); GetCounterB();	−	○	○	○	○
		GetCounterP();	−	1.04	1.02	0.21	0.21
		GetCounterMatte(); GetCounterM();	−	1.30	1.30	0.31	0.31
		GetCounterDuplex();	-	×	×	×	×
Set printer counter value (P)	SetCounterP();		−	1.04	1.02	0.21	0.21
Clear printer counter value	CvSetClearCounterA(); CvSetClearCounterB();	SetClearCounterA(); SetClearCounterB();	−	○	○	○	○
Clear printer counter value(Matte)	SetClearCounterM();		−	1.30	1.30	0.31	0.31
Get the number of free image buffers	CvGetFreeBuffer();	GetFreeBuffer();	−	○	○	○	○
Get remaining print quantity	CvGetPQTY();	GetPQTY();	−	○	○	○	○
Get the media counter of remaining sheets	CvGetMediaCounter();	GetMediaCounter();	−	○	○	○	○
Half Size Conversion Media Counter of Remaining Sheets	GetMediaCounterH();		−	×	×	×	×
Get media color offset value of the lot	CvGetMediaColorOffset();	GetMediaColorOffset();	−	○	○	○	○
Get media lot information	CvGetMediaLotNo();	GetMediaLotNo();	−	○	○	○	○
Get printer serial number	CvGetSerialNo();	GetSerialNo();	−	○	○	○	○
Set printer update mode	CvSetFirmwUpdateMode();	SetFirmwUpdateMode();	−	○	○	○	○
Write firmware data	CvSetFirmwDataWrite();	SetFirmwDataWrite();	−	○	○	○	○
Clear color data	CvSetColorDataClear();	SetColorDataClear();	−	×	×	×	×
Write color data	CvSetColorDataWrite();	SetColorDataWrite();	−	×	×	×	×
Set color data version	CvSetColorDataVersion();	SetColorDataVersion();	−	×	×	×	×
Write color data (temporary)	SetColorDataWriteTmp();		−	×	×	×	×
Set color data version(temporary)	SetColorDataVersionTmp();		−	×	×	×	×
Get color data version	CvGetColorDataVersion();	GetColorDataVersion();	−	○	○	○	○
Get Color Data Version <Type designation>	GetColorDataVersionRes ();		-	×	×	×	×
Get Color Data Version <Type Media designation>	GetColorDataVersionResEX ();		−	×	×	×	×
Get color data checksum	CvGetColorDataChecksum();	GetColorDataChecksum();	−	○	○	○	○

API function	Direct Dll(CSP Stat.Dll) function name		Note	Standalone		For Tower	
				DS40	DS80	DS40T	DS80T
Get Color Data Checksum <Type designation>	GetColorDataChecksumRes();		-	×	×	×	×
Get Color Data Checksum Ex<Type、Media designation>	GetColorDataChecksumResEX();		−	×	×	×	×
Cutter control command	SetCutterMode();		−	○	○	○	○
Get media ID setting value	GetMediaIdSetInfo();		−	○	○	○	○
Set 1-image retention control command	SetRetentionMode();		−	○	×	○	×
Overcoat finish control command	SetOvercoatFinish()		−	1.30	1.30	0.31	0.31
Print retry control command	SetRetryControl()		−	1.30	1.30	×	×
Get media class	GetRfidMediaClass()		−	×	×	×	×
Get RF-ID reserve data	GetRfidReserveData()		−	×	×	×	×
Get media counter default	GetInitialMediaCount()		−	2.00	×	×	×
Get media offset count	GetMediaCountOffset()		−	2.00	×	×	×
Get Turning unit firmware version	GetVersionDuplex()		−	×	×	×	×
Full Cutter Set-up	SetFullCutterSetUp();		−	1.60	×	×	×
Full Cutter Set-up Extended Settings	SetFullCutterSetUpEX()		−	1.60	×	×	×
Set Standby Mode Transition Time	SetStandbyTime();		−	×	×	×	×
Get Standby Mode Transition Time	GetStandbyTime();		−	×	×	×	×
Set Media End Keep Mode	SetEndKeepMode ();		−	×	×	×	×
Get Media End Keep Mode	GetEndKeepMode ();		−	×	×	×	×
Set USB iSerialNumber availability setting	SetUSBSerialEnable();		−	2.00	×	×	×
Get USB iSerialNumber availability setting	GetUSBSerialEnable();		−	2.00	×	×	×
Set Rewind Mode setting	SetRewindMode();		−	×	×	×	×
Get Rewind Mode setting	GetRewindMode();		−	×	×	×	×
Panorama Print Start Check	GetPanoramaPrintable()		−	×	×	×	×
Continuous Panoramic Prints settings	SetContPanorama()		−	×	×	×	×
Set Print Speed	SetPrintSpeed()		−	×	×	×	×
Set Gamma Table	SetGammaTable()		−	×	×	×	×
Get Gamma Table Checksum	GetGammaTableChecksum()		−	×	×	×	×
Clear Data Table	SetDataTableClear();		−	×	×	×	×
Set Turning unit rewrite mode	SetDunitFirmwUpdateMode();		−	×	×	×	×
Write Turning unit firmware data	SetDunitFirmwDataWrite();		−	×	×	×	×
Get Turning unit firmware rewriting status	GetDunitUpdateStatus();		−	×	×	×	×
Request Creation of Turning Unit Error Log	DunitErrorLogCreateReq();		−	×	×	×	×
Get Turning Unit Error Log Data Length	GetDunitErrorLogDataLength();		−	×	×	×	×
Receive the Turning Unit Error Log Data	GetDunitErrorLogData();		−	×	×	×	×
Get the Turning Unit Error Log Creation Status	GetDunitErrorLogStatus();		−	×	×	×	×
Set decurl control	SetDecurlCtrl();		−	×	×	×	×
Current time notification	SetSysTime();		−	×	×	×	×
Get Supported Media Information	GetSupportedMediaInfo();		−	×	×	×	×
Setting crease line control	SetCreaseLine();		−	×	×	×	×
Setting slitter control	SetSlitterCtrl();		−	×	×	×	×
Creaser unit disabling settings	SetCreaserUnitDisable();		−	×	×	×	×
Get the value of creaser unit disable setting	GetCreaserUnitDisable();		−	×	×	×	×
Slitter unit disabling settings	SetSlitterUnitDisable();		−	×	×	×	×
Get the value of slitter unit disable setting	GetSlitterUnitDisable();		−	×	×	×	×
Band unevenness prevention operation setting	SetPreventBandUnevenness();		−	×	×	×	×
Common set command	CvSetCommand();	SetCommand();	−	○	○	○	○
Common get command	CvGetCommandEX();	GetCommandEX();	−	○	○	○	○

9-3. Status API function list for DS-RX1, DP-TC10, DP-DS80D, DP-DS620, DP-DS820, DP-DS820DX, DP-QW410

○: Firmware compatible　×: Not firmware compatible(False or (-1))　Value: Compatible firmware version (Also compatible with subsequent versions)

API function	Direct Dll(CSP Stat.Dll) function name	Note	Standalone						
			DS-RX1	DP-TC10	Duplex DP-DS80D	DP-DS620	DP-DS820	DP-DS820DX	DP-QW410
Printer ID Number	GetPrinterPortNum();	-	○	○	○	○	○	○	○
Media size setting	SetMediaSize();	-	○	○	○	○	○	○	○
Send image data	SendImageData();	*3	○	○	○	○	○	○	○
Printer resolution setting	SetResolution();	-	○	○	○	○	○	○	○
Designating print quantity	SetPQTY();	*3	○	○	○	○	○	○	○
Start printing	PrintImageData();	*3	○	○	○	○	○	○	○
Start page layout	StartPageLayout();	-	○	○	○	○	○	○	○
End page layout	EndPageLayout();	-	○	○	○	○	○	○	○
USB timeout setting	SetUSBTimeout();	-	○	○	○	○	○	○	○
CXM(for Tower) Function									
Get minilab tower status	GetMiniLabTowerStatis();	-	×	×	×	×	×	×	×
Minilab print data group no. Serial no. initial setting	SetMiniLabGroupNoIni();	-	×	×	×	×	×	×	×
Setting the miniLab print data group number	SetMiniLabGroupNo();	-	×	×	×	×	×	×	×
Setting the miniLab print data serial number	SetMiniLabSerialNo();	-	×	×	×	×	×	×	×
Setting miniLab back print data	SetMiniLabBackprintData();	-	×	×	×	×	×	×	×
Stop printing command	StopPrint();	-	×	×	×	×	×	×	×
Designation of paper feed destination	SetMiniLabOutputDest();	-	×	×	×	×	×	×	×
Set sorter paper max position detection sensor ON/OFF	SetMiniLabSortSensor();	-	×	×	×	×	×	×	×
Get sorter paper max position detection sensor ON/OFF	GetMiniLabSortSensor();	-	×	×	×	×	×	×	×
Get minilab group No.	GetMiniLabGroupNo();	-	×	×	×	×	×	×	×
Get minilab serial No.	GetMiniLabSerialNo();	-	×	×	×	×	×	×	×
Get printer address	GetMiniLabPrinterAdd();	-	×	×	×	×	×	×	×

API function	Direct Dll(CSP Stat.Dll) function name		Note	Standalone						
				DS-RX1	DP-TC10	Duplex DP-DS80D	DP-DS620	DP-DS820	DP-DS820DX	DP-QW410
Get printer version information	CvGetVersion();	GetFirmwVersion();	*1	○	○	○	○	○	○	○
Get printer sensor information	CvGetSensorInfo();	GetSensorInfo();	*2	○	○	○	○	○	○	○
Get printer resolution	CvGetResolutionH(); CvGetResolutionV();	GetResolutionH(); GetResolutionV();	*2	○	○	○	○	○	○	○
Get printer media code	CvGetMedia();	GetMedia();	*1	○	○	○	○	○	○	○
Get printer status	CvGetStatus();	GetStatus();	*1	○	○	○	○	○	○	○
Get printer counter value	CvGetCounterL(); CvGetCounterA() CvGetCounterB();	GetCounterL(); GetCounterA(); GetCounterB();	*2	○	○	○	○	○	○	○
		GetCounterP();	*2	○	○	○	○	○	○	○
		GetCounterMatte(); GetCounterM();	*2	○	○	○	○	○	○	○
		GetCounterDuplex();	-	×	×	○	×	×	0.52	×
Get Life Counter Value Extended	GetCounterLifeEX();		*3	×	×	×	4.00	×	×	×
Get Matte Counter Value Extended	GetCounterMatteEX();		*3	×	×	×	4.00	×	×	×
Set printer counter value (P)	SetCounterP();		*3	○	○	○	○	○	○	○
Clear printer counter value	CvSetClearCounterA(); CvSetClearCounterB();	SetClearCounterA(); SetClearCounterB();	*3	○	○	○	○	○	○	○
Clear printer counter value(Matte)	SetClearCounterM();		*3	○	○	○	○	○	○	○
Get the number of free image buffers	CvGetFreeBuffer();	GetFreeBuffer();	*2	○	○	○	○	○	○	○
Get remaining print quantity	CvGetPQTY();	GetPQTY();	*2	○	○	○	○	○	○	○
Get the media counter of remaining sheets	CvGetMediaCounter();	GetMediaCounter();	*1	○	○	○	○	○	○	○
Half Size Conversion Media Counter of Remaining Sheets	GetMediaCounterH();		*1	×	×	×	○	○	○	×
Get media color offset value of the lot	CvGetMediaColorOffset();	GetMediaColorOffset();	*2	○	○	○	○	○	○	○
Get media lot information	CvGetMediaLotNo();	GetMediaLotNo();	*1	○	○	○	○	○	○	○
Get printer serial number	CvGetSerialNo();	GetSerialNo();	*1	○	○	○	○	○	○	○
Set printer update mode	CvSetFirmwUpdateMode();	SetFirmwUpdateMode();	*3	○	○	○	○	○	○	○
Write firmware data	CvSetFirmwDataWrite();	SetFirmwDataWrite();	-	○	○	○	○	○	○	○
Clear color data	CvSetColorDataClear();	SetColorDataClear();	*3	×	×	×	○	○	○	○
Write color data	CvSetColorDataWrite();	SetColorDataWrite();	*3	×	×	×	○	○	○	○
Set color data version	CvSetColorDataVersion();	SetColorDataVersion();	*3	×	×	×	○	○	○	○
Write color data (temporary)	SetColorDataWriteTmp();		-	×	×	×	×	×	×	×
Set color data version(temporary)	SetColorDataVersionTmp();		-	×	×	×	×	×	×	×
Get color data version	CvGetColorDataVersion();	GetColorDataVersion();	*2	○	○	○	○	○	○	○
Get Color Data Version <Type designation>	GetColorDataVersionRes();		*2	×	×	×	○	○	○	○
Get Color Data Version <Type Media designation>	GetColorDataVersionResEX();		*2	×	×	×	3.00	○	○	○
Get color data checksum	CvGetColorDataChecksum();	GetColorDataChecksum();	*2	○	○	○	○	○	○	○
Get Color Data Checksum <Type designation>	GetColorDataChecksumRes();		*2	×	×	×	○	○	○	○
Get Color Data Checksum <Type、Media designation>	GetColorDataChecksumResEX();		*2	×	×	×	3.00	○	○	○
Cutter control command	SetCutterMode();		*3	○	○	○※1	○	○	○※1	○

API function	Direct Dll(CSP Stat.Dll) function name	Note	Standalone						
			DS-RX1	DP-TC10	Duplex DP-DS80D	DP-DS620	DP-DS820	DP-DS820DX	DP-QW410
Get media ID setting value	GetMediaIdSetInfo();	*2	○	○	○	○	○	○	○
Set 1-image retention control command	SetRetentionMode();	-	×	×	×	×	×	×	×
Overcoat finish control command	SetOvercoatFinish();	*3	○	○	○	○	○	○	○
Print retry control command	SetRetryControl();	*3	○	○	○※1	○	○	○※1	○
Get media class	GetRfidMediaClass();	*5	○	○	×	○	○	○	○
Get RF-ID reserve data	GetRfidReserveData();	*2	○	○	×	○	○	○	○
Get media counter default	GetInitialMediaCount();	*1	○	○	×	○	○	○	○
Get media offset count	GetMediaCountOffset();	-	2.00	×	×	×	×	×	×
Get Turning unit firmware version	GetVersionDuplex();	-	×	×	○	×	×	×	×
Full Cutter Set-up	SetFullCutterSetUp();	*3	×	×	×	○	○	×	○
Full Cutter Set-up Extended Settings	SetFullCutterSetUpEX();	*3	×	×	×	1.20	○	×	○
Set Standby Mode Transition Time	SetStandbyTime();	*3	×	×	×	○	○	○	×
Get Standby Mode Transition Time	GetStandbyTime();	*2	×	×	×	○	○	○	×
Set Media End Keep Mode	SetEndKeepMode();	*3	×	×	×	○	○	○	○
Get Media End Keep Mode	GetEndKeepMode();	*2	×	×	×	○	○	○	○
Set USB iSerialNumber availability setting	SetUSBSerialEnable();	*3	2.00	×	×	○	○	○	○
Get USB iSerialNumber availability setting	GetUSBSerialEnable();	*2	2.00	×	×	○	○	○	○
Set Rewind Mode setting	SetRewindMode();	-	×	×	×	○	○	×	×
Get Rewind Mode setting	GetRewindMode();	-	×	×	×	○	○	×	×
Panorama Print Start Check	GetPanoramaPrintable();	*2	×	×	×	1.20	○	×	×
Continuous Panoramic Prints settings	SetContPanorama();	*3	×	×	×	1.31	0.50	×	×
Set Print Speed	SetPrintSpeed();	*3	×	×	×	×	○	○	○
Set Gamma Table	SetGammaTable();	*3	×	×	×	×	0.50	○	×
Get Gamma Table Checksum	GetGammaTableChecksum();	*2	×	×	×	×	0.50	○	×
Clear Data Table	SetDataTableClear();	*3	×	×	×	×	0.50	○	×
Set Turning unit rewrite mode	SetDunitFirmwUpdateMode();	-	×	×	1.11	×	×	×	×
Write Turning unit firmware data	SetDunitFirmwDataWrite();	-	×	×	1.11	×	×	×	×
Get Turning unit firmware rewriting status	GetDunitUpdateStatus();	-	×	×	1.11	×	×	×	×
Request Creation of Turning Unit Error Log	DunitErrorLogCreateReq();	-	×	×	1.17	×	×	×	×
Get Turning Unit Error Log Data Length	GetDunitErrorLogDataLength();	-	×	×	1.17	×	×	×	×
Receive the Turning Unit Error Log Data	GetDunitErrorLogData();	-	×	×	1.17	×	×	×	×
Get the Turning Unit Error Log Creation Status	GetDunitErrorLogStatus();	-	×	×	1.17	×	×	×	×
Set decurl control	SetDecurlCtrl();	-	×	×	×	×	×	×	○
Current time notification	SetSysTime();	-	×	×	×	×	×	×	○
Get Supported Media Information	GetSupportedMediaInfo();	*2	×	×	×	4.00	×	×	×
Setting crease line control	SetCreaseLine();	*3	×	×	×	×	×	○	×
Setting slitter control	SetSlitterCtrl();	*3	×	×	×	×	×	0.52	×
Creaser unit disabling settings	SetCreaserUnitDisable();	*3	×	×	×	×	×	0.92	×
Get the value of creaser unit disable setting	GetCreaserUnitDisable();	*2	×	×	×	×	×	0.92	×

API function	Direct Dll(CSP Stat.Dll) function name		Note	Standalone						
				DS-RX1	DP-TC10	Duplex DP-DS80D	DP-DS620	DP-DS820	DP-DS820DX	DP-QW410
Slitter unit disabling settings	SetSlitterUnitDisable();		*3	×	×	×	×	×	0.92	×
Get the value of slitter unit disable setting	GetSlitterUnitDisable();		*2	×	×	×	×	×	0.92	×
Band unevenness prevention operation setting	SetPreventBandUnevenness();		*3	×	×	×	×	×	1.05	×
Common set command	CvSetCommand();	SetCommand();	*4	○	○	○	○	○	○	○
Common get command	CvGetCommandEX();	GetCommandEX();	*4	○	○	○	○	○	○	○

※1: Only compatible when printing on roll media.

Note (Compatible model= DP-DS620, DP-DS820, DP-DS820DX)

*1: The printer can send return data even in Standby mode.

*2: During Standby mode, the printer will return to normal mode before carrying out the command and sending the return data (sends after approx. 10 sec).
The Host will have to wait for the printer to send the return data.

*3: During Standby mode, the printer will return to normal mode before carrying out the command (performs command after approx. 10 sec).

*4: During Standby mode, the behavior of the printer depends on the type of command to be issued.

*5: DP-DS820 printers, DP-DS820DX printers and the firmware version of the DP-DS620 printer is 3.00 or later, the same operation as * 1. If the firmware version of the DP-DS620 printer is less than 3.00, the behavior is the same as * 2.

(Refer to8.Standby Mode)

10.About the API function movement of the KIOSK model

Movement when KIOSK F/W was written in at a tower controller, and an API function for towers was called.

	Command	CSP Printer Status API function	DS40T	DS80T
Setting	Setting the MiniLab Print Data Group Number	SetMiniLabGroupNo( )	○	○
	Minilab Print Data Group No., Serial No.Initial Setting	SetMiniLabGroupNoIni( )	○	○
	Setting the MiniLab Print Data Serial Number	SetMiniLabSerialNo( )	○	○
	Setting MiniLab Back Print Data	SetMiniLabBackprintData( )	●	●
	Designation of Paper Feed Destination	SetMiniLabOutputDest( )	●	●
	Stop Printing Command	StopPrint( )	○	○
	Set Sorter Paper MAX Position Detection Sensor ON/OFF	SetMiniLabSrtSensor( )	●	●
	Set Printer Counter Value	SetCounterP( )	○	○
Getting	Get Minilab Tower Status	GetMiniLabTowerStatus ( )	○	○
	Get Minilab Group No.	GetMiniLabGroupNo( )	○	○
	Get Minilab Serial No.	GetMiniLabSerialNo( )	○	○
	Get Printer Address	GetMiniLabPrinterAdd( )	○	○
	Get Sorter Paper MAX Position Detection Sensor ON/OFF	GetMiniLabSrtSensor( )	○	○
	Get Printer Counter Value	GetCounterP( )	○	○
The command that movement is different				
Getting	Get Printer Status	GetStatus( )	△	△

○ = There is no difference.
● = Setting is ignored.
△ = When a printer output a seal image("Shooting" status) , the status does not change in "Backprint".

11.Procedure to firmware update when using API

11-1 Procedure to update.

If you write the firmware using StatusAPI, you can do it in the following procedures. Then check that the writing occurred properly by checking the version and checksum.

Using API)

		Reference
1 SetFirmwUpdateMode()	Switch to firmware update mode.	P67
2 SetFirmwDataWrite()	Write firmware data.	P68
3 GetFirmwVersion()	Check the firmware version.	P43
4 SetColorDataClear()	Clear the control data.	P74
5 SetColorDataVersion()	Set the 300dpi control data version.	P75
6 SetColorDataWrite()	Write the 300dpi control data.	P74
7 GetColorDataVersion()	Check the 300dpi control data version.	P75
8 GetColorDataChecksum()	Check the 300dpi control data checksum.	P79
9 SetColorDataVersion()	Set the 600dpi control data version.	P75
10 SetColorDataWrite()	Write the 600dpi control data.	P74
11 GetColorDataVersion()	Check the 600dpi control data version.	P75
12 GetColorDataChecksum()	Check the 600dpi control data checksum.	P79

■DP-DS620(Firmware version less than 3.00), DP-DS820DX

Using API)

		Reference
1 SetFirmwUpdateMode()	Switch to firmware update mode.	P67
2 SetFirmwDataWrite()	Write firmware data.	P68
3 GetFirmwVersion()	Check the firmware version.	P43
4 SetColorDataClear()	Clear the control data.	P74
5 SetColorDataVersion()	Set the 300dpi control data version.	P75
6 SetColorDataWrite()	Write the 300dpi control data.	P74
7 GetColorDataVersionRes()	Check the 300dpi control data version.	P76
8 GetColorDataChecksumRes()	Check the 300dpi control data checksum.	P79
9 SetColorDataVersion()	Set the 600dpi control data version.	P75
10 SetColorDataWrite()	Write the 600dpi control data.	P74
11 GetColorDataVersionRes()	Check the 600dpi control data version.	P76
12 GetColorDataChecksumRes()	Check the 600dpi control data checksum.	P79
13 SetColorDataVersion()	Set the Low Speed control data version.	P75
14 SetColorDataWrite()	Write the Low Speed control data.	P74
15 GetColorDataVersionRes()	Check the Low Speed control data version.	P76
16 GetColorDataChecksumRes()	Check the Low Speed control data checksum.	P80

■DP-DS820, DP-QW410, DP-DS620 (Firmware version 3.00 or later)

In the DP-DS820, DP-QW410 and DP-DS620(Firmware version 3.00 or later) multiple media is mounted. There is a need to write the media type for each color control data.

(Excluding steps 9-12 and 17-20 because DP-QW410 does not support resolution 600 dpi and high density.

Excluding steps 17-20 because DP-DS620(Firmware version 3.00 or later) does not support high density.)

Using API)

		Reference
1 SetFirmwUpdateMode()	Switch to firmware update mode.	P67
2 SetFirmwDataWrite()	Write firmware data.	P68
3 GetFirmwVersion()	Check the firmware version.	P43
4 SetColorDataClear()	Clear the control data.	P74
Media Type 1		
5 SetColorDataVersion()	Set the 300dpi control data version.	P75
6 SetColorDataWrite()	Write the 300dpi control data.	P74
7 GetColorDataVersionResEX()	Check the 300dpi control data version.	P77
8 GetColorDataChecksumResEX()	Check the 300dpi control data checksum.	P80
9 SetColorDataVersion()	Set the 600dpi control data version.	P75
10 SetColorDataWrite()	Write the 600dpi control data.	P74
11 GetColorDataVersionResEX()	Check the 600dpi control data version.	P77
12 GetColorDataChecksumResEX()	Check the 600dpi control data checksum.	P80
13 SetColorDataVersion()	Set the Low Speed control data version.	P75
14 SetColorDataWrite()	Write the Low Speed control data.	P74
15 GetColorDataVersionResEX()	Check the Low Speed control data version.	P77

16	GetColorDataChecksumResEX()	Check the Low Speed control data checksum.	P80
17	SetColorDataVersion()	Set the High Density control data version.	P75
18	SetColorDataWrite()	Write the High Density control data.	P74
19	GetColorDataVersionraResEX()	Check the High Density control data version.	P77
20	GetColorDataChecksumResEX()	Check the High Density control data checksum.	P80

Media Tyep 2
Repeat 5-20

11-2 Procedure to update for Turning unit.

If you write the Turning unit firmware using StatusAPI, you can do it in the following procedures. Then check that the writing occurred properly by checking the version.

Using API)

			Reference
1	SetDunitFirmwUpdateMode()	Set Turning unit rewrite mode.	P69
2	GetDunitUpdateStatus()	Get Turning unit firmware rewriting status	P70
3	SetDunitFirmwDataWrite()	Write Turning unit firmware data	P69
4	GetDunitUpdateStatus()	Get Turning unit firmware rewriting status	P70
5	GetVersionDuplex()	Get Turning unit firmware version	P89

11-3 Regarding the version and checksum.

By reading the control data version, after writing the 300dpi control data, the version and checksum 300dpi will be read, and after writing the 600dpi control data, the version and checksum 600dpi will be read

The relationship between printer status and the version and checksum that are read are shown below.

Printer status	Control Data version and checksum
After writing 300dpi control data	300dpi
After writing 600dpi control data	600dpi
After turning power ON	300dpi
300dpi After printing	300dpi
600dpi After printing	600dpi

■DP-DS620, DP-DS820, DP-DS820DX, DP-QW410

The relationship between printer status and the version and checksum that are read are shown below.

<table>
<tr><th rowspan="2">Printer status</th><th rowspan="2">Type</th><th colspan="2">Control Data version and checksum</th></tr>
<tr><th>Type designation none</th><th>Type designation valid</th></tr>
<tr><td rowspan="4">After writing control data</td><td>300dpi</td><td>300dpi</td><td>300dpi</td></tr>
<tr><td>600dpi *2</td><td>600dpi</td><td>600dpi</td></tr>
<tr><td>Low Speed</td><td>Low Speed</td><td>Low Speed</td></tr>
<tr><td>High Density *1</td><td>High Density</td><td>High Density</td></tr>
<tr><td rowspan="4">After turning power ON</td><td>300dpi</td><td>300dpi</td><td>300dpi</td></tr>
<tr><td>600dpi *2</td><td rowspan="3">Unreadable
(only 300dpi)</td><td>600dpi</td></tr>
<tr><td>Low Speed</td><td>Low Speed</td></tr>
<tr><td>High Density *1</td><td>High Density</td></tr>
<tr><td rowspan="4">After printing</td><td>300dpi</td><td>300dpi</td><td>300dpi</td></tr>
<tr><td>600dpi *2</td><td>600dpi</td><td>600dpi</td></tr>
<tr><td>Low Speed</td><td rowspan="2">Unreadable
(specified resolution to print only 300dpi or 600dpi)</td><td>Low Speed</td></tr>
<tr><td>High Density *1</td><td>High Density</td></tr>
</table>

*1: DP-DS820 only
*2: Excluding DP-QW410

12.Command Specification

Printer ID number, getting the quantity of connections

[Format] long GetPrinterPortNum(char *pArray, int arraysize);

[Argument]
pArray: Header pointer for array storage
arraysize: Designates the size of the array

[Return]
Successful: number of printers connected
Failure: 0

[Explanation] This returns the device and units ID's currently connected.
If successful, it returns the number of printers connected.

Device ID numbers	Model
1	CV
2	CW
3	DS40
4	DS80
5	DS-RX1
6	CW02
7	DP-TC10
8	DP-DS80D
10	TOWER_CONTROLER
20	DP-DS620
22	DS-622
30	DP-DS820
35	DP-QW410
40	DP-DS820DX
90-99	Reserved

[Sample Coding]

< Visual C >

```
long Pnum;
char PnumArray[3][2];
if(( Pnum = GetPrinterPortNum( PnumArray , sizeof(PnumArray) )) < 0 ){
        // error
}

//In order of the device and unit ID
//PnumArray[0][0]    Device type
//PnumArray[0][1]    Unit ID number
//PnumArray[1][0]    Device type
//PnumArray[1][1]    Unit ID number
//PnumArray[2][0]    Device type
//PnumArray[2][1]    Unit ID number
```

< Visual Basic >

```
Dim Pnum As Long
Dim PnumArray(2,3) As Char
Pnum = GetPrinterPortNum(PnumArray(0,0), 6);
If Pnum < 0 Then GoTo Error

'The order is different from in Visual C.
'PnumArray(0,0)    Device type
'PnumArray(1,0)    Unit ID number
'PnumArray(0,1)    Device type
'PnumArray(1,1)    Unit ID number
'PnumArray(0,2)    Device type
'PnumArray(1,2)    Unit ID number
```

■Attention when connecting to the Tower:
Only one tower can be connected.
Execute GetPrinterPortNum( ) , then the Tower recognizes the Tower controller and return UnitID No10.
Direct communication to the Tower is possible as well as to the Printer, but in the Tower case only following API function can be used;
GetFirmwVersion();
GetSerialNo();

Printer Filter Function Setting

[Format] long SetPrinterFilter(long Filter_Sw);

[Argument] Filter_Sw: Filter function setting

PFILTER_OFF (0) Filter function OFF(initial value)
PFILTER_ON (1) Filter function ON

[Return] Filter function ON 1
Filter function OFF 0

[Explanation] This turns the Filter function ON/OFF.
With the filter, the DLL determines if the API are compatible or not with the printer model and firmware, and controls the transmission and execution of commands to the printer. For details of the functions, refer to the list "6.CSP Status API functions".

Filter	Function
OFF (initial value)	DLL does not control command execution to the printer. If the application uses printer-incompatible API, the printer may stop. This is the case with old DLL with no filters.
ON	If API functions are not printer-compatible, the DLL controls the execution of commands to the printer. Use this setting when using a stand-alone printer.

[Sample Coding]

< Visual C >

```
if( SetPrinterFilter( PFILTER_ON ) == 1 ){
        // Filter ON
}
```

< Visual Basic >

```
Dim Result As Long
Result = SetPrinterFilter( PFILTAER_ON )
```

USB Transmission Timeout Time Setting

[Format] long SetUSBTimeout(long lPortNum, long lTime);

[Argument]

Time:	Transmission Timeout setting(in seconds)
0:	Transmission Timeout OFF(initial value)
10-120:	Time between when USB transmission (reading data) stops and API starts timeout. The setting is in 1 second increments.

[Return]

Successful:	1
Failure:	0

[Explanation] This sets the time until the USB transmission timeout starts.
USB transmission timeout occurs when the API stops reading the return data from the printer.

Setting value	Function
Timeout OFF 0 (initial)	If USB transmission stops, the API waits until it can read data without using timeout. If transmission to 1 printer stops, the API keeps waiting, so transmission to other printers dosen't start, and the system stops. This is true for older DLL without the Timeout setting.
Timeout time 10-120 seconds	If the USB transmission stops, when the Timeout time is exceeded, the API sends a Timeout error, and stops transmitting to that printer. Even if 1 printer stops, transmission to other printers is possible.

The following chart shows reference times until the timeout.
For the DS40T/80T, with Firmware Ver0.21 and earlier, after the application sends the print data, when there is a return request like Get Status, there is a timeframe where transmission data cannot be read. Please set the Timeout setting time longer than that.

Printer operation	DS40T/80T Firm Ver0.21	Timeout setting reference times
Normal printing	Less than 1 second	60 seconds
For media initialization (During initialize: next image trans-status get)	Approx. 50 seconds	90 seconds
During recovery process after power failure (During recovery: next image trans-status get)	Approx. 100 seconds	120 seconds

[Sample Coding]

< Visual C >

```
if( SetUSBTimeout( PortNo, 60 ) == 1 ){
        // TIMEOUT 60s
}
```

< Visual Basic >

```
Dim Result As Long
Result = SetUSBTimeout( PortNo, 60 )
```

Media Size Setting

[Format] long SetMediaSize(long lPortNum,long Media);

[Argument] lPortNum: Designates the port number
Media: Designates paper size and image layout (defined in header: see below)

[Return] Successful: 1
Failure: 0 Not within media range (for example, when 8-inch size layout is given for a 6-inch printer)

[Explanation] This designated the output page size for use in print processing.
Please compare the output page size with the media size set in the printer, and select an equivalent or smaller size.
When sending the image with SendImageData as on P36, be sure to use this command to select the output size.

[Sample Coding]

< Visual C >

```
if( SetMediaSize(PortNo, CSP_PCx2) < 0 ){
        // error
}
```

< Visual Basic >

```
Dim Result As Long
Result = SetMediaSize(PortNo, CSP_PCx2)
If Result <= 0 Then GoTo Error
```

■Page Definition Contents

Media	Header Definition	Media	Size Layout	Media	Size Layout
01	CSP_L	26	CSP_8x7P1 *4	54	CSP_4x4
02	CSP_2L	27	CSP_8x11 *4	55	CSP_4x6
03	CSP_PC			56	CSP_4x8
04	CSP_A5	31	CSP_8x10	57	CSP_4P5x4P5
05	CSP_A5W	32	CSP_8x12	58	CSP_4P5x6
06	CSP_PCx2 (PC 2-image layout)	33	CSP_8x4	59	CSP_4P5x8
07	CSP_Lx2 (L 2-image layout)	34	CSP_8x5	60	CSP_4x3
08	CSP_PC_REWIND (PC Rewind) *1	35	CSP_8x6	61	CSP_4x4P5
09	CSP_L_REWIND (L Rewind) *1	36	CSP_8x8	62	CSP_4P5x3
10	CSP_5x5 *1	37	CSP_8x4x2	63	CSP_4P5x4
11	CSP_6x6 *3	38	CSP_8x5x2	64	CSP_4x3x2
12	CSP_6x4P5 *2	39	CSP_8x6x2	65	CSP_4P5x3x2
13	CSP_6x4P5x2 *2	40	CSP_8x5_8x4	66	CSP_4P5x4x2
14	CSP_6x4P5_REWIND (6x4.5 Rewind) *2	41	CSP_8x6_8x4	71	CSP_A5FORMAT
		42	CSP_8x6_8x5	72	CSP_A5x2
21	CSP_TC86x61	43	CSP_8x8_8x4	73	CSP_A5_REWIND
22	CSP_TC127x61	44	CSP_8x4x3	74	CSP_A4x5
		45	CSP_A4_LENGTH	75	CSP_A4x6
		46	CSP_8x7	76	CSP_A4x8
		47	CSP_8x9	77	CSP_A4x10
		48	CSP_8x4_REWIND	78	CSP_A4FORMAT
		49	CSP_8x5_REWIND	79	CSP_A4x5x2
		50	CSP_8x6_REWIND	80	CSP_A4x5_REWIND

*1: DP-DS620 and DS-RX1 firmware version 2.06 or later.

*2: DP-DS620 firmware version 1.10 or later.

*3: DP-DS620 and DS40 firmware version 1.60 or later and DS-RX1 firmware version 2.06 or later.

*4: DP-DS820DX firmware version 1.06 or later.

■Header definition contents（2-sided DP-DS80D, DP-DS820DX cut sheet setting）

Media	Size Layout	Media	Size Layout
Single-side print setting (*1)		Duplex print front print setting	
131	CSP_8x10_CutPaper	225	CSP_8x10_35_DuplexCutPaper_F (*4)
132	CSP_8x12_CutPaper	226	CSP_8x7_10_DuplexCutPaper_F (*3)
133	CSP_8x4_CutPaper	227	CSP_8x11_DuplexCutPaper_F (*3)
134	CSP_8x5_CutPaper	228	CSP_8x5_90_DuplexCutPaper_F (*3)
135	CSP_8x6_CutPaper	229	CSP_8x5_90x2_DuplexCutPaper_F (*3)
136	CSP_8x8_CutPaper	230	CSP_8x6_50_DuplexCutPaper_F (*2)
137	CSP_8x4x2_CutPaper	231	CSP_8x10_DuplexCutPaper_F
138	CSP_8x5x2_CutPaper	232	CSP_8x12_DuplexCutPaper_F
139	CSP_8x6x2_CutPaper	233	CSP_8x4_DuplexCutPaper_F
151	CSP_8x10_50_CutPaper	234	CSP_8x5_DuplexCutPaper_F
152	CSP_8x10_75_CutPaper	235	CSP_8x6_DuplexCutPaper_F
153	CSP_8x4x3_CutPaper	236	CSP_8x8_DuplexCutPaper_F
		237	CSP_8x4x2_DuplexCutPaper_F
		238	CSP_8x5x2_DuplexCutPaper_F
		239	CSP_8x6x2_DuplexCutPaper_F
		245	CSP_8xA4_DuplexCutPaper_F (*4)
		251	CSP_8x10_50_DuplexCutPaper_F
		252	CSP_8x10_75_DuplexCutPaper_F
		253	CSP_8x4x3_DuplexCutPaper_F
		Duplex print back print setting	
		325	CSP_8x10_35_DuplexCutPaper_R (*4)
		326	CSP_8x7_10_DuplexCutPaper_R (*3)
		327	CSP_8x11_DuplexCutPaper_R (*3)
		328	CSP_8x5_90_DuplexCutPaper_R (*3)
		329	CSP_8x5_90x2_DuplexCutPaper_R (*3)
		330	CSP_8x6_50_DuplexCutPaper_R (*2)
		331	CSP_8x10_DuplexCutPaper_R
		332	CSP_8x12_DuplexCutPaper_R
		333	CSP_8x4_DuplexCutPaper_R
		334	CSP_8x5_DuplexCutPaper_R
		335	CSP_8x6_DuplexCutPaper_R
		336	CSP_8x8_DuplexCutPaper_R
		337	CSP_8x4x2_DuplexCutPaper_R
		338	CSP_8x5x2_DuplexCutPaper_R
		339	CSP_8x6x2_DuplexCutPaper_R
		345	CSP_8xA4_DuplexCutPaper_R (*4)
		351	CSP_8x10_50_DuplexCutPaper_R
		352	CSP_8x10_75_DuplexCutPaper_R
		353	CSP_8x4x3_Dupl;exCutPaper_R

*1:DP-DS80D only.
*2:DP-DS820DX only.
*3:DP-DS820DX firmware version 1.06 or later.
*4:DP-DS820DX firmware version 1.11 or later.

Sending Image Data

[Format] long SendImageData(long lPortNum,LPSTR Buff,
long XPos,long YPos,long Width,long Height);

[Argument]

lPortNum:	Port number
Buff:	Header pointer for RGB bitmap and raw data (The data is RGB 24bit color data)
Xpos:	Designates the output coordinates(X axis, Dot)
Ypos:	Designates the output coordinates(Y axis, Dot)
Width:	Image width(Dot)
Height:	Image height(Dot)

[Return]

Successful:	1
Failure:	0

[Explanation] This sends the image data.
When sending the image data using this command, be sure to set the output size with the SetMediaSize() command as on P34. Please compare the image size with the media size set in the printer, and select an equivalent or smaller size. Refer to "4. Print Process Flow" for details.

[Sample Coding]

< Visual C >

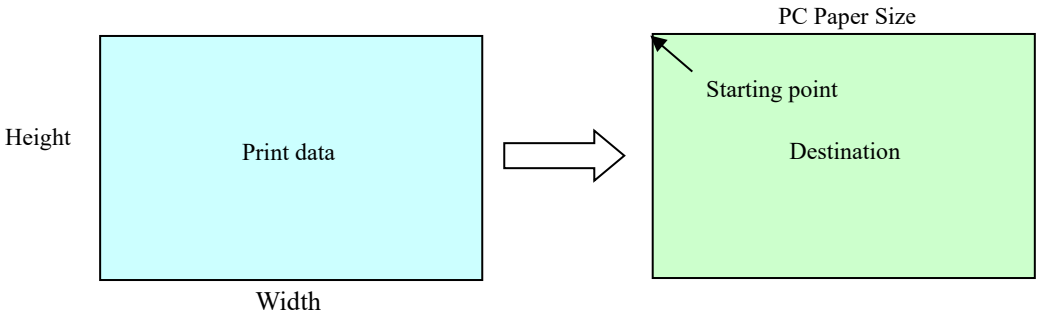
```
char RGBData[8478720]          //Buffer for print data
                               //Ex)the numbers are for PC image size 2048 x 1380 x 2
if( SetMediaSize(PortNo, CSP_PCx2) > 0 ){
        if( SendImageData(PortNo, RGBData, 0, 0, 2048, 1380) <= 0 ){
                // error
        }
}
```

< Visual Basic >

```
Dim Result As Long
Dim RGBData(8478720) As Byte  'Print data buffer, numbers referring to PC image size
                              'Ex)the numbers are for PC image size 2048 x 1380 x 2
Result = SetMediaSize(PortNo, CSP_PCx2)
If Result <= 0 Then GoTo Error

Result = SendImageData(PortNo,VarPtr(RGBData(0)), 0, 0, 2048, 1380)
If Result <= 0 Then GoTo Error
```

*The SendImageData function allows height, width, and destination coordinate designation.

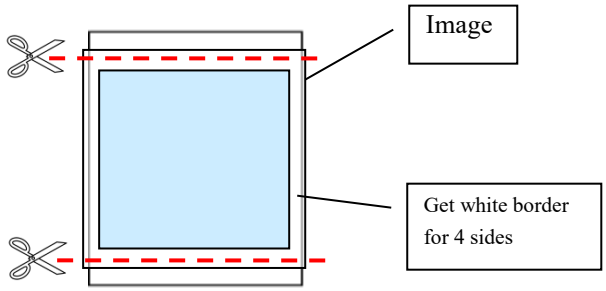


■ Improved image data transmission method
[After Version 0.5.9.1 and 0.6.0.1 or later]
With this command, the API sends data so that the end of the plane data is a short packet (less than 512 bytes), in order to make the transmission operation more stable.

■ Image pixel size (CV, CW)

Model	Paper type	Print area size W (Head width) x H (Paper feed direction)	Image pixel size	
			Resolution setting: 334x600dpi	Resolution setting: 334x334dpi
CV	DSC	121.5x92.0 mm	1598x2176 pixel	-
	L	130.0x92.0 mm	1710x2176 pixel	-
	PC	155.0x105.0 mm	2038x2480 pixel	-
	2DSC	121.5x181.0 mm	1598x4276 pixel	-
	2L	130.0x181.0 mm	1710x4276 pixel	-
	A5 (6x8)	155.0x206.0 mm	2038x4870 pixel	-
	A5W (6x9)	155.0x232.0 mm	2038x5480 pixel	-
CW	DSC	121.5x92.0 mm	1598x2176 pixel	1598x1210 pixel
	L	130.0x92.0 mm	1710x2176 pixel	1710x1210 pixel
	PC	155.0x105.0 mm	2038x2480 pixel	2038x1380 pixel
	2DSC	121.5x181.0 mm	1598x4276 pixel	1598x2380 pixel
	2L	130.0x181.0 mm	1710x4276 pixel	1710x2380 pixel
	A5 (6x8)	155.0x206.0 mm	2038x4870 pixel	2038x2710 pixel
	A5W (6x9)	155.0x232.0 mm	2038x5480 pixel	2038x3050 pixel
White border T/B		Top & Bottom 4.5 mm each	Top and Bottom 120 pixel each	Top and Bottom 60 pixel each
White border L/R		Left & Right 4.5 mm each	Left and Right 60 pixel each	Left and Right 60 pixel each

If you want to add a border to the 4 sides of the print, a white border can be imported for the 4 sides of the image. Refer to the values in the chart for border size.



■ Image pixel size (DP-QW410)

Model	Paper type	Print area size W (Head width) x H (Paper feed direction)	Image pixel size	
			Resolution setting: 300x600dpi	Resolution setting: 300x300dpi
DP-QW410	4x3 (*1)	107.2x79.3 mm		1266x936 pixel
	4x4	107.2x104.7 mm		1266x1236 pixel
	4x4.5 (*1)	107.2x117.4 mm		1266x1386 pixel
	4x6	107.2x155.5 mm		1266x1836 pixel
	4.5x3 (*1)	119.3x79.3 mm		1408x936 pixel
	4.5x4 (*1)	119.3x104.7 mm		1408x1236 pixel
	4.5x4.5	119.3x117.4 mm		1408x1386 pixel
	4.5x6	119.3x155.5 mm		1408x1836 pixel
	4.5x8	119.3x206.3 mm		1408x2436 pixel
White border T/B		Top & Bottom 4.5 mm each		Top and Bottom 54 pixel each
White border L/R		Left & Right 5.54 mm each		Left and Right 66 pixel each

*1: firmware version 1.09 or later

■ Image pixel size (DS40/80, DS40T/80T, DS-RX1, CW02, DP-DS80D, DP-DS620, DP-DS820, DP-DS820DX)

Model	Paper type	Print area size W (Head width) x H (Paper feed direction)	Image pixel size	
			Resolution setting: 300x600dpi	Resolution setting: 300x300dpi
DS40/40T DS-RX1, CW02 DP-DS620	5x3.5(L)	131.1x 92.1 mm	1548x2176 pixel	1548x1088 pixel
	6x4(PC)	156.1x105.0 mm	1844x2480 pixel	1844x1240 pixel
	5x7(2L)	131.1x181.0 mm	1548x4276 pixel	1548x2138 pixel
	6x8(A5)	156.1x206.2 mm	1844x4872 pixel	1844x2436 pixel
DS40/40T CW02 DP-DS620 (*1)	6x9(A5W)	156.1x232.0 mm	1844x5480 pixel	1844x2740 pixel
DS-RX1 (*3), DP-DS620	5x5	131.1x130.4 mm	1548x3080 pixel	1548x1540 pixel
DP-DS620 (*1)	6x4.5	156.1x117.3mm	1844x2772 pixel	1844x1386 pixel
DS40 (*2), DS-RX1 (*3), DP-DS620	6x6	156.1x155.4 mm	1844x3672 pixel	1844x1836 pixel
DS80/80T DP-DS80D DP-DS820 DP-DS820DX	(8x4)	207.3x104.6 mm	2448x2472 pixel	2448x1236 pixel
	(8x5)	207.3x130.0 mm	2448x3072 pixel	2448x1536 pixel
	(8x6)	207.3x155.4 mm	2448x3672 pixel	2448x1836 pixel
	(8x7.1) (*5)	207.3x183.0 mm	2448x2162 pixel	2448x4324 pixel
	(8x8)	207.3x206.2 mm	2448x4872 pixel	2448x2436 pixel
	(8x10)	207.3x257.0 mm	2448x6072 pixel	2448x3036 pixel
	(8x11) (*5)	207.3x283.0 mm	2448x3343 pixel	2448x6686 pixel
	(8x12)	207.3x307.8 mm	2448x7272 pixel	2448x3636 pixel
	A4 Length (*4)	207.3x300.0 mm	2448x7088 pixel	2448x3544 pixel
DP-DS820	(8x7)	207.3×180.8 mm	2448×4272 pixel	2448×2136 pixel
	(8x9)	207.3×231.6 mm	2448×5472 pixel	2448×2736 pixel
	A5 Format	214.0×151.0 mm	2528×3568 pixel	2528×1784 pixel
	A4x5	214.0×130.0 mm	2528×3072 pixel	2528×1536 pixel
	A4x6	214.0×155.4 mm	2528×3672 pixel	2528×1836 pixel
	A4x8	214.0×206.2 mm	2528×4872 pixel	2528×2436 pixel
	A4x10	214.0×257.0 mm	2528×6072 pixel	2528×3036 pixel
	A4 Format	214.0×300.0 mm	2528×7088 pixel	2528×3544 pixel
White border T/B		Top & Bottom 4.5 mm each	Top and Bottom 108 pixel each (PC:120 Pixel)	Top and Bottom 54 pixel each (PC:60Pixel)
White border L/R		Left & Right 5.0 mm each	Left and Right 60 pixel each	Left and Right 60 pixel each

*1 DP-DS620 firmware version 1.10 or later

*2 DS40 firmware version 1.60 or later

*3 DS-RX1 firmware version 2.06 or later

*4 Excluding DP-DS820DX

*5 DP-DS820DX firmware version 1.06 or later.

■Image pixel size (DP-TC10)

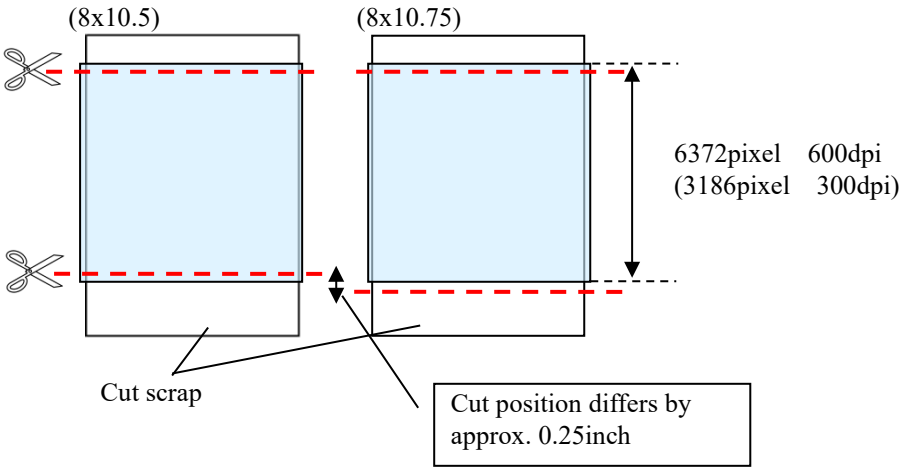
Model	Paper type	Print area size W (Head width) x H (Paper feed direction)	Image pixel size	
			Resolution setting: 300x600dpi	Resolution setting: 300x300dpi
DP-TC10	TC86x61	90.1x 64.0 mm	1064x1512 pixel	1064x 756 pixel
	TC127x61	131.1x 64.0 mm	1548x1512 pixel	1548x 756 pixel
White border T/B		Top & Bottom 4.5 mm each	Top and Bottom 108 pixel each	Top and Bottom 54 pixel each
White border L/R		Left & Right 5.0 mm each	Left and Right 60 pixel each	Left and Right 60 pixel each

■Image pixel size (DP-DS80D/DP-DS820DX, Cut sheet)

Model	Paper type	Print area size W (Head width) x H (Paper feed direction)	Image pixel size	
			Resolution setting: 300x600dpi	Resolution setting: 300x300dpi
DP-DS80D DP-DS820DX	(8x4)	207.3x104.6 mm	2448x2472 pixel	2448x1236 pixel
	(8x5)	207.3x130.0 mm	2448x3072 pixel	2448x1536 pixel
	(8x5.9) (*2)	207.3x153.1 mm	2448x3616 pixel	2448x1808 pixel
	(8x6)	207.3x155.4 mm	2448x3672 pixel	2448x1836 pixel
	(8x6.5) (*1)	207.3x155.4 mm	2448x3972 pixel	2448x1986 pixel
	(8x7.1) (*2)	207.3x183.0 mm	2448x4324 pixel	2448x2162 pixel
	(8x8)	207.3x206.2 mm	2448x4872 pixel	2448x2436 pixel
	(8x10)	207.3x257.0 mm	2448x6072 pixel	2448x3036 pixel
	(8x10.35) (*3)	207.3x265.9 mm	2448x6282 pixel	2448x3141 pixel
	(8x10.5) *	207.3x269.7 mm	2448x6372 pixel	2448x3186 pixel
	(8x10.75) *	207.3x269.7 mm	2448x6372 pixel	2448x3186 pixel
	(8x11) (*2)	207.3x283.0 mm	2448x3343 pixel	2448x6686 pixel
	(8xA4) (*3)	207.3x300.0 mm	2448x7088 pixel	2448x3544 pixel
	(8x12)	207.3x307.8 mm	2448x7272 pixel	2448x3636 pixel
White border T/B		Top & Bottom 4.5 mm each	Top and Bottom 108 pixel each	Top and Bottom 54 pixel each
White border L/R		Left & Right 5.0 mm each	Left and Right 60 pixel each	Left and Right 60 pixel each

*1 DP-DS820DX only.
*2 DP-DS820DX firmware version 1.06 or later.
*3 DP-DS820DX firmware version 1.11 or later.

*The image pixel sizes for (8x10.5) and (8x10.75) are the same. The prints that are output differ as follows.



■ Change in Output Size Limitations according to Printer Location

As detailed below, there have been some changes in the output limitations for large sizes according to DS40T location, and location limitations for DS80T.

Printer	Existing Specifications	New Specifications
DS40T	1st and 3rd levels are limited to only L-size and PC size prints	1st level is limited to only L-size and PC size prints
DS80T	Can't be set on 1st - 3rd levels	Can't be set on 1st level

Chart of sizes for each DS40T printer location (○= available, ×= N/A, ☐ Ver0.32 or laternewly available)

Location	Print Size				
	5x3.5(L)	5x7(2L)	6x4(PC) *	6x8(A5)	6x9(A5W)
1st level	○	×	○	×	×
2nd level	○	○	○	○	○
3rd level	○	○	○	○	○
4th level	○	○	○	○	○

*Including multi-cut prints

If a non-available print size is designated on the 1st level, the printer will return "Paper Definition Error".

Chart of sizes for each DS80T printer location (○= available, ×= N/A, ☐ Ver0.32 or later available)

Location	Print Size						
	8x4 *	8x5 *	8x6 *	8x8 *	8x10	8x11.7	8x12
1st level	×	×	×	×	×	×	×
2nd level	○	○	○	○	○	○	○
3rd level	○	○	○	○	○	○	○
4th level	○	○	○	○	○	○	○

*Including multi-cut prints

If printer is set on the 1st level, when designated to print, the printer will return "Paper Definition Error".

Setting the Resolution

[Format] long SetResolution(long lPortNum, long lResolution);

[Argument]

lPortNum:	Port Number
Media:	Resolution designation defined in the header RESOLUTION300 or RESOLUTION600

[Return]

Successful:	1
Failure:	0

[Explanation]
This designates the image resolution.
DP-QW410 is valid only for RESOLUTION300.
This function is invalid for CV and CW.

[Sample Coding]

< Visual C >

```
if(( SetResolution( PortNo, RESOLUTION300)) <= 0 ){
        //Error
}
```

< Visual Basic >

```
Dim Result As Long
Result = SetResolution(PortNo, RESOLUTION300)
If Result <= 0 Then GoTo Error
```

Designating Print Quantity

[Format] long SetPQTY(long lPortNum, long pqty);

[Argument]

lPortNum:	Designates the port number
pqty:	Designates the number of prints

[Return]

Successful:	1
Failure:	0

[Explanation]
This designates the number of prints for the printer.
Valid only for printing on roll paper.

[Sample Coding]

< Visual C >

```
if( SetPQTY(PortNo, 2) <= 0 ){
        // error
}
```

< Visual Basic >

```
Dim Result As Long
Result = SetPQTY(PortNo, 2)
If Result <= 0 Then GoTo Error
```

Start Printing

[Format] long PrintImageData(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: 1
Failure: 0

[Explanation] Starts printing.(This prints a partial page for multiple layout pages)

[Sample Coding]

< Visual C >

```
if( PrintImageData(PortNo) <= 0 ){
        // error
}
```

< Visual Basic >

```
Dim Result As Long
Result = PrintImageData(PortNo)
If Result <= 0 Then GoTo Error
```

Start Page Layout

[Format] long StartPageLayout(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: 1
Failure: 0

[Explanation] Starts page layout.
This is always used in conjunction with End Page Layout.

[Sample Coding]

< Visual C >

```
if( StartPageLayout(PortNo) <= 0 ){
        // error
}
/*Image processing*/
EndPageLayout(PortNo);
```

< Visual Basic >

```
Dim Result As Long
Result = StartPageLayout(PortNo)
If Result <= 0 Then GoTo Error
'Image processing
EndPageLayout(PortNo)
```

End Page Layout

[Format] long EndPageLayout(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: 1
Failure: 0

[Explanation] Ends page layout.
This is always used in conjunction with Start Page Layout.

[Sample Coding]

< Visual C >

```
StartPageLayout(PortNo);
/* Image processing */
if( EndPageLayout(PortNo) <= 0 ){
        // error
}
```

< Visual Basic >

```
Dim Result As Long
StartPageLayout(PortNo)
'Image processing
Result = EndPageLayout(PortNo)
If Result <= 0 Then GoTo Error
```

Get Printer Version Information

[Format] long GetFirmwVersion(long lPortNum, LPSTR p);
long CvGetVersion(long lPortNum, LPSTR p);

[Argument] lPortNum: Port number
p: Pointer to the receiving buffer

[Return] Successful: The number of characters received in buffer p
Failure: -1

[Explanation] Receives the printer version information in the buffer.

[Sample Coding]

< Visual C >

```
char rbuf[256];
if( GetFirmwVersion( PortNo, (LPSTR)rbuf ) > 0 ){
        // Next process
}
```

< Visual Basic >

```
Dim rb As String * 255, Ver As String
Dim n As Long
n = GetFirmwVersion(PortNo, rb)
If n > 0 Then Ver = Left(rb, n)      'Retrieves as a character string
```

[Version Information]

DS *.** Firmware version information

Get Printer Sensor Information

[Format] long GetSensorInfo(long lPortNum, LPSTR p);
long CvGetSensorInfo(long lPortNum, LPSTR p);

[Argument] lPortNum: Port number
p: Pointer to the receiving buffer

[Return] Successful: The number of characters received in buffer p
Failure: -1

[Explanation] Receives value of each sensor (through AD converter) in the buffer.

[Sample Coding]

< Visual C >

```
char rbuf[256];
if( GetSensorInfo( PortNo, (LPSTR)rbuf ) > 0 ){
        // Next process
}
```

< Visual Basic >

```
Dim rb As String * 255, Sen As String
Dim n As Long
n = GetSensorInfo(PortNo, rb)
If n > 0 Then Sen = Left(rb, n)      'Retrieves as a character string
```

[Sensor Information for]

HDT-***;	Head temperature	
MDT-***;	Media temperature	
PMK-***;	Paper mark	
RML-***;	Ribbon mark left	(Except CV and CW are unused."000"always returns to values)
RMC-***;	Ribbon mark center	
RMR-***;	Ribbon mark right	
PSZ-***;	Paper size	(Except CV and CW are unused."000"always returns to values)
PNT-***;	Paper notch	(Except CV and CW are unused."000"always returns to values)
PJM-***;	Paper jam	(Except CV and CW are unused."000"always returns to values)
PED-***;	Paper end	
PET-***;	Paper empty	(Except CV and CW are unused."000"always returns to values)
HDV-***;	Head voltage	
HMD-***;	Humidity	(CV is excluded)
RP1-***;	Roll media paper end detection sensor-1	(Only compatible for DP-DS80D)
RP2-***;	Roll media paper end detection sensor-2	(Only compatible for DP-DS80D)
CSR-***;	Color Sensor (Red)	(Compatible for DP-DS620, DP-DS820, DP-DS820DX)
CSG-***;	Color Sensor (Green)	(Compatible for DP-DS620, DP-DS820, DP-DS820DX)
CSB-***;	Color Sensor (Blue)	(Compatible for DP-DS620, DP-DS820, DP-DS820DX)
DC5-***;	USB power supply voltage	(Only compatible for DP-QW410)
PSS-***;	Cut paper feed sensor	(Only compatible for DP-DS820DX)
PE2-***;	Paper end detection sensor 2	(Only compatible for DP-DS820DX)
EXI-***;	Exit Sensor	(Only compatible for DP-DS820DX)
REV-***;	Cut paper reversal detection sensor	(Only compatible for DP-DS820DX)
WPD-***;	Cut paper detection sensor	(Only compatible for DP-DS820DX)
PPD-***;	Protective Paper Detection Sensor	(Only compatible for DP-DS820DX)

Get Printer Resolution

[Format]
```
long GetResolutionH( long lPortNum );
long CvGetResolutionH( long lPortNum );
long GetResolutionV( long lPortNum );
long CvGetResolutionV( long lPortNum );
```

[Argument] lPortNum: Port number

[Return] Successful: Horizontal, or Vertical resolution (dpi)
Failure: -1

[Explanation] GetResolutionH(), CvGetResolutionH() returns Horizontal resolution (dpi).
GetResolutionV(), CvGetResolutionV() returns Vertical resolution (dpi).

[Sample Coding]

< Visual C >
```
long rh;
if(( rh = GetResolutionH( PortNo )) >= 0 ){
        // Returns Horizontal resolution to rh
}
```

< Visual Basic >
```
Dim rh As Long
rh = GetResolutionH(PortNo)
```

Get Printer Media Code

[Format] long GetMedia(long lPortNum, LPSTR p);
long CvGetMedia(long lPortNum, LPSTR p);

[Argument] lPortNum: Port number
p: Pointer to the receiving buffer

[Return] Successful: The number of characters received by buffer p
Failure: -1

[Explanation] Receives media code in the buffer.

[Sample Coding]

< Visual C >

```
char rbuf[256];
if( GetMedia( PortNo, (LPSTR)rbuf ) > 0 ){
        // Next process
}
```

< Visual Basic >

```
Dim rb As String * 255, MediaCode As String
Dim n As Long

n = GetMedia(PortNo, rb)
If n > 0 Then MediaCode = Left(rb, n)          ' Retrieves as a character string
```

[Media code]

Media code is defined by 5 decimal bytes. The return value for the Get Media Code command is referred to as the following 5 bytes (ASCII numeric).

5th, 4th byte (0n000) Reserved	3rd, 2nd byte (00nn0) Paper Size	1st byte (0000n)
nn = 00	00150 4inch width	0:4x6
	00160 4.5inch width	1:4.5x8 3:4.5x6
	00200 5x3.5 (L) 00210 5x7 (2L) 00300 6x4 (PC) 00310 6x8 (A5) 00400 6x9 (A5W) 00500 8x10 00510 8x12 00600 A4 00120 TC86x61 00130 TC127x61	n = 0 or 1 (1:6x4 only)

(Information on media other than media size will be supported by get media extension code command)

Example

Paper Size	(Size: Width x Height)	Media Code
5x3.5 (L)	(127.0 x 89.0 mm)	00200
6x4 (PC)	(152.0 x 102.0 mm)	00301
5x7 (2L)	(127.0 x 178.0 mm)	00210
6x8 (A5)	(152.0 x 203.0 mm)	00310
6x9 (A5W)	(152.0 x 229.0 mm)	00400

Get Printer Status

[Format] long GetStatus(long lPortNum);
long CvGetStatus(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: Status
Failure: STATUS_ERROR or CSPSTATUS_TIMEOUT

[Explanation] This returns the printer status.
Bit position of the status is defined in CspStat.h by macro.
Meanings of the symbols are as follows:

GROUP_USUALLY	[Usual Operation]	Group identification bit
GROUP_SETTING	[Setting Error]	Group identification bit
GROUP_HARDWARE	[Hardware Error]	Group identification bit
GROUP_SYSTEM	[System Error]	Group identification bit
GROUP_FLSHPROG	[Rewriting Mode]	Group identification bit
GROUP_DUNIT	[Turning unit error]	Group identification bit
GROUP_HARDWARE2	[Hardware Error 2]	Group identification bit

STATUS_USUALLY_STANDBY_MODE	Standby Mode
STATUS_ERROR	Status Receiving Error
CSPSTATUS_TIMEOUT	Timeout Error (not able to receive the command)
STATUS_USUALLY_IDLE	Idle
STATUS_USUALLY_PRINTING	Printing
STATUS_USUALLY_PAPER_END	Paper End
STATUS_USUALLY_RIBBON_END	Ribbon End
STATUS_USUALLY_COOLING	Head Cooling Down
STATUS_USUALLY_MOTCOOLING	Main Motor Cooling Down
STATUS_USUALLY_SHOOTING	Tower Shooting
STATUS_USUALLY_BACKPRINT	Tower Back Printing
STATUS_SETTING_COVER_OPEN	Cover Open
STATUS_SETTING_PAPER_JAM	Paper Jam
STATUS_SETTING_RIBBON_ERR	Ribbon Error (Detect Error, Ribbon break)
STATUS_SETTING_PAPER_ERR	Paper Definition Error
STATUS_SETTING_DATA_ERR	Data Error (Illegal command)
STATUS_SETTING_SCRAPBOX_ERR	Scrap Box Error
STATUS_SETTING_TRAY_OPEN	Paper Tray Open
STATUS_SETTING_TRASHBOX_FULL	Trash Box Full
STATUS_HARDWARE_ERR01	Head Voltage Error
STATUS_HARDWARE_ERR02	Head Position Error
STATUS_HARDWARE_ERR03	Power Supply Fan Stopped
STATUS_HARDWARE_ERR04	Cutter Error (Cut jamming etc)
STATUS_HARDWARE_ERR05	Pinch Roller Position Error
STATUS_HARDWARE_ERR06	Abnormal Head Temperature
STATUS_HARDWARE_ERR07	Abnormal Media Temperature
STATUS_HARDWARE_ERR08	Ribbon Tension Error
STATUS_HARDWARE_ERR09	RFID Module Error
STATUS_HARDWARE_ERR10	Paper Rewind Motor Abnormal Temperature
STATUS_CXM_RS422_STOP	CXM Tower RS422 Connection Stop
STATUS_HARDWARE_ERR12	USB Power Supply Voltage Error
STATUS_SYSTEM_ERR01	System Error

STATUS_FLSHPROG_IDLE	Idling for receiving rewriting data
STATUS_FLSHPROG_DATA_ERR1	Transmitted Data Error
STATUS_FLSHPROG_DEVICE_ERR1	Device Error
STATUS_FLSHPROG_OTHERS_ERR1	Other Error

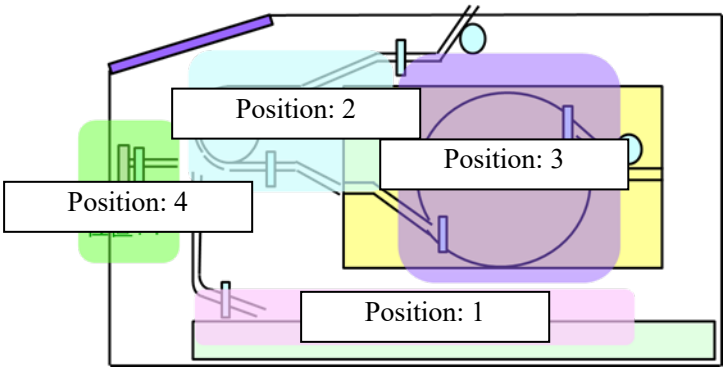
STATUS_HARDWARE2_ERR01	Paper feed lever position error
STATUS_HARDWARE2_ERR02	Turn roller position error
STATUS_HARDWARE2_ERR03	Turn unit error
STATUS_HARDWARE2_ERR04	Creaser unit error
STATUS_HARDWARE2_ERR05	Slitter unit error
STATUS_HARDWARE2_ERR06	Too many protective sheets
STATUS_HARDWARE2_ERR07	Unusable paper detected

Error of a Turning unit

UNIT_ERROR_JAMMING_SUPPLY	Paper is jammed in the supply slot（position:1）
UNIT_ERROR_JAMMING_PASS	Paper is jammed in the upper feed area（position:2）
UNIT_ERROR_JAMMING_SHELL	Paper is jammed in the shell area（position:3）
UNIT_ERROR_JAMMING_EJECT	Paper is jammed in the ejection slot area（position:4）

UNIT_ERROR_CAPSTAN_MOTOR	Paper delivery motor malfunction
UNIT_ERROR_SHELL_MOTOR	Delivery motor malfunction in the reversing block (shell)
UNIT_ERROR_PINCH	Pinch operation malfunction
UNIT_ERROR_PASS_GUIDE	Active pass guide operation malfunction
UNIT_ERROR_SKEW_GUIDE	Side guide operation malfunction
UNIT_ERROR_SKEW_REJECT	Side adjustment malfunction
UNIT_ERROR_SHELL_ROTATE	Reversing block (shell) operation malfunction
UNIT_ERROR_LEVER	Paper feed lever operation malfunction
UNIT_ERROR_CUTTER	Cut operation malfunction

UNIT_ERROR_TRAY_OUT	Tray out error
UNIT_ERROR_COVER_OUT	Maintenance cover is open
UNIT_ERROR_SYSTEM	System error



[Sample Coding]

< Visual C >

```
include "CspStat.h"
long stat;
stat = GetStatus( PortNo );
if( stat & GROUP_USUALLY ){ // Usual Operation Status Group
        switch( stat ){
                case STATUS_USUALLY_IDLE: ;
                case STATUS_USUALLY_PRINTING: ;
                case STATUS_USUALLY_PAPER_END: ;
                :
        }
}
if( stat & GROUP_SETTING ){ // Setting Error Status Group
        switch( stat ){
                case STATUS_SETTING_COVER_OPEN: ;
                :
        }
}
```

< Visual Basic >

```
Dim stat As Long

stat = GetStatus(PortNo)
If stat And GROUP_USUALLY Then          ' Usual Operation Status Group
        Select Case stat
                Case STATUS_USUALLY_IDLE: Text1.Text = "IDLE"
                Case STATUS_USUALLY_PRINTING: Text1.Text = "PRINTING"
                Case STATUS_USUALLY_PAPER_END: Text1.Text = "PAPER_END"
                Case STATUS_USUALLY_RIBBON_END: Text1.Text = "RIBBON_END"
                Case STATUS_USUALLY_COOLING: Text1.Text = "COOLING"
        End Select
ElseIf stat And GROUP_SETTING Then      ' Setting Error Status Group
        ' Operation Error ( )
ElseIf stat And GROUP_HARDWARE Then     ' Hardware Error Status Group
        ' Hardware Error ( )
ElseIf stat And GROUP_SYSTEM Then       ' System Error Status Group
        ' System Error ( )
End If
```

Getting Turning unit Status (DP-DS80D)

[Format] long GetDunitStatus(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: Turning unit Status
Failure: STATUS_ERROR or CSPSTATUS_TIMEOUT

[Explanation] This returns the Turning unit status.

Bit position of the status is defined in CspStat.h by macro. Meanings of the symbols are as follows:

Status of a Turning unit

UCS_JAMMING_SUPPLY_SENS_ON	The print didn't pass the supply sensor, so the sensor didn't switch ON (no paper) (paper feed operation)
UCS_JAMMING_SUPPLY_SENS_OFF	The print didn't reach the supply sensor, so the sensor didn't switch OFF (paper present) (paper feed operation)
UCS_JAMMING_SLOT_SENS_ON	The print didn't pass the delivery slot sensor, so the sensor didn't switch ON (no paper) (paper feed / reversing operation)
UCS_JAMMING_SLOT_SENS_OFF	The print didn't reach the delivery slot sensor, so the sensor didn't switch OFF (paper pres.) (paper feed/reversing operation)
UCS_JAMMING_PASS_SENS_ON	The print didn't pass the pinch pass sensor, so the sensor didn't switch ON (no paper) (reversing / ejecting operation)
UCS_JAMMING_PASS_SENS_OFF	The print didn't reach the pinch pass sensor, so the sensor didn't switch OFF (paper present) (reversing / ejecting operation)
UCS_JAMMING_SHELL_SENS1_ON	The print didn't pass shell pass sensor-1, so the sensor didn't switch ON (no paper) (reversing / ejecting operation)
UCS_JAMMING_SHELL_SENS1_OFF	The print didn't reach shell pass sensor-1, so the sensor didn't switch OFF (paper present) (reversing / ejecting operation)

UCS_JAMMING_SHELL_SENS2_ON	The print didn't pass shell pass sensor-2, so the sensor didn't switch ON（no paper）（reversing operation）
UCS_JAMMING_SHELL_SENS2_OFF	The print didn't reach shell pass sensor-2, so the sensor didn't switch OFF（paper present）（reversing operation）
UCS_JAMMING_EJECT_SENS_ON	The print didn't pass the eject sensor, so the sensor didn't switch ON（no paper）（ejecting operation）
UCS_JAMMING_EJECT_SENS_OFF	The print didn't reach the eject sensor, so the sensor didn't switch OFF（paper present）（ejecting operation）
UCS_JAMMING_SLOT_FG_SENS	During printing, the printer transfer FG sensor didn't detect the print（printing）
UCS_JAMMING_SHELL_FG_SENS	While delivering the print out of the shell roller, the FG sensor in the shell didn't detect the print（reversing operation）
UCS_JAMMING_PAPER_JAM	When the power is turned ON, paper is in the shell（power ON）
UCS_CAPSTAN_TRAPZ_ERROR	The capstan drive control didn't end within the set time
UCS_SHELL_ROLLER_ERROR	Shell roller malfunction（slow or stuck）
UCS_PINCH_OPEN_ERROR	The pinch roller could not be moved to the release position
UCS_PINCH_CLOSE_ERROR	The pinch roller could not be moved to the pinch position
UCS_PINCH_INIT_ERROR	The pinch roller initialization could not be completed
UCS_PINCH_POS_UNKNOWN	The pinch roller position is unknown（during trials only）
UCS_PASS_GUIDE_SUPPLY_ERROR	The pass guide could not be moved to the supply position
UCS_PASS_GUIDE_SHELL_ERROR	The pinch roller could not be moved to the retracted position
UCS_PASS_GUIDE_EJECT_ERROR	The pinch roller could not be moved to the eject position
UCS_PASS_GUIDE_INIT_ERROR	The pass guide initialization could not be completed
UCS_PASS_GUIDE_POS_UNKNOWN	The pass guide position is unknown（during trials only）
UCS_SIDE_GUIDE_HOME_ERROR	The side guide could not be moved to the HOME position
UCS_SIDE_GUIDE_POS_ERROR	The side guide could not be moved to the prescribed position
UCS_SIDE_GUIDE_INIT_ERROR	The side guide initialization could not be completed
UCS_ACT_GUIDE_HOME_ERROR	The act guide could not be moved to the HOME position
UCS_SHELL_ROTATE_HOME_ERROR	The shell roller didn't rotate to the HOME position
UCS_SHELL_ROTATE_REV_ERROR	The shell roller didn't rotate to the REV position
UCS_LEVER_DOWN_ERROR	The paper feed lever could not be moved to the DOWN position
UCS_LEVER_LOCK_ERROR	The paper feed lever could not be moved to the LOCK position
UCS_LEVER_UP_ERROR	The paper feed lever could not be moved to the UP position
UCS_CUTTER_HOME_ERROR	The cutter could not be returned to the HOME position
UCS_CUTTER_AWAY_ERROR	The cutter could not be moved to the AWAY position
UCS_CUTTER_INIT_ERROR	The cutter initialization could not be completed
UCS_CUTTER_POS_UNKNOWN	The cutter position is unknown（during trials only）
UCS_TRAY_OUT	The paper tray was removed during printing
UCS_TOP_COVER_OPEN	The maintenance cover opened during printing

Getting Turning unit and cut sheet information

[Format] long GetCutPaper(long lPortNum, LPSTR p);

[Argument]
lPortNum: Port number
p: Pointer to the receiving buffer

[Return]
Successful: Number of characters received at buffer p
Failure: -1

[Explanation] Receives a character string for paper size (tray guide position) and paper status

Turning unit status

4、3 digits (nn00) Paper size (tray guide position)	2、1 digits (00nn) Paper status
0000 Unit not connected 0100 8 x 10.75 0200 8 x 12	0000 No cut sheet 0001 Protective sheet 0002 Cut sheet set

Sample code

Paper size	Paper status	Media code
8x10.75	Yes	0102
8x10.75	Protective sheet	0101
8x10.75	No	0100

Turning unit, Cancelling the 2nd-side printing and ejecting the media

[Format] BOOL SetCancelDuplexPrint(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: TRUE
Failure: FALSE

[Explanation] During duplex printing, this command cancels the 2nd-side print after printing the 1st side, and ejects the media.
This command is used so that, in situations such as the ribbon ending after printing the 1st side, etc., the media on hold in the printer is forcibly ejected.

The relation between this command and the printer operation differs depending on the printer status and the status of the 2nd-side print data transmission.
Refer to the chart below for the operations.

Relation between commands and printer operation

Printer status	2nd-side print data transmission status	
	After2nd-side data trans.	Before 2nd-side data trans.
When the ribbon ends after completing 1st-side printing	1st side is printed and media is ejected. Printer's print buffer is cleared	1st side is printed and media is ejected.
Other than above	After completion of duplex printing, the media is ejected. Command is invalid	1st side is printed and media is ejected.

[Sample Coding]

< Visual C >
```
SetCancelDuplexPrint ( PortNo );
```

< Visual Basic >
```
SetCancelDuplexPrint (PortNo)
```

Get Printer Counter Value

[Format]

```
long GetCounterL( long lPortNum );
long CvGetCounterL( long lPortNum );

long GetCounterA( long lPortNum );
long CvGetCounterA( long lPortNum );

long GetCounterB( long lPortNum );
long CvGetCounterB( long lPortNum );

long GetCounterP( long lPortNum );

long GetCounterMatte( long lPortNum );
long GetCounterM( long lPortNum );
long GetCounterDuplex( long lProtNum );
```

[Argument] lPortNum: Port number

[Return] Successful: Counter Value
Failure: -1

[Explanation]

GetCounterL(),CvGetCounterL()	Returns Life Counter Value.
GetCounterA(),CvGetCounterA()	Returns Counter A Value.
GetCounterB(),CvGetCounterB()	Returns Counter B Value.
GetCounterP(long lPortNum)	Returns Counter P Value
GetCounterMatte()	Returns Matte Counter Value.
GetCounterM()	Returns Counter M Value.
GetCounterDuplex()	Returns Counter Duplex Value.

■Life Counter, Counter A, Counter B

- For the 6-inch width size, it counts up 1 for each sheet. However, for the 6x9(A5W), 6x8(A5) and 5x7(2L) sizes, it counts up 2 for each sheet (converting to PC/L size). For multi-cut prints, it counts up 2 after every 2nd sheet.
- For the 8-inch width size stand-alone model, it counts up 1 for every sheet. For multi-cut printing, it counts up 1 after every 2nd sheet (every 3rd sheet for 3-image layout)
- For the 8-inch width size tower model, it counts up 1-3 depending on print size (see the chart on the next page).
- Counters A and B can be cleared with SetClearCounterA() and SetClearCounterB() functions respectively.

■Counter P

- Counter P is initialized when power on. Random setting is OK with SetCounterP().
- Counter P value is countered according to each discharge of image.

■Matte Counter, Counter M

- When overcoat finish is matte print, Matte Counter and Counter M will be counted up (Life Counter and Counter A/B are counted up also).
- Counter M is clearable with SetClearCounterM() function.
- These counters are effective to the printer (DS40/80 Ver.1.30 or later. DS40T/80T Ver.0.31 or later.), which supported matte printing.
- Specification of count up is the same as that of the above-mentioned Life Counter and Counter A/B.

■Counter Duplex

- When printing using cut sheet (using Turning unit) , Counter Duplex will be counted up.
- Duplex counter 1 count up when printing one side of the cut sheet, and for 2 count up when printing on both sides

[Sample Coding]

< Visual C >

```
long counter;
if(( counter = GetCounterL( PortNo )) >= 0 ){
        // Returns Life Counter Value to counter
}
```

< Visual Basic >

```
Dim counter As Long
counter = GetCounterL(PortNo)
```

Counter L/A/B/P Count-up Action

■Tower:

Timing of a count-up is after a print picture is correctly conveyed to the belt sorter or the upper tray of a tower.
Count-up is not performed when an error occurred. Counter P is cleared to 0, if a power supply is turned off.

DS40T	Print Size		Counter L/A/B	Counter P
Single-cut	5x3.5 (L)		+1	+1
	6x4 (PC)		+1	+1
	5x7 (2L)		+2	+1
	6x8 (A5)		+2	+1
	6x9 (A5W)		+2	+1
Multi-cut	6x4x2	1st image	---	+1
	(PC 2-image layout)	2nd image	+2	+1
	5x3.5x2 (*1)	1st image	---	+1
	(L 2-image layout)	2nd image	+2	+1

(*1) DS40T firmware version 0.33 or later

DS80T	Print Size		Counter L/A/B	Counter P
Single-cut	8x10		+3	+1
	8x12		+3	+1
	8x4		+1	+1
	8x5		+1	+1
	8x6		+1	+1
	8x8		+2	+1
	A4_Length		+3	+1
Multi-cut	8x4x2	1st image	---	+1
		2nd image	+2	+1
	8x5x2	1st image	---	+1
		2nd image	+2	+1
	8x6x2	1st image	---	+1
		2nd image	+2	+1
	8x5+8x4	1st image	---	+1
		2nd image	+2	+1
	8x6+8x4	1st image	---	+1
		2nd image	+2	+1
	8x6+8x5	1st image	---	+1
		2nd image	+2	+1
	8x8+8x4	1st image	---	+1
		2nd image	+2	+1
	8x4x3	1st image	---	+1
		2nd image	---	+1
		3rd image	+3	+1

■Stand-alone:

Timing of a count-up is after performing a cut of a print picture correctly.

Count-up is not performed when an error occurred. Counter P is cleared to 0, if a power supply is turned off.

DS40,DS-RX1, CW02,DP-DS620	Print Size		Counter L/A/B	Counter P (*1)
Single-cut	5x3.5 (L)		+1	+1
	6x4 (PC)		+1	+1
	6x4.5 (*3)		+1	+1
	5x5 (*5)		+2	+1
	6x6 (*4)		+2	+1
	5x7 (2L)		+2	+1
	6x8 (A5)		+2	+1
	6x9 (A5W) *Except for DS-RX1		+2	+1
Multi-cut	6x4x2	1st image	---	+1
	(PC 2-image layout)	2nd image	+2	+1
	6x4.5x2 (*3)	1st image	---	+1
	(6x4.5 2-image layout)	2nd image	+2	+1
	5x3.5x2 (*2)	1st image	---	+1
	(L 2-image layout)	2nd image	+2	+1
2inch cut	6x4	1st sheet	---	+1
*Only correspondence model		2nd sheet	+1	+1
Refet to cutter control	6x8	1st sheet	---	+1
command		2nd sheet	---	+1
		3rd sheet	---	+1
		4th sheet	+2	+1

(*1) DS40 firmware version 1.04 or later

(*2) DS40 firmware version 1.50 or later and DS-RX1 firmware version 1.20 or later and DP-DS620

(*3) DP-DS620 firmware version 1.10 or later

(*4) DS40 firmware version 1.60 or later and DS-RX1 firmware version 2.06 or later and DP-DS620

(*5) DS-RX1 firmware version 2.06 or later and DP-DS620

DS80	Print Size		Counter L/A/B	Counter P (*1)
Single-cut	8x10		+1	+1
	8x12		+1	+1
	8x4		+1	+1
	8x5		+1	+1
	8x6		+1	+1
	8x8		+1	+1
	A4_Length		+1	+1
Multi-cut	8x4x2	1st image	---	+1
		2nd image	+1	+1
	8x5x2	1st image	---	+1
		2nd image	+1	+1
	8x6x2	1st image	---	+1
		2nd image	+1	+1
	8x5+8x4	1st image	---	+1
		2nd image	+1	+1
	8x6+8x4	1st image	---	+1
		2nd image	+1	+1
	8x6+8x5	1st image	---	+1
		2nd image	+1	+1
	8x8+8x4	1st image	---	+1
		2nd image	+1	+1
	8x4x3	1st image	---	+1
		2nd image	---	+1
		3rd image	+1	+1

(*1)DS80 firmware version 1.02 or later

DP-TC10	Print Size	Counter L/A/B	Counter P
Single-cut	TC86x61	+1	+1
	TC127x61	+1	+1
2-image layout continuous printing,	TC127x61	+2	+1

DP-DS820	Print Size		Counter L/A/B	Counter P
Single-cut	8x4		+1	+1
	8x5		+1	+1
	8x6		+1	+1
	8x7		+1	+1
	8x8		+1	+1
	8x9		+1	+1
	8x10		+1	+1
	8x12		+1	+1
	A4_Length		+1	+1
	A5 Format		+1	+1
	A4x5		+1	+1
	A4x6		+1	+1
	A4x8		+1	+1
	A4x10		+1	+1
	A4 Format		+1	+1
Multi-cut	8x4x2	1st image	---	+1
		2nd image	+1	+1
	8x5x2	1st image	---	+1
		2nd image	+1	+1
	8x6x2	1st image	---	+1
		2nd image	+1	+1
	8x5+8x4	1st image	---	+1
		2nd image	+1	+1
	8x6+8x4	1st image	---	+1
		2nd image	+1	+1
	8x6+8x5	1st image	---	+1
		2nd image	+1	+1
	8x8+8x4	1st image	---	+1
		2nd image	+1	+1
	8x4x3	1st image	---	+1
		2nd image	---	+1
		3rd image	+1	+1
	A5x2	1st image	---	+1
		2nd image	+1	+1
	A4x5x2	1st image	---	+1
		2nd image	+1	+1
2inch cut	8x4	1st image	---	+1
		2nd image	+1	+1
	8x6	1-2nd image	---	+1/image
	A4x6	3rd image	+1	+1
	8x8	1-3rd image	---	+1/image
	A4x8	4th image	+1	+1
	8x10	1-4th image	---	+1/image
	A4x10	5th image	+1	+1
	8x12	1-5th image	---	+1/image
		6th image	+1	+1

CV, CW	Print Size		Counter L/A/B
Single-cut	5x3.5 (L)		+1
	6x4 (PC)		+1
	5x7 (2L)		+2
	6x8 (A5)		+2
	6x9 (A5W)		+2
Multi-cut	6x4x2	1st image	---
	(PC 2-image layout)	2nd image	+2

DP-QW410	Print Size		Counter L/A/B	Counter P
Single-cut	4x3		+1	+1
	4x4		+1	+1
	4x4.5		+1	+1
	4x6		+1	+1
	4.5x3		+1	+1
	4.5x4		+1	+1
	4.5x4.5		+1	+1
	4.5x6		+1	+1
	4.5x8		+1	+1
Multi-cut	4x3x2	1st sheet	---	+1
		2nd sheet	+1	+1
	4.5x3x2	1st sheet	---	+1
		2nd sheet	+1	+1
	4.5x4x2	1st sheet	---	+1
		2nd sheet	+1	+1
2inch cut	4x4	1st sheet	---	+1
		2nd sheet	+1	+1
	4x6, 4.5x6	1st sheet	---	+1
		2nd sheet	---	+1
		3rd sheet	+1	+1
	4.5x8	1st sheet	---	+1
		2nd sheet	---	+1
		3rd sheet	---	+1
		4th sheet	+1	+1

■Duplex

The counter is increased after a print image is cut normally.
The counter doesn't increase when an error occurs. Counter P is cleared to 0 when the power is turned OFF.

When using roll media

DP-DS80D DP-DS820DX	Print size		Counter L/A/B	Counter P	Counter Duplex
Single cut	8x10		+1	+1	---
	8x12		+1	+1	---
	8x4		+1	+1	---
	8x5		+1	+1	---
	8x6		+1	+1	---
	8x7.1 (*2)		+1	+1	---
	8x8		+1	+1	---
	8x11 (*2)		+1	+1	---
	A4_Length (*1)		+1	+1	---
Multi-cut	8x4x2	1st image	---	+1	---
		2nd image	+1	+1	---
	8x5x2	1st image	---	+1	---
		2nd image	+1	+1	---
	8x6x2	1st image	---	+1	---
		2nd image	+1	+1	---
	8x5+8x4	1st image	---	+1	---
		2nd image	+1	+1	---
	8x6+8x4	1st image	---	+1	---
		2nd image	+1	+1	---
	8x6+8x5	1st image	---	+1	---
		2nd image	+1	+1	---
	8x8+8x4	1st image	---	+1	---
		2nd image	+1	+1	---
	8x4x3	1st image	---	+1	---
		2nd image	---	+1	---
		3rd image	+1	+1	---

*1 DP-DS80D only

When using cut media

DP-DS80D DP-DS820DX	Print size			Counter L/A/B	Counter P	Counter Duplex
Single Cut	8x10	1 side		+1	+1	+1
		Both sides		+2	+1	+2
	8x12	1 side		+1	+1	+1
		Both sides		+2	+1	+2
	8x4	1 side		+1	+1	+1
		Both sides		+2	+1	+2
	8x5	1 side		+1	+1	+1
		Both sides		+2	+1	+2
	8x5.9 (*2)	Both sides		+2	+1	+2
	8x6	1 side		+1	+1	+1
		Both sides		+2	+1	+2
	8x6.5 (*2)	Both sides		+2	+1	+2
	8x7.1 (*2)	Both sides		+2	+1	+2
	8x8	1 side		+1	+1	+1
		Both sides		+2	+1	+2
	8x10.35 (*2)	Both sides		+2	+1	+2
	8x10.5	1 side		+1	+1	+1
		Both sides		+2	+1	+2
	8x10.75	1 side		+1	+1	+1
		Both sides		+2	+1	+2
	8x11 (*2)	Both sides		+2	+1	+2
	8xA4 (*2)	Both sides		+2	+1	+2
Multi Cut	8x4x2	1 side (*1)	1st sheet	---	---	---
			2nd sheet	+1	+2	+1
		Both side	1st sheet	---	---	---
			2nd sheet	+2	+2	+2
	8x5x2	1 side (*1)	1st sheet	---	---	---
			2nd sheet	+1	+2	+1
		Both side	1st sheet	---	---	---
			2nd sheet	+2	+2	+2
	8x5.9x2 (*2)	Both side	1st sheet	---	---	---
			2nd sheet	+2	+2	+2
	8x6x2	1 side (*1)	1st image	---	---	---
			2nd image	+1	+2	+1
		Both side	1st image	---	---	---
			2nd image	+2	+2	+2
	8x4x3	1 side (*1)	1st sheet	---	---	---
			2nd sheet	---	---	---
			3rd sheet	+1	+3	+1
		Both side	1st sheet	---	---	---
			2nd sheet	---	---	---
			3rd sheet	+2	+3	+2

*1 DP-DS80D only

*2 DP-DS820DX only

Get Life Counter Value Extended (DP-DS620)

[Format] long GetCounterLifeEX(long lPortNum, DWORD dwMedia);

[Argument]
lPortNum: Port number
dwMedia: Specifying read target

[Return]
Successful: Counter Value
Failure: -1

[Explanation] Read the number of Life Counter value.

Specifying read target
COUNT_ALL : Life Counter for All Media
COUNT_SD : Life Counter printed by SD Media
COUNT_PD : Life Counter printed by PD Media
When you set the Other than those above, it is returned Life Counter for all media.

[Sample Coding]

< Visual C >

```
long counter;
if(( counter = GetCounterLifeEX ( PortNo, COUNT_ALL )) >= 0 ){
        // Returns Life Counter Value to counter
}
```

< VB.NET >

```
Dim c As Integer
c = GetCounterLifeEX (PortNo, COUNT_ALL)
If c >= 0 Then Text1.Text = Str(c) Else Text1.Text = "ERROR!"
```

Get Matte Counter Value Extended (DP-DS620)

[Format] long GetCounterMatteEX(long lPortNum, DWORD dwMedia);

[Argument]
lPortNum : Port number
dwMedia : Specifying read target

[Return]
Successful: Counter Value
Failure: -1

[Explanation] Read the number of Matte Counter value.

Specifying read target
COUNT_ALL : Matte Counter for All Media
COUNT_SD : Matte Counter printed by SD Media
COUNT_PD : Matte Counter printed by PD Media
When you set the Other than those above, it is returned Matte Counter for all media

[Sample Coding]

< Visual C >

```
long counter;
if(( counter = GetCounterMatteEX ( PortNo, COUNT_ALL )) >= 0 ){
        // Returns Matte Counter Value to counter
}
```

< VB.NET >

```
Dim c As Integer
c = GetCounterMatteEX (PortNo, COUNT_ALL)
If c >= 0 Then Text1.Text = Str(c) Else Text1.Text = "ERROR!"
```

Clear Printer Counter Value

[Format]
BOOL SetClearCounterA(long lPortNum);
BOOL CvSetClearCounterA(long lPortNum);

BOOL SetClearCounterB(long lPortNum);
BOOL CvSetClearCounterB(long lPortNum);

BOOL SetClearCounterM(long lPortNum);

[Argument]
lPortNum: Port number

[Return]
Successful: TRUE
Failure: FALSE

[Explanation]
SetClearCounterA(), CvSetClearCounterA() Clears Counter A.
SetClearCounterB(), CvSetClearCounterB() Clears Counter B.
SetClearCounterM() Clears Counter M.
Counter M is effective to the printer (DS40/80 Ver.1.30 or later. DS40T/80T Ver.0.31 or later.), which supported matte printing.

[Sample Coding]
< Visual C >
```
if( SetClearCounterA( PortNo )){
        // Counter A was cleared
}
```

< Visual Basic >
```
If SetClearCounterA(PortNo) <> False Then
        ' Counter A was cleared
End If
```

Set Printer Counter Value

[Format]
BOOL SetCounterP(long lPortNum, long lCounter);

[Argument]
lPortNum: Port number
lCounter: Set counter value

[Return]
Successful: TRUE
Failure: FALSE

[Explanation]
Sets Counter value P. Set random number to Counter P. When power off, it is initialized to 0.
If a negative value is specified, 0 is set.

[Sample Coding]
< Visual C >
```
if( SetCounterP( PortNo, 100)){
        // Set Counter P
}
```

< Visual Basic >
```
If SetCounterP(PortNo, 100) <> False Then
        ' Set Counter P
End If
```

Get the Number of Free Image Buffers

[Format] long GetFreeBuffer(long lPortNum);
long CvGetFreeBuffer(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: The number of free print buffers
Failure: -1

[Explanation] Printer will return the number of free image buffers.

[Sample Coding]

< Visual C >

```
long bn;
if(( bn = GetFreeBuffer( PortNo )) >= 0 ){
        // Returns the number of free buffers to bn
}
```

< Visual Basic >

```
Dim bn As Long
bn = GetFreeBuffer(PortNo)
```

Get Remaining Print Quantity

[Format] long GetPQTY(long lPortNum);
long CvGetPQTY(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: Remaining number of images to be printed
Failure: -1

[Explanation] Returns the number of images remaining to be printed.

[Sample Coding]

< Visual C >

```
long number;
if(( number = GetPQTY( PortNo )) >= 0 ){
        // Returns the number of remaining prints to number
}
```

< Visual Basic >

```
Dim number As Long
number = GetPQTY(PortNo)
```

Get the Media Counter of Remaining Sheets

[Format] long GetMediaCounter(long lPortNum);
long CvGetMediaCounter(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: The number of remaining sheets
Failure: -1

Machine	Media	Number of Printable sheets	Number of sheets default value
CV	5x3.5 (L)	400	500
	6x4 (PC)	400	450
	5x7 (2L)	230	280
	6x9 (A5W)	180	230
CW CW02	5x3.5 (L)	660	710
	6x4 (PC)	600	650
	5x7 (2L)	350	400
	6x8 (A5)	300	350
	6x9 (A5W)	280	330
DS40 DS40T	5x3.5 (L)	400	450
	6x4 (PC)	400	450
	5x7 (2L)	230	280
	6x8 (A5)	200	250
	6x9 (A5W)	180	230
DS-RX1	5x3.5 (L)	700	750
	6x4 (PC)	700	750
		350	400
	5x7 (2L)	350	400
	6x8 (A5)	350	400
DS80 / DS80T DP-DS80D	8x10	130	180
	8x12	110	160
DP-TC10	TC86x61	600	650
	TC127x61	600	650
DP-DS620	5x3.5 (L) *2	420	420
	6x4 (PC)	400	400
	5x7 (2L)	230	230
	6x8 (A5)	200	200
		208	208
	6x9 (A5W) *1	180	180
DP-DS820	8x10	130	130
	8x12	110	110
	A4	110	110
DP-DS820DX	8x10	130	130
		180	180
	8x12	110	110
		150	150
DP-QW410	4x6	150	150
	4.5x6 *3	150	150
	4.5x8	110	110

The above table shows the initial number of typical medias for each model.
*1 DP-DS620 firmware version 1.10 or later
*2 DP-DS620 firmware version 1.60 or later
*3 DP-QW410 firmware version 1.07 or later

When paper initialization is performed due to media replacement, etc., the remaining number of printed media may be consumed depending on the model. See below for consumption.

Media status	DP-TC10(ver.1.20 only) DP-DS620 (excluding IAM product media) DP-DS820	DP-QW410 DP-DS620 IAM product media DP-DS820DX	Others
Unused media	0	0	-1
Media in use	0	-1	-1

[Explanation]

Printer will return the number of sheets remaining in the printer.
The number of remaining sheets is greater than the number of printable sheets.
(For DP-QW410, this value is 0 and ribbon end. For the DP-DS620/DP-DS820/DP-DS820DX, if this value is 0 it signals ribbon end, but it runs ribbon check before this point. Refer to "(4) About Ribbon End Check Operation" for details.)

[Sample Coding]

< Visual C >

```
long number;
if(( number = GetMediaCounter( PortNo )) >= 0 ){
        // Returns the number of remaining sheets to number
}
```

< Visual Basic >

```
Dim number As Long
number = GetMediaCounter(PortNo)
```

Half Size Conversion Media Counter of Remaining Sheets

[Format] long GetMediaCounterH(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: The Media Counter of Remaining Sheets which has been converted to half size of loaded ribbon.
Failure: -1

Model	Media Size	Half Size	Half size conversion media default counter of sheets
DP-DS620	5x7 (2L)	5x3.5	460
	6x8 (A5)	6x4	400
			416
	6x9 (A5W) *1	6x4.5	360
DP-DS820	8x10	8x5	260
	8x12	8x6	220
	A4	A5	220
DP-DS820DX	8x10	8x5	360
	8x12	8x6	300

*1 DP-DS620 firmware version 1.10 or later

[Explanation] When you load the ribbon size media to printer, it will return the number of remaining sheets which has been converted to half size.

[Sample Coding]

< Visual C >

```
long number;
if(( number = GetMediaCounterH ( PortNo )) >= 0 ){
        // Next process
}
```

< Visual Basic >

```
Dim number As Long
number = GetMediaCounterH (PortNo)
```

Get Media Color Offset Value of the Lot

[Format] long GetMediaColorOffset(long lPortNum);
long CvGetMediaColorOffset(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: Media color offset value of the lot
Failure: -1

[Explanation] Printer will return the offset value.

[Sample Coding]

< Visual C >

```
long offset;
if(( offset = GetMediaColorOffset( PortNo )) >= 0 ){
        // Returns the value to offset
}
```

< Visual Basic >

```
Dim offset As Long
offset = GetMediaColorOffset(PortNo)
```

[Example] In the case where offset = 169082895 (decimal) --> 0x0A14000F (hex),
the offset values for each color are defined as below.

Color	Offset value
Yellow	10(0x0A)
Magenta	20(0x14)
Cyan	0 (0x00)
Op	15(0x0F)

Get Media Lot Information

[Format] long GetMediaLotNo(long lPortNum, LPSTR p);
long CvGetMediaLotNo(long lPortNum, LPSTR p);

[Argument] lPortNum: Port number
p: Pointer to the receiving buffer

[Return] Successful: The number of characters received by buffer p
Failure: -1

[Explanation] Printer will return the information stored in RFID tag of the media.

[Sample Coding]

< Visual C >

```
char rbuf[256];
if( GetMediaLotNo( PortNo, (LPSTR)rbuf ) > 0 ){
// Next process
}
```

< Visual Basic >

```
Dim rb As String * 255, Ver As String
Dim n As Long
n = GetMediaLotNo(PortNo, rb)
If n > 0 Then Ver = Left(rb, n)       ' Retrieves as a character string
```

[Media tag information]
ML********** User-specific information<16Bytes>

Get Printer Serial Number

[Format] long GetSerialNo(long lPortNum, LPSTR p);
long CvGetSerialNo(long lPortNum, LPSTR p);

[Argument] lPortNum: Port number
p: Pointer to the receiving buffer

[Return] Successful: The number of characters received by buffer p
Failure: -1

[Explanation] Printer will return the printer serial number.

[Sample Coding]

< Visual C >

```
char rbuf[256];
if( GetSerialNo( PortNo, (LPSTR)rbuf ) > 0 ){
        // Next process
}
```

< Visual Basic >

```
Dim rb As String * 255, Ver As String
Dim n As Long
n = GetSerialNo(PortNo, rb)
If n > 0 Then Ver = Left(rb, n)       ' Retrieves as a character string
```

[Example of serial number]
DS4Xnnnnnnnn

Set Firmware Update Mode

[Format] BOOL SetFirmwUpdateMode(long lPortNum);
BOOL CvSetFirmwUpdateMode(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: TRUE
Failure: FALSE

[Explanation] Changes the printer mode to firmware update mode.
Refer to "11.Procedure to firmware update when using API" for details.

[Sample Coding]

< Visual C >

```
if( SetFirmwUpdateMode(PortNo) == TRUE ){
        // Next process
}
```

< Visual Basic >

```
If ( SetFirmwUpdateMode(PortNo) = True ) Then
        ' Next process
End If
```

Write Firmware Data

[Format] BOOL SetFirmwDataWrite(long lPortNum, LPSTR lpData, DWORD dwDataLen);
BOOL CvSetFirmwDataWrite(long lPortNum, LPSTR lpData, DWORD dwDataLen);

[Argument]

lPortNum:	Port number
lpData:	Pointer to the buffer where the data is to be rewritten
dwDataLen:	The number of characters of the data

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation] Sends data to rewrite firmware to the printer. The data is supplied by Motorola S format file.
When this command is issued, a buffer (approx. 10M byte) is necessary for reading the file and for storing the data temporarily.
After completion of data writes, the printer is rebooted automatically, and Update Mode is reset.
Refer to "11.Procedure to firmware update when using API" for details.

[Sample Coding] < Visual Basic >

```
Dim fd(10000000)
Dim c As Long, n As Long

C=0
Open fname For Binary Access Read As #1
FileLength = LOF(1)
For n = 0 To FileLength - 1
        Get #1, , fd(c)
        c = c + 1
Next n
Close #1

SetFirmwDataWrite( PortNo, VarPtr(fd(0)), c )
```

Set Turning unit Rewrite Mode

[Format] BOOL SetDunitFirmwUpdateMode(long lPortNum, DWORD dwBaudRate);

[Argument]

lPortNum:	Port number
dwBaudRate	Baud Rate Setting 0 : 38400bps 1 : 57600bps 2 : 115200bps 3 : 230400bps

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation]
Changes the printer mode to Turning unit firmware rewrite mode.
Refer to "11.Procedure to firmware update when using API"for details.
The setting may fail to update. In this case, please use 2:115200 bps.

[Sample Coding]

< Visual C >

```
if( SetDunitFirmwUpdateMode(PortNo,3) == TRUE ){
        // Next process
}
```

< Visual Basic >

```
If ( SetDunitFirmwUpdateMode(PortNo,3) = True ) Then
        ' Next process
End If
```

Write Turning unit Firmware Data

[Format] BOOL SetDunitFirmwDataWrite(long lPortNum, LPSTR lpData, DWORD dwDataLen);

[Argument]

lPortNum:	Port number
lpData:	Pointer to the buffer where the data is to be rewritten
dwDataLen:	The number of characters of the data

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation]
Sends data to rewrite Turning unit firmware to the printer. The data is supplied by binary format file.
When this command is issued, a buffer (approx. 10M byte) is necessary for reading the file and for storing the data temporarily.
After completion of data writes, the printer is rebooted automatically, and Update Mode is reset.
Refer to "11.Procedure to firmware update when using API" for details.

[Sample Coding]

< Visual Basic >

```
Dim fd(10000000)
Dim c As Long, n As Long

C=0
Open fname For Binary Access Read As #1
FileLength = LOF(1)
For n = 0 To FileLength - 1
        Get #1, , fd(c)
        c = c + 1
Next n
Close #1

SetDunitFirmwDataWrite( PortNo, VarPtr(fd(0)), c )
```

Get Turning unit Firmware Rewriting Status

[Format] long GetDunitUpdateStatus (long lPortNum, LPSTR p);

[Argument]
lPortNum: Port number
p: Pointer to the receiving buffer

[Return]
Successful: The number of characters received in buffer p
Failure: -1

[Explanation] Receives the firmware rewriting information into the buffer.

[Sample Coding]

< Visual C >

```
char rbuf[256];
if(GetDunitUpdateStatus ( PortNo, (LPSTR)rbuf ) > 0 ){
        // Next process
}
```

< Visual Basic >

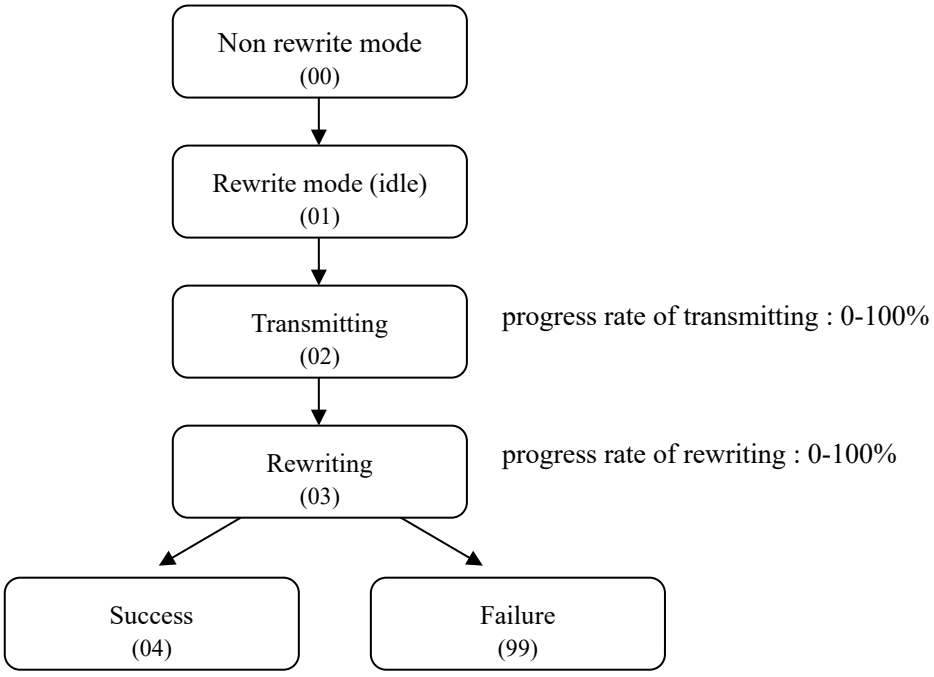
```
Dim rb As String * 255, Ver As String
Dim n As Long
n = GetDunitUpdateStatus (PortNo, rb)
If n > 0 Then Ver = Left(rb, n)       'Retrieves as a character string
```

[Firmware Rewriting Information]

STS-**; Rewriting Status
00:non rewrite mode
01:Turning unit firmware rewrite mode (idle)
02:Turning unit firmware is transmitting.
03:Turning unit firmware is rewriting.
04:Rewriting the Turning unit firmware is success.
99:Rewriting the Turning unit firmware is failure.

PRG-***; Progress Rate (000-100)

EXT-********; Extended Information (00000000-99999999)



Status transition diagram of Turning unit firmware rewrite mode

Request Creation of Turning Unit Error Log（DP-DS80D）

[Format] BOOL DunitErrorLogCreateReq(long lPortNum, DWORD dwBaudRate);

[Argument]

lPortNum:	Port number
dwBaudRate:	Baud rate setting 0: 38400 bps 1: 57600 bps 2: 115200 bps

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation] This requests the printer to create a Turning Unit Error Log.
For details, please refer to “P73 Getting the Turning Unit Error Log Data”.

[Sample Coding]

< Visual C >

```
if( DunitErrorLogCreateReq(PortNo,2) == TRUE ){
        // Process
}
```

< Visual Basic >

```
If DunitErrorLogCreateReq(PortNo,2) = True Then
        'Process
End if
```

Get Turning Unit Error Log Data Length（DP-DS80D）

[Format] long GetDunitErrorLogDataLength(long lPortNum);

[Argument]

lPortNum:	Port number

[Return]

Successful:	Data length
Failure:	-1

[Explanation] This gets the data length of the Turning Unit Error Log.
For details, please refer to “P73 Getting the Turning Unit Error Log Data”.

[Sample Coding]

< Visual Basic >

```
Dim length As Long

length = GetDunitErrorLogDataLength ( PortNo )
```

Receive the Turning Unit Error Log Data（DP-DS80D）

[Format] long GetDunitErrorLogData(long lPortNum, LPSTR lpData, DWORD dwBufSize);

[Argument]

lPortNum:	Port number
lpData:	Pointer for the buffer to receive the log data
dwBufSize:	Size of the buffer the application selected to receive the data

[Return]

Successful:	Data length
Failure:	-1

[Explanation] This receives the Turning Unit Error Log data.
The Error Log Data is in CSV format text. When using this command, a buffer (about 10 Kbyte) is needed to temporarily hold the Error Log data.
When the receiving is complete, the printer will end the Get Error Log mode.
For details, please refer to “P73 Getting the Turning Unit Error Log Data”.

[Sample Coding]

< Visual Basic >

```
Dim fd(10000)
Dim c As Long, n As Long

GetDunitErrorLogData ( PortNo, VarPtr(fd(0)), 10000 )
```

Get the Turning Unit Error Log Creation Status（DP-DS80D）

[Format] long GetDunitErrorLogStatus(long lPortNum, LPSTR p, DWORD dwBufSize);

[Argument]

lPortNum:	Port number
p:	Pointer for receiving buffer
dwBufSize:	Size of the buffer that the application selected to receive the data

[Return]

Successful:	The number of characters received at buffer p
Failure:	-1

[Explanation] This receives the Error Log Creation Status at the character string buffer.

[Sample Coding]

< Visual C >

```
char rbuf[256];
if(GetDunitErrorLogStatus ( PortNo, (LPSTR)rbuf, 256 ) > 0 ){
        // Process
}
```

< Visual Basic >

```
Dim rb As String * 255, Sen As String
Dim n As Long
n = GetDunitErrorLogStatus (PortNo, rb, 256)
If n > 0 Then Sen = Left(rb, n)      'When importing a character string.
```

[Getting Error Log Creation Status]

STS-**;	Rewriting Status
	00: Turning Unit Error Log incomplete
	01: Creating Turning Unit Error Log
	02: Turning Unit Error Log complete
	99: Error
PRG-***;	Progress status (000～100)
EXT-********;	Extension status (00000000～99999999)
	00005401: Turning Unit Log Data Forwarding Preparation error
	00005402: Turning Unit Log Forwarding Byte-count Reception error
	00005403: Turning Unit Log Data Reception error
	00005404: Turning Unit Log Data Overflow
	00005405: Turning Unit Log Forwarding End Reception error
	00005406: Turning Unit Log Transmit Mode End error
	00005407: Time-out Error
	00005420: Other Error

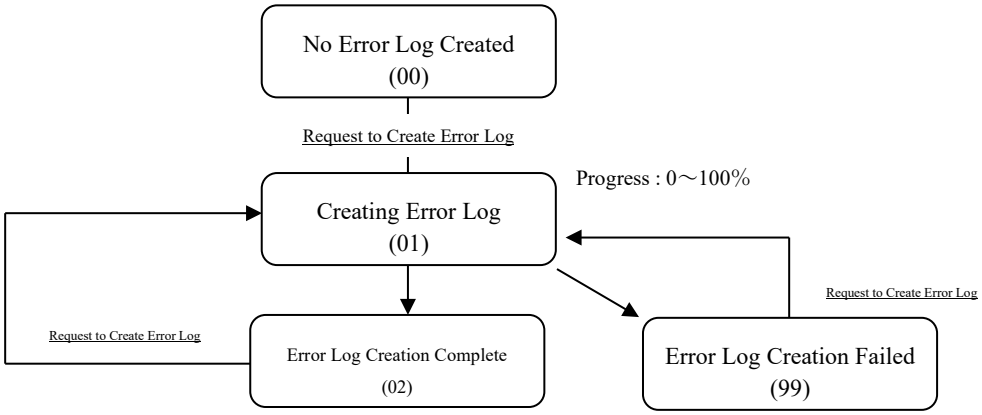


Diagram of the Succession of the Turning Unit Error Log Mode Status

Getting the Turning Unit Error Log Data

When using the API to get the Error Log, please use the following procedure.

Reference page

Example of API use)

1	DunitErrorLogCreateReq()	Request Creation of Turning Unit Error Log	P71
2	GetDunitErrorLogStatus()	Get Turning Unit Error Log Creation Status	P72
3	GetDunitErrorLogDataLength()	Get Turning Unit Error Log Data Length	P71
4	GetDunitErrorLogData()	Receive Turning Unit Error Log Data	P72

Clear Color Data

[Format] BOOL SetColorDataClear(long lPortNum);
BOOL CvSetColorDataClear(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: TRUE
Failure: FALSE

[Explanation] Clear color control data stored in the printer.
When rewriting color control data, please first clear color control data with this command.
Refer to "11.Procedure to firmware update when using API" for details.

[Sample Coding]

< Visual C >

```
if(SetColorDataClear( PortNo )){
        // Color control data successfully cleared
}
```

< Visual Basic >

```
If SetColorDataClear(PortNo) = True Then
        ' Color control data successfully cleared
End If
```

Write Color Data

[Format] BOOL SetColorDataWrite(long lPortNum, LPSTR lpData, DWORD dwDataLen);
BOOL CvSetColorDataWrite(long lPortNum, LPSTR lpData, DWORD dwDataLen);

[Argument] lPortNum: Port number
lpData: Pointer to the buffer where the color data is to be rewritten
dwDataLen: The number of characters of the data

[Return] Successful: TRUE
Failure: FALSE

[Explanation] Sends color data to rewrite.
Before sending the new color data with this command, please first clear existing color data.

Color data is provided in an original format binary file, and when this command is issued, a buffer (approx. 10M byte) is necessary for reading the file and for storing the data temporarily.
Refer to "11.Procedure to firmware update when using API" for details.

[Sample Coding]

< Visual Basic >

```
Dim fd(10000000)
Dim c As Long, n As Long

C=0
Open fname For Binary Access Read As #1
FileLength = LOF(1)
For n = 0 To FileLength - 1
        Get #1, , fd(c)
        c = c + 1
Next n
Close #1

SetColorDataWrite( PortNo, VarPtr(fd(0)), c )
```

Set Color Data Version

[Format] BOOL SetColorDataVersion(long lPortNum, LPSTR lpData, DWORD dwDataLen);
BOOL CvSetColorDataVersion(long lPortNum, LPSTR lpData, DWORD dwDataLen);

[Argument]

lPortNum:	Port number
lpData:	Pointer to the buffer where the color data version is stored
dwDataLen:	The number of characters of the data

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation] Sets the color control data version.
After rewriting color control data, please set the version name as the file name of the color control data supplied.
Refer to "11.Procedure to firmware update when using API" for details.

[Sample Coding] < Visual Basic >

```
Dim fname As String
' Store version name in fname
SetColorDataVersion( PortNo, fname, Len(fname) )
```

Get Color Data Version

[Format] long GetColorDataVersion(long lPortNum, LPSTR p);
long CvGetColorDataVersion(long lPortNum, LPSTR p);

[Argument]

lPortNum:	Port number
p:	Pointer to the receiving buffer

[Return]

Successful:	The number of characters received by buffer p
Failure:	-1

[Explanation] Printer will return the version name in the buffer.
Refer to "11.Procedure to firmware update when using API" for details.

[Sample Coding] < Visual C >

```
char rbuf[256];
if(GetColorDataVersion( PortNo, (LPSTR)rbuf ) > 0 ){
        // Next process
}
```

< Visual Basic >

```
Dim rb As String * 255, Ver As String
Dim n As Long

n = GetColorDataVersion(PortNo, rb)
If n > 0 Then Ver = Left(rb, n)       ' Retrieves as a character string
```

Get Color Data Version <Type Designation>

[Format] long GetColorDataVersionRes(long lPortNum, LPSTR p , long lType);

[Argument]
lPortNum: Port number
p: Pointer to the receiving buffer
lType: Type designation setting value : 0 - 4

[Return]
Successful: The number of characters received by buffer p
Failure: -1

[Explanation] Printer will return checksum of the version name of specified type in the buffer.
For details of the functions, refer to "11.Procedure to firmware update when using API".

Acquisition data for each model and valid setting value

acquired value	setting value		
	DP-DS620 DP-DS820DX	DP-DS820	DP-QW410
300dpi	0	0	0
600dpi	1	1	(not applicable)
slow speed	2	2	4
high density	(not applicable)	3	(not applicable)

[Sample Coding]

< Visual C >

```
long lType;
char rbuf[256];
if(GetColorDataVersionRes( PortNo, (LPSTR)rbuf , lType ) > 0 ){
        // Next process
}
```

< Visual Basic >

```
Dim lType As Long
Dim rb As String * 255, Ver As String
Dim n As Long
n = GetColorDataVersionRes(PortNo, rb, lType)
If n > 0 Then Ver = Left(rb, n)
```

Get Color Data Version <Type, Media Designation>

[Format] long GetColorDataVersionResEX(long lPortNum, LPSTR p , long lType, DWORD dwMedia);

[Argument]

lPortNum:	Port number	
p:	Pointer to the receiving buffer	
lType:	Type designation	setting value : 0 - 4
dwMedia:	Media type designation	setting value : 1(SD) , 2(PD), 3(PP)

[Return]

Successful:	The number of characters received by buffer p
Failure:	-1

[Explanation] Printer will return checksum of the version name of specified type in the buffer.
For details of the functions, refer to "11.Procedure to firmware update when using API".

Acquisition data for each model and valid setting value

acquired value	setting value		
	DP-DS620 DP-DS820DX	DP-DS820	DP-QW410
300dpi	0	0	0
600dpi	1	1	(not applicable)
slow speed	2	2	4
high density	(not applicable)	3	(not applicable)

When you set '1' to the Media Type designation, printer will return version name of the color data for SD media in the buffer. *3
When you set '2' to the Media Type designation, printer will return version name of the color data for PD media in the buffer. *1
When you set '3' to the Media Type designation, printer will return version name of the color data for PP media in the buffer. *2

*1: DP-DS820DX, DP-QW410 and DP-DS620(ver.3.00 or later)
*2: DP-DS820 only
*3: Excluding DP-DS820DX

[Sample Coding]

< Visual C >

```
char rbuf[256];
if(GetColorDataVersionResEX( PortNo, (LPSTR)rbuf , 1, 3 ) > 0 ){
        // Next process
}
```

< Visual Basic >

```
Dim rb As String * 255, Ver As String
Dim n As Long
n = GetColorDataVersionResEX(PortNo, rb, 0, 3)
If n > 0 Then Ver = Left(rb, n)
```

Set the Color Control Data Version in Temporary Buffer (CW)

[Format] BOOL SetColorDataVersionTmp(long lPortNum, LPSTR lpData, DWORD dwDataLen);

[Argument]

lPortNum:	Port number
lpData:	Pointer to the buffer where the color control data is stored
dwDataLen:	Data character length

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation] This sets the Color Control Data Version in the printer RAM (temporary buffer).
When sending the color control data to the temporary file, use this command to set the color control data version.
Use the file name of the color control data file that is provided when setting the color control data version.

[Sample Coding] Refer to SetColorDataWriteTmp for the sample

Send Color Control Data to Temporary Buffer (CW)

[Format] BOOL SetColorDataWriteTmp(long lPortNum, LPSTR lpData, DWORD dwDataLen);

[Argument]

lPortNum:	Port number
lpData:	Pointer to the buffer where the color control data is stored
dwDataLen:	Data character length

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation] This sends the color control data to the printer RAM (temporary buffer). By sending the color control data, the previous data will be written over.

This command rewrites the CW printer color control data, and is used when the resolution is changed frequently. The temporary file will be erased when the power is turned OFF. Rebooting will return it to the default value.
(The default value can be rewritten using SetColorDataWrite().)

Color control data is provided in an original format binary file. When using this command, a buffer for temporary storage (approximately 10 Mbyte) is necessary to read the file and rewrite the data.

[Sample Coding] < Visual Basic >

```
Dim fd(10000000)
Dim c As Long, n As Long

C=0
Open fname For Binary Access Read As #1
FileLength = LOF(1)
For n = 0 To FileLength - 1
        Get #1, , fd(c)
        c = c + 1
Next n
Close #1

SetColorDataVersionTmp( PortNo, fname, Len(fname) )
SetColorDataWriteTmp( PortNo, VarPtr(fd(0)), c )
```

Get Color Data Checksum

[Format] long GetColorDataChecksum(long lPortNum, LPSTR p);
long CvGetColorDataChecksum(long lPortNum, LPSTR p);

[Argument]
lPortNum: Port number
p: Pointer to the receiving buffer

[Return]
Successful: The number of characters received by buffer p
Failure: -1

[Explanation] Printer will return checksum of the color data in the buffer.
Refer to "11.Procedure to firmware update when using API" for details.

[Sample Coding]

< Visual C >

```
char rbuf[256];
if(GetColorDataChecksum( PortNo, (LPSTR)rbuf ) > 0 ){
        // Next process
}
```

< Visual Basic >

```
Dim rb As String * 255, Ver As String
Dim n As Long
n = GetColorDataChecksum(PortNo, rb)
If n > 0 Then Ver = Left(rb, n)       ' Retrieves as a character string
```

Get Color Data Checksum <Type Designation>

[Format] long GetColorDataChecksumRes(long lPortNum, LPSTR p, long lType);

[Argument]
lPortNum: Port number
p: Pointer to the receiving buffer
lType: Type designation setting value : 0 - 4

[Return]
Successful: The number of characters received by buffer p
Failure: -1

[Explanation] Printer will return checksum of the color data of specified type in the buffer.
For details of the functions, refer to the "11.Procedure to firmware update when using API".

Acquisition data for each model and valid setting value

acquired value	setting value		
	DP-DS620 DP-DS820DX	DP-DS820	DP-QW410
300dpi	0	0	0
600dpi	1	1	(not applicable)
slow speed	2	2	4
high density	(not applicable)	3	(not applicable)

[Sample Coding]

< Visual C >

```
long lType;
char rbuf[256];
if(GetColorDataChecksumRes( PortNo, (LPSTR)rbuf , lType ) > 0 ){
        // Next process
}
```

< Visual Basic >

```
Dim lType As Long
Dim rb As String * 255, Ver As String
Dim n As Long
n = GetColorDataChecksumRes(PortNo, rb, lType)
If n > 0 Then Ver = Left(rb, n)
```

Get Color Data Checksum <Type, Media Designation>

[Format] long GetColorDataChecksumResEX(long lPortNum, LPSTR p, long lType, DWORD dwMedia);

[Argument]

lPortNum:	Port number	
p:	Pointer to the receiving buffer	
lType:	Type designation	setting value : 0 - 4
dwMeida	Media designation	setting value : 1 (SD) , 2(PD), 3(PP)

[Return]

Successful:	The number of characters received by buffer p
Failure:	-1

[Explanation] Printer will return checksum of the color data of specified type in the buffer.
For details of the functions, refer to the "11.Procedure to firmware update when using API".

Acquisition data for each model and valid setting value

acquired value	setting value		
	DP-DS620 DP-DS820DX	DP-DS820	DP-QW410
300dpi	0	0	0
600dpi	1	1	(not applicable)
slow speed	2	2	4
high density	(not applicable)	3	(not applicable)

When you set '1' to the Media Type designation, printer will return checksum of the color data for SD media in the buffer. *3
When you set '2' to the Media Type designation, printer will return checksum of the color data for PD media in the buffer. *1
When you set '3' to the Media Type designation, printer will return checksum of the color data for PP media in the buffer. *2

*1: DP-DS820DX, DP-QW410 and DP-DS620(ver.3.00 or later)
*2: DP-DS820 only
*3: Excluding DP-DS820DX

[Sample Coding]

< Visual C >

```
char rbuf[256];
if(GetColorDataChecksumResEX( PortNo, (LPSTR)rbuf , l, 3 ) > 0 ){
        // Next process
}
```

< Visual Basic >

```
Dim rb As String * 255, Ver As String
Dim n As Long
n = GetColorDataChecksumResEX(PortNo, rb, l, 3)
If n > 0 Then Ver = Left(rb, n)
```

Cutter Control Command

[Format] BOOL SetCutterMode(long lPortNum, DWORD ctMode);

[Argument]
lPortNum: Port number
ctMode: Cutter mode selection

[Return]
Successful: TRUE
Failure: FALSE

[Explanation] This is to designate the cutter operation.
The cutter operation designation has a macro definition of CspStat.h.
The symbols have the following meanings:

CUTTER_MODE_STANDARD	(0)	Standard cutter operation
CUTTER_MODE_NONSCRAP	(1)	Non-scrap cutter operation
CUTTER_MODE_SINGLECUT	(2)	Single cut mode operation (DP-TC10)
CUTTER_MODE_NONSCRAP_1	(100)	2 image layout and both sheets non-scrap cutter operation *4
CUTTER_MODE_NONSCRAP_2	(101)	2 image layout and 1st sheet non-scrap cutter operation*4
CUTTER_MODE_NONSCRAP_3	(102)	2 image layout and 2nd sheet non-scrap cutter operation *4
CUTTER_MODE_2INCHCUT	(120)	2inch cut operation *1
CUTTER_MODE_ L2IMAGEPRINT	(130)	L card size 2-image layout continuous printing designation (for DP-TC10 only) *2
CUTTER_MODE_PANORAMA	(1000)	Panorama printing operation (with blank between images)*3

Note) The cutter control command sets the operation before the image data is sent. The command is valid for 1 image. After performing the designated cut for the printed image, the printer will return to its standard cut operation.
2inch cut operation is effective in these paper size.

DS40	:	6x4, 6x8
DS-RX1	:	6x4, 6x8, 6x4x2
DP-DS620	:	6x4, 6x8, 6x4x2
DP-DS820	:	8x4, 8x6, 8x8, 8x10, 8x12, A4x6, A4x8, A4x10
DP-QW410	:	4x4, 4x6, 4.5x4, 4.5x6, 4.5x8, 4.5x4x2

Panorama printing supports these sizes.

DP-DS620	:	6x16(two 6x8 images), 6x24(two 6x8 images)
DP-DS820	:	8x20(two 8x10 images), 8x24(two 8x12 images), A4x2(two A4 images), 8x30(three 8x10 images), 8x24(three 8x12images), A4x3(three A4 images)

*1 Effective at the firmware version of the model in the following table.

Model	DS40	DS-RX1	DP-DS620	DP-DS820	DP-QW410
Firmware Ver.	1.40 or later	1.10 or later	○	○	○

*2 For details on the DP-TC10 L card size 2-image layout continuous printing, please refer to "13.Appendix (2) About DP-TC10 L card size 2-image layout continuous printing".

*3 Supported by DP-DS620 firmware version 01.20 or later or DP-DS820.
Refer to "13.Appendix (5) Panorama Printing with White Border", for more information about panorama printing.

*4 Supported by DP-DS620 firmware version 04.00 or later.
The Non-scrap cutter operation when the 2image layout and Non-scrap cutter operation parameter (CUTTER_MODE_NONSCRAP_1, CUTTER_MODE_NONSCRAP_2, CUTTER_MODE_NONSCRAP_3) is selected is shown in the table below.

Specified value	2image layout		1image layout
	First print	Second print	
CUTTER_MODE_NONSCRAP_1	Non-scrap cutter operation	Non-scrap cutter operation	Non-scrap cutter operation
CUTTER_MODE_NONSCRAP_2	Non-scrap cutter operation	Normal operation	Non-scrap cutter operation
CUTTER_MODE_NONSCRAP_3	Normal operation	Non-scrap cutter operation	Non-scrap cutter operation

[Sample Coding]

< Visual C >

```
SetCutterMode(PortNo, (DWORD)CUTTER_MODE_NONSCRAP);
```

< Visual Basic >

```
SetCutterMode(PortNo, CUTTER_MODE_NONSCRAP)
```

Get Media ID Set Info

[Format] long GetMediaIdSetInfo(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: Media ID set info
Failure: -1

[Explanation] Return of the Media ID set info.

Model	Return Value	The contents of a Media ID set info
CW, CW02	0	ID setup is not completed.
	1	CW-standard ID
	2	CW-PMI ID
Others	0	

[Sample Coding]

< Visual C >

```
long idval;
if(( idval = GetMediaIdSetInfo( PortNo )) >= 0 ){
        // Returns the Media ID set info
}
```

< Visual Basic >

```
Dim number As Long
idval = GetMediaIdSetInfo(PortNo)
```

1-image Retention Control Command (DS40)

[Format] BOOL SetRetentionMode(long lPortNum, DWORD rtMode);

[Argument] lPortNum: Port number
rtMode: 1-image retention mode selection

[Return] Succesful: TRUE
Failure: FALSE

[Explanation] This is to designate the 1-Image retention operation for printer (DS40).
The 1-Image retention mode designation has a macro definition of CspStat.h.
The symbols have the following meanings:

RETENTION_MODE_OFF	(0)	Normal Mode(Default)
RETENTION_MODE_ON	(1)	1-Image retention Mode

Note) If the printer is turned off, it will revert to the normal mode the next time it is started up.

[Sample Coding]

< Visual C >

```
SetRetentionMode(PortNo, (DWORD)RETENTION_MODE_ON);
```

< Visual Basic >

```
SetRetentionMode(PortNo, RETENTION_MODE_ON)
```

Overcoat Finishing Control Command

[Format] BOOL SetOvercoatFinish(long lPortNum, DWORD ovcoat);

[Argument] lPortNum: Port number
ovcoat: Overcoat finishing mode selection

[Return] Successful: TRUE
Failure: FALSE

[Explanation] This is to designate the overcoat finishing operation.
The overcoat finishing mode designation has a macro definition of CspStat.h. The symbols have the following meanings:

OVERCOAT_FINISH_GLOSSY	(0)	glossy(default)
OVERCOAT_FINISH_MATTE1	(1)	matte 1
OVERCOAT_FINISH_PMATTE1	(11)	Partial matte1 (Partial matte :) HR3000 recommended
OVERCOAT_FINISH_FINEMATTE	(21)	Fine matte
OVERCOAT_FINISH_LUSTER	(22)	Luster
OVERCOAT_FINISH_HSMATTE	(31)	Light matte
OVERCOAT_FINISH_PMATTE11	(101)	Partial matte (Matte)
OVERCOAT_FINISH_PMATTE12	(121)	Partial matte (Fine matte)
OVERCOAT_FINISH_PMATTE13	(122)	Partial matte (Luster)

Note) The overcoat finishing control command sets the operation before the image data is sent.
The command is valid for 1 image.
After performing the designated overcoat finishing for the printed image, the printer will return to its glossy finishing operation.

The table below shows the status of support for this command by model.

Model	Version	Return
CV <MPⅢ> / CW	-	FALSE
DS40/80 DS40T/80T	Compatible	TRUE
	Not Compatible	FALSE
Others	Compatible	TRUE

・Overcoat finish <overall type>
○:Support -:No support Value:Effective firmware version (This or later)

	Model								
Overcoat finish	DS40T DS80T	DS40 DS80	CW2 DS-RX1	DP-DS80D	DP-DS620	DP-DS820	DP-DS820DX	DP-QW410	Others
OVERCOAT_FINISH_GLOSSY	○	○	○	○	○	○	○	○	○
OVERCOAT_FINISH_MATTE1	0.31	1.30	○	○	○	○	○	○	-
OVERCOAT_FINISH_FINEMATTE	-	-	-	-	1.20	○	○	-	-
OVERCOAT_FINISH_LUSTER	-	-	-	-	1.30	○	○	-	-
OVERCOAT_FINISH_HSMATTE	-	-	-	-	-	0.61	○	-	-

・Overcoat finish <partial matte type> *1

○:Support -:No support Value:Effective firmware version (This or later)

Overcoat finish	Model				
	DP-TC10	DP-DS620	DP-DS820	DP-QW410	Others
OVERCOAT_FINISH_PMATTE1	○	-	-	-	-
OVERCOAT_FINISH_PMATTE11	-	1.20	○	○	-
OVERCOAT_FINISH_PMATTE12	-	1.30	○	-	-
OVERCOAT_FINISH_PMATTE13	-	1.30	○	-	-

*1 Refer to “13.Appendix(3) About Partial Matte Printing”.

[Sample Coding]

< Visual C >

```
SetOvercoatFinish (PortNo, (DWORD) OVERCOAT_FINISH_MATTE2);
```

< Visual Basic >

```
SetOvercoatFinish (PortNo, OVERCOAT_FINISH_MATTE2)
```

Print Retry Control command

[Format] BOOL SetRetryControl (long lPortNum, DWORD retry);

[Argument]

lPortNum:	Port number	
retry:	Selects Print Retry ON/OFF	
	PRINT_RETRY_OFF (0)	Print retry [OFF]
	PRINT_RETRY_ON (1)	Print retry [ON]

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation] With this function setting, when there is a printer error, this makes a new attempt to print the data that was in the printer buffer after the error is resolved.
The operation designation for Retry Control is macro defines as CspStat.h.
This command is valid for DS40/80(Ver.1.30 or later.) and CW02, DS-RX1, DP-DS80D, DP-DS620, DP-DS820, DP-DS820DX, DP-QW410.

Note) The Retry Control command is sent before the image data. Each command is valid for 1 image. The normal initial status of the printer has the Retry Control OFF, so if this command is not executed, the operation is the same as always. Details of operation are as follows.

With the DP-DS80D, the retry function cannot be used when using cut paper.
The same is true for the DP-DS820DX.

Setting Value:	Disable（Default）	Enable
Function:	Printing data in a printer buffer is cleared in the case of a printer error.After an error recovery, the cleared image data must be re-sent.	Printing data in a printer buffer is held in the case of a printer error.After an error recovery, the print process resumes with the saved data. However, if the power is turned OFF, the data will not be saved.

For retry printing operation in each error

Contents of an error	Print Re-try setup is "Disable"	Print Re-try setup is "Enable"
Paper End	Please resend picture data not printed after error recovery.	Printing processing is resumed from data currently held in a printer buffer after error recovery. When paper initialization operation is performed, it is printed after paper initialization operation.
Ribbon End		
Cover Open		
Scrap Box Error		
Paper Jam	Please resend picture data not printed after error recovery.	DP-DS820, DP-QW410: Even if the printing retry function has been set to "Enable", print data in a printer buffer is cleared. Please resend picture data not printed after error recovery. Other model: Printing processing is resumed from data currently held in a printer buffer after error recovery. When paper initialization operation is performed, it is printed after paper initialization operation.
Ribbon Error		
Paper Definition Error		
Data Error	For error recovery, the power supply of a printer once needs to be turned OFF. Therefore, print data in a printer buffer is cleared by power-supply OFF of a printer regardless of a setup of the Print Re-try function. Please resend picture data not printed after printer re-starting.	For error recovery, the power supply of a printer once needs to be turned OFF. Therefore, print data in a printer buffer is cleared by printer power-supply OFF even if the Print Re-try function is set to "Enable". Please resend picture data not printed after printer re-starting.
Head Voltage Error		
Head Position Error		
Power Supply Fan Stop		
Cutter Error		
Abnormal Head Temp.		
Abnormal Media Temp.		
Ribbon Tension Error		
RFID Module Error		
Abnormal motor Temp.		
USB Power Supply Voltage Error		
System Error		

Get Media Class

[Format] long GetRfidMediaClass(long lPortNum, LPSTR p);

[Argument]
lPortNum: Port number
p: Pointer to the receiving buffer

[Return]
Successful: The number of characters received by buffer p
Failure: -1

[Explanation] The media class data recorded in the RF-ID tag is returned. (ASCII character of four digits)

・Media Type

Acquired string	Description
(None)	Default media
0001	Digital (SD) media.
0002	Premium(PD) media
0003	Pure Premium (PP) media.
1004	Metallic paper
1023	Silverpearl paper

Models other than DP-DS620, DP-DS820, DP-DS820DX and DP-QW410: Not set. (No String)
* The default media depends on the model.

[Sample Coding]

< Visual C >

```
char rbuf[256];
if(GetRfidMediaClass( PortNo, (LPSTR)rbuf ) > 0 ){
        // Next process
}
```

< VB.NET >

```
Dim s As String = New String(“”, 255)
Dim i As Integer
i = GetRfidMediaClass(PortNo, s)
If i > 0 Then Text1.Text = VB.Left(s, i) Else Text1.Text = "ERROR!"
```

Get RF-ID reserve data

[Format] long GetRfidReserveData(long lPortNum, LPSTR p, DWORD dwPage);

[Argument]

lPortNum:	Port number
p:	Pointer to the receiving buffer
dwPage:	Page of reserve data

[Return]

Successful:	The number of characters received by buffer p
Failure:	-1

[Explanation] The reserve data recorded in the RF-ID tag is returned. (ASCII character of four digits)

[Sample Coding]

< Visual C >

```
char rbuf[256];
if(GetRfidReserveData( PortNo, (LPSTR)rbuf ,0x01 ) > 0 ){ // Page 1
        // Next process
}
```

< VB.NET >

```
Dim s As String = New String(“”, 255)
Dim i As Integer
i = GetRfidReserveData(PortNo, s, &h2 ) 'Page 2
If i > 0 Then Text1.Text = VB.Left(s, i) Else Text1.Text = "ERROR!"
```

Get Initial Media Count

[Format] long GetInitialMediaCount(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: Initial media count
Failure: -1

[Explanation] This returns the initial media count.
The returned for the initial media count is the actual number +50, so when using a function, be sure to factor in the +50 count. (DP-DS620, DP-DS820 and DP-QW410 are excluded.)

[Sample Coding]

< Visual C >

```
long number;
if(( number = GetInitialMediaCount(PortNo) >= 0 ){
        // Next process
}
```

< VB.NET >

```
Dim number As Integer
number = GetInitialMediaCount(PortNo)
```

Get Turning unit Firmware Version (DP-DS80D)

[Format] long GetVersionDuplex(long lPortNum, LPSTR p);

[Argument] lPortNum: Port number
p: Pointer the reciving buffer

[Return] Successful: The number of characters received by buffer p
Failure: -1

[Explanation] Receive the Turning unit firmware version to the character string buffer.
When a printer is not DP-DS80D, returns -1.

[Sample Coding]

< Visual C >

```
char rbuf[256];
if ( GetVersionDuplex( PortNo, (LPSTR)rbuf ) > 0 ){
        // Process
}
```

< Visual Basic >

```
Dim rb As String * 255, Ver As String
Dim n As Long
n = GetVersionDuplex(PortNo, rb)
If n > 0 Then Ver = Left(rb, n)    'Retrieves as a character string
```

[Version Infomation]

*.** Frimware version
△△△△△△ When Turning unit firmware is not support function or Turning unit is not connected, version Information is blank (△ 0x20).

Full Cutter Set-up

[Format] long SetFullCutterSetUp(long lPortNum, long lCutSize1, long lCutSize2, long lCutSize3, long lCutSize4);

[Argument]

lPortNum:	Port number	
lCutSize1:	Image1 cut size	setting value:0, 20 -
lCutSize2:	Image2 cut size	setting value:0, 20 -
lCutSize3:	Image3 cut size	setting value:0, 20 -
lCutSize4:	Image4 cut size	setting value:0, 20 -

[Return]

Succesful:	1
Failure:	0

[Explanation]

With the media sizes shown in the table below, it will cut to the designated sizes to make from 1 to the corresponding maximum cutting sheet number.
The cut size can be set in 0.1-inch increments in the 2 inches to the maximum size range. The total size should not exceed the maximum size.
When you set '0' to the cutting size of the image 1, it will print in usually size (6x8). In that case, the cut size of the image 2-4 is ignored.

When the cutting sheet number is one sheet, please set '0' to the cut size of the image 2 - 4.
When the cutting sheet number is two sheets, please set '0' to the cut size of the image 3 - 4.
When the cutting sheet number is three sheets, please set '0' to the cut size of the image 4.

If you set the out-of-range cut size, it is returned "0".

■ Media size and cut size setting range

Media size	Maximum cutting sheet number	Maximum size	No. of images	Cut size setting range	
				Minimum	Maximum
5x3.5 (L) *1	1 sheet	3.5 inches	1	20	35
6x4 (PC) *1	2 sheets	4 inches	1	20	40
4x4 *4 4.5x4 *5			2	20	20
4x4.5 *5	2 sheets	4.5 inches	1	20	45
4.5x4.5 *5			2	20	25
4x6 *4	3 sheets	6 inches	1	20	60
4.5x6 *4			2	20	40
			3	20	20
5x7 (2L) *1	3 sheets	7 inches	1	20	70
8x7			2	20	50
			3	20	30
6x8 (A5) *2	4 sheets	8 inches	1	20	80
4.5x8 *4			2	20	60
			3	20	40
			4	20	20
6x9 (A5W) *1	4 sheets	9 inches	1	20	90
8x9			2	20	70
			3	20	50
			4	20	30
8x10 *3	4 sheets	10inches	1	20	100
			2	20	80
			3	20	60
			4	20	40
A4 *3	4 sheets	11.7inches	1	20	117
			2	20	97
			3	20	77
			4	20	57
8x12 *3	4 sheets	12inches	1	20	120
			2	20	100
			3	20	80
			4	20	60

*1 Supported by DP-DS620 firmware version 01.20 or later.
*2 Supported by DS40 firmware version 01.60 or later and DP-DS620
*3 Supported by DP-DS820.
*4 Supported by DP-QW410.
*5 Supported by DP-QW410 firmware version 01.09 or later.

[Attention] If you specify both this command and cutter control command, this command will be given priority.
If the total cut size set using this command is larger than the image size sent to the printer, this command is disabled and printing is done with the media size (normal size) sent to the printer.

[Sample Coding]

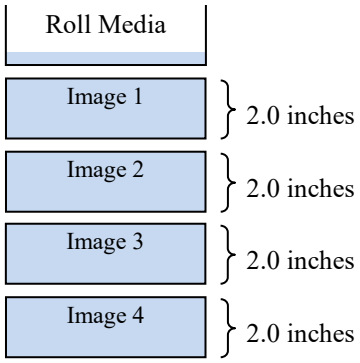
< Visual C >

```
if((SetFullCutterSetUp( PortNo, 20, 20, 20, 20 )) == 0 ){
        //Error
}
```

< Visual Basic >

```
Dim Result As Long
Result = SetFullCutterSetUp(PortNo, 20, 20, 20, 20)
If Result = 0 Then GoTo Error
```

*It will print the following result, when using the sample coding above.



Full Cutter Set-up Extended Settings

[Format] long SetFullCutterSetUpEX(long lPortNum, long lCutNum, long lScrapSize, long lCutSize1, ...);

[Argument]

lPortNum:	Port number	
lCutNum:	Number of image cut size settings	Setting value: 1-6
lScrapSize:	Intermediate scrap cut size	Setting value: 0, 12-22 (*1)
lCutSize1:	Image1 cut size	Setting value: 0, 20-
lCutSize2:	Image2 cut size	Setting value: 0, 20-
lCutSize3:	Image3 cut size	Setting value: 0, 20-
lCutSize4:	Image4 cut size	Setting value: 0, 20-
lCutSize5:	Image5 cut size	Setting value: 0, 20-
lCutSize6:	Image6 cut size	Setting value: 0, 20

[Return]

Successful:	1
Failure:	0

[Explanation]

Full Cutter Set-up Extended command that makes it possible to set the size of an intermediate scrap cut.
The scrap size can be set in 0.01-inch increments in the 0.12 to 0.22 inches range. (*1)
When setting the intermediate scrap cut size to a valid value other than 0, pay attention that the total cut size calculated with the formula below does not exceed the media maximum size.

Total cut size = (Total of each image cut size) + ((Intermediate scrap cut size/10) x (Number of images-1))

Example: The following formula is obtained when the media size is 6x9 inches (90), there are 4 images of 2 inches each (setting value = 20), and the intermediate scrap is 0.22 inches (setting value = 22).

(20+20+20+20+0+0)+((22/10)x(4-1)) = 86.6 ≦ 90

When intermediate scrap cut size is set to 0, refer to the table in Full Cutter Set-up for the media size and cut size setting ranges for each image.

If values outside the setting range above are set for the cut sizes, “0” is returned.

Effective at the firmware version of the model in the following table.

Model	DP-DS620	DS40	DP-DS820	DP-QW410
Firmware Ver.	1.20 or later	1.60 or later	○	○

For the paper size and cut size setting range, see the table " Media size and cut size setting range " in the full cutter setup.

(*1) DP-QW410 can only be set to 0 (no intermediate scrap cut) or 22.

[Attention]

If you specify both this command and the cutter control command, this command will be given priority. If the total cut size set using this command is larger than the image size sent to the printer, this command is disabled and printing is done with the media size (normal size) sent to the printer.

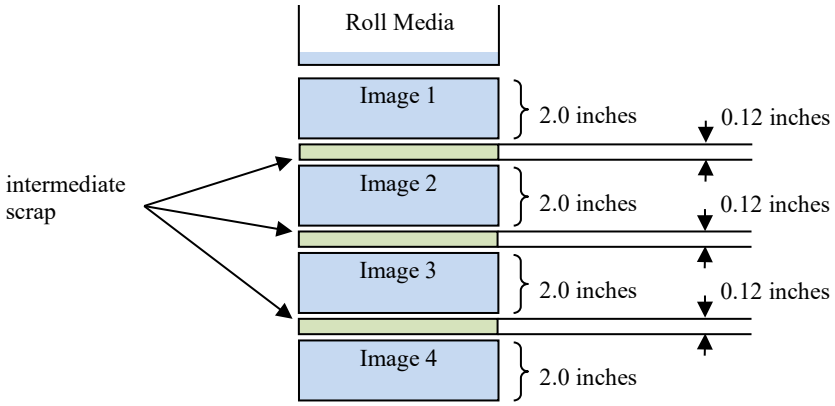
[Sample Coding] < Visual C >

```
if((SetFullCutterSetUpEX( PortNo, 4, 12, 20, 20, 20, 20 )) == 0 ){
        //Error
}
```

< Visual Basic >

```
Dim Result As Long
Result = SetFullCutterSetUpEX (PortNo, 4, 12, 20, 20, 20, 20)
If Result = 0 Then GoTo Error
```

* Using the sample coding above will print the following result.



Set Standby Mode Transition Time

[Format] long SetStandbyTime(long lPortNum, long lTime);

[Argument]

lPortNum:	Port number	
lTime:	Standby mode transition time	Setting value: 0, 1 - 99

[Return]

Succesful:	1
Failure:	0

[Explanation]

Set the time to transition to standby mode in 1 minute increments. (Up to 99 minutes maximum)
When the idle state continues for a set time, it transitions to the standby mode.
Settings are stored in the printer. (It is valid even when you turn off the printer.)

It becomes the specification the printer does not transition to the standby mode, if you set 00.
If you set the out-of-range transition time, it is returned "0".

[Attention]

When setting the Standby Mode transition time to less than the 10-minute default, if the time between the switch to Standby and reversion to Normal Mode is repeatedly short cycles of less than 10 minutes, the lifespan of the power unit in the printer could be decreased, so please try to avoid this situation.

(The lifespan of the power relay is expected to be 11 years, with transitions between the power-saving Standby Mode and Normal Mode every 10 minutes, for 12 hours a day, 20 days a month. If the cycle time between the two modes is shorter than 10 minutes, the lifespan will be shorter than that.)

[Sample Coding]

< Visual C >

```
if((SetStandbyTime( PortNo, 0 )) == 0 ){
      //Error
}
```

< Visual Basic >

```
Dim Result As Long
Result = SetStandbyTime(PortNo, 99)
If Result = 0 Then GoTo Error
```

Get Standby Mode Transition Time

[Format] long GetStandbyTime(long lPortNum);

[Argument]

lPortNum:	Port number	

[Return]

Succesful:	Standby mode transition time
Failure:	-1

[Explanation]

This returns the Standby mode transition time.
The default value for this setting is "10 minutes".

[Sample Coding]

< Visual C >

```
long time;
if(( time = GetStandbyTime( PortNo )) >= 0 ){
            // Next process
}
```

< VB.NET >

```
Dim time As Integer
time = GetStandbyTime(PortNo)
```

Set Media End Keep Mode

[Format] long SetEndKeepMode(long lPortNum, long lMode);

[Argument]

lPortNum:	Port number	
lMode:	Keep Mode Setting	Setting value:0, 1

[Return]

Succesful:	1
Failure:	0

[Explanation]

After the Media End has occurred, when you open and close the door or you restart the printer, it is set to determine whether or not to hold the Media End.
Media End (ribbon end, paper end) and Media Errors (Ribbon error, paper jam) are eligible.
Settings are stored in the printer. (It is valid even when you turn off the printer.)

When you set '0' to the keep mode setting, it becomes the mode not to keep media end.
When you set '1' to the keep mode setting, it becomes the mode to keep media end. (factory preset value)
When you set the out-of-range keep mode, it is returned "0".

[Sample Coding]

< Visual C >

```
if((SetEndKeepMode ( PortNo, 0 )) == 0 ){
        //Error
}
```

< Visual Basic >

```
Dim Result As Long
Result = SetEndKeepMode (PortNo, 1)
If Result = 0 Then GoTo Error
```

Get Media End Keep Mode

[Format] long GetEndKeepMode (long lPortNum);

[Argument]

lPortNum:	Port number	

[Return]

Succesful:	Keep Mode Setting
Failure:	-1

[Explanation]

This returns the Media End Keep Mode setting value.
The factory preset value for this setting is "the mode to keep media end (1)".

[Sample Coding]

< Visual C >

```
long mode;
if((mode = GetEndKeepMode ( PortNo )) > 0 ){
        // Next process
}
```

< VB.NET >

```
Dim mode As Integer
mode = GetEndKeepMode (PortNo)
```

Set USB iSerialNumber Availability Setting

[Format] long SetUSBSerialEnable(long lPortNum, long lEnable);

[Argument]
lPortNum: Port number
lEnable: Availability Setting

[Return]
Succesful: 1
Failure: 0

[Explanation]
Sets the USB iSerialNumber availability setting.
Settings are stored in the printer. (It is valid even when you turn off the printer.)
This configuration change will be effective when the power of the printer is turned on again.

When you set '0' to the USB iSerialNumber availability setting, it becomes disabled. (default)
When you set '1' to the USB iSerialNumber availability setting, it becomes enabled.
Also, when you enable this setting, serial number is the product serial number of the product-specific information.

[Sample Coding]

< Visual C >

```
if( SetUSBSerialEnable(PortNo, 1) == 1 ){     // Enable the USB serial number
        // Successful
}
```

< VB.NET >

```
If SetUSBSerialEnable(PortNo, 0) = 1 Then    ' Disable the USB serial number
        ' Successful
End If
```

Get USB iSerialNumber Availability Setting

[Format] long GetUSBSerialEnable(long lPortNum);

[Argument]
lPortNum: Port number

[Return]
1 : The USB iSerialNumber is enabled.
0 : The USB iSerialNumber is disabled.
-1 : Fail

[Explanation]
printer will return the USB iSerialNumber availability setting.
The default value for this setting is "The USB iSerialNumber is disabled (0)".

[Sample Coding]

< Visual C >

```
if( GetUSBSerialEnable(PortNo) == 1 ){
        // use
}
```

< VB.NET >

```
If GetUSBSerialEnable (PortNo) = 0 Then
        ' not use
End If
```

Set Rewind Mode setting (DP-DS620, DP-DS820)

[Format] long SetRewindMode(long lPortNum, int iMode);

[Argument]
lPortNum: Port number
iMode: Rewind Mode Setting

[Return]
Successful: 1
Failure: -1

[Explanation]
Set the Rewind Mode.
When you set '0' to the rewind mode setting, it releases the rewind mode. (default)
When you set '1' to the rewind mode setting, it becomes the rewind mode.

When the Rewind mode is set, if you specify the paper size for multiple screen allocation and send only the image data for one screen and execute printing, the paper size is replaced with the paper size for rewind printing, and the print Will be done.

Specified paper size	Replaced paper size
CSP_Lx2	CSP_L_REWIND
CSP_PCx2	CSP_PC_REWIND
CSP_6x4P5x2	CSP_6x4P5_REWIND
CSP_8x4x2	CSP_8x4_REWIND
CSP_8x5x2	CSP_8x5_REWIND
CSP_8x6x2	CSP_8x6_REWIND
CSP_8x4x3	CSP_8x4_REWIND
CSP_A5x2	CSP_A5_REWIND
CSP_A4x5x2	CSP_A4x5_REWIND

*However, the following the paper size for multiple screen allocation are not applicable to Rewind mode.
CSP_8x5_8x4, CSP_8x6_8x4, CSP_8x6_8x5, CSP_8x8_8x4

[Sample Coding]

< Visual C >

```
if( SetRewindMode( PortNo, 1 ) == 1 ){        // Set to the Rewind Mode
            // Successful
}
```

< VB.NET >

```
If SetRewindMode( PortNo,0 ) = 1 Then        ' Release the Rewind Mode
            ' Successful
End If
```

Get Rewind Mode setting (DP-DS620, DP-DS820)

[Format] long GetRewindMode (long lPortNum);

[Argument] lPortNum: Port number

[Return]
1 : The Rewind Mode is active.
0 : The Rewind Mode is inactive.
-1 : Fail

[Explanation] This returns the Rewind Mode Status.
The default value for this setting is "The Rewind Mode is inactive (0)".

[Sample Coding]

< Visual C >

```
if(GetRewindMode ( PortNo ) == 1 ){
        // use
}
```

< VB.NET >

```
If GetRewindMode ( PortNo ) = 0 Then
        ' not use
End If
```

Panorama Print Start Check (DP-DS620, DP-DS820)

[Format] DWORD GetPanoramaPrintable(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: Printer status indicating whether or not panorama printing is possible
Failure: STATUS_COMM_ERROR

[Explanation] This command returns the printer status indicating whether or not panorama printing is possible.

Effective at the firmware version of the model in the following table.

Model	DP-DS620	DP-DS820
Firmware Ver.	01.20 or later	○

The values for each status are defined in CspStat.h by a macro.
The status wordings have the following meanings.

Return value	value[HEX]	Status
STATUS_PRINTABLE	0x00000000	Panoramic prints can be started.
STATUS_HIGHTEMP_HEAD	0x00000001	High head temperature.
STATUS_HIGHHUMIDITY *1	0x00000100	High humidity.
STATUS_LOWTEMP_MEDIA *2	0x00000020	Low media temperature.
STATUS_OTHER_STATE	0x00010000	Other status (printing, cooling, error, etc.) Details can be obtained with the Get Printer Status command.
STATUS_COMM_ERROR	0xFFFFFFFF	Communication error

*1: Supported by DP-DS620 firmware version 01.40 or later and DP-DS820 01.01 or later.
*2: Supported by DP-DS620 firmware version 01.40 or later and DP-DS820 01.03 or later.

[Attention]

・When the head temperature is high, send the panoramic prints data without waiting for the status to change to show printing is possible. (The same as for normal printing, it will start cooling, and printing will start when the head temperature has dropped.)

・When the humidity is high, the print quality of the panoramic prints may deteriorate, so we don't recommend printing in high-humidity situations.

・When the media temperature is low, don't perform the continuous panoramic prints. (It may occur the problem such as a paper jam according to the image.)

(For details regarding panorama printing with a white border, please refer to "13.Appendix (5) Panorama Printing with White Border".)
(Please refer to the continuous panorama SDK for continuous panorama printing)

[Sample Coding]

< Visual C >

```
DWORD   stat;
stat = GetPanoramaPrintable( PortNo );
if(stat == STATUS_LOWTEMP_MEDIA){
        // please do not run the continuous panoramic prints
}else if(stat == STATUS_HIGHHUMIDITY){
        // run of the continuous panoramic prints is not recommended
}else if((stat == STATUS_PRINTABLE) || (stat == STATUS_HIGHTEMP_HEAD)){
        // please start the panoramic prints
}else{
        // checking printer status, and others
}
```

< VB.NET >

```
Dim stat As UInt32
stat = GetPanoramaPrintable( PortNo )
If stat = STATUS_LOWTEMP_MEDIA Then
        ' please do not run the continuous panoramic prints
Else If stat = STATUS_HIGHHUMIDITY Then
        ' run of the continuous panoramic prints is not recommended
Else If (stat = STATUS_PRINTABLE) Or (stat = STATUS_HIGHTEMP_HEAD) Then
        ' please start the panoramic prints
Else
        ' checking printer status, and others
End If
```

Get Media Offset Count (DS40, DS-RX1)

[Format] long GetMediaCountOffset(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: Media Offset Count
Failure: -1

[Explanation] This returns the media offset count (a value already added to the actual sheet count).

[Sample Coding]

< Visual C >

```
long        qtr_offset;
qtr_offset = GetMediaCountOffset( PortNo );
```

< Visual Basic >

```
Dim qtr_offset As Integer
qtr_offset = GetMediaCountOffset( PortNo )
```

Continuous Panoramic Prints Settings (DP-DS620, DP-DS820)

[Format] BOOL SetContPanorama(long lPortNum, DWORD dwContinuous, DWORD dwOverlap);

[Argument]

lPortNum:	Port number
dwContinuous:	Continuous Panoramic Prints specification.
dwOverlap:	Overlap width.

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation] This function sets the operation of continuous panoramic prints.

Effective at the firmware version of the model in the following table.

Model	DP-DS620	DP-DS820
Firmware Ver.	01.31 or later	00.50 or later

The continuous panoramic prints specification has a macro definition of CspStat.h.
The symbols have the following meanings.

CONT_PANORAMA_PRINT :	Continuous Panoramic Prints has specified.
CONT_PANORAMA_LAST :	Last image of Continuous Panoramic Prints, or normal printing. (Continuous Panoramic Prints cancelation)

[Attention] Continuous Panoramic Prints specification is only valid for the first image and second image.
If you set this parameter for the third image, the specified parameter will be ignored, and the paper will be cut.

The overlap width for the first image printed with Continuous Panoramic Prints is also applied to the second image and later on, but is disabled if a different value is set for the second or later images. Also, if you set this value to less than the recommended value, the print quality of the overlapped area will deteriorate.

[Sample Coding]

```
< Visual C >
if( SetContPanorama( PortNo, CONT_PANORAMA_PRINT, 200 ) { // Continuous Panoramic Prints settings
        // Successful
}

< VB.NET >
If SetContPanorama( PortNo, CONT_PANORAMA_PRINT, 200 ) Then ‘Continuous Panoramic Prints settings
        ‘ Successful
End If
```

Set Print Speed

[Format] BOOL SetPrintSpeed(long lPortNum, long lPrintSpeed);

[Argument]
lPortNum: Port Number
lPrintSpeed: Print Speed
0: Normal printing (Default)
1: Printing of 600dpi resolution (*2)
2: low speed printing (*2) or High quality printing (*3)
3: High density printing (*1)

*1: DP-DS820 only
*2: Excluding DP-QW410
*3: DP-QW410 only

[Return]
Successful: TRUE
Failure: FALSE

[Explanation]
Printing is carried out at the designated print speed.
Specify this when you want to reduce the printing speed and aim for higher image quality.

[Attention]
The printer will revert to the conventional operation after each image is printed.
This command must be designated for each image.

[Sample Coding]

< Visual C >

```
if( SetPrintSpeed(PortNo, 3) ){      // High density printing
        // Successful
}
```

< VB.NET >

```
If SetPrintSpeed(PortNo, 0) Then  // Normal printing
        ' Successful
End If
```

Set Gamma Table

[Format] BOOL SetGammaTable(long lPortNum, LPBYTE lpGammaTable, long lDataLen);

[Argument]
lPortNum： Port number
lpGammaTable： Pointer to the buffer where the gamma table data
lDataLen： Gamma table data size.(Unit:Byte)

[Return]
Successful: TRUE
Failure: FALSE

[Explanation] Sets the gamma table data.
For information on the format of the gamma table data, please refer to the table below.

Type	Item	Explanation
Header (8byte)	Data unit size	0002h fixed
	Row	0100h fixed
	Plane	0004h fixed
	Checksum	xxxxh（sum value of expansion and gamma table data）
Expansion Data (8byte)	Mode	0000h fixed
	Rsv1	0000h fixed
	Rev2	0000h fixed
	Rev	0000h fixed
Gamma Table Data (2048byte)	Y[0]…Y[255] M[0]…M[255] C[0]…C[255] OP[0]…OP[255]	0000h～FFFFh

[Attention] Because the gamma table data is lost when the printer is turned OFF or reverts to stand-by mode, it should be sent just before printing.

[Sample Coding]

< Visual C >
```
BYTE tableData[2064];
…
if(SetGammaTable( PortNo, tableData, 2064 )){
        // successful
}
```

< VB.NET >
```
Dim tableData(2064) As Byte
…
If SetGammaTable( PortNo, tableData, 2064 ) Then
        ' successful
End If
```

Get Gamma Table Checksum

[Format] long GetGammaTableChecksum(long lPortNum, LPSTR p);

[Argument]
lPortNum : Port number
lpRcvBuf : Pointer to the receiving buffer

[Return]
Successful: The number of characters received by buffer p
Failure: -1

[Explanation]
Printer will return checksum of the gamma table data in the buffer.
However, if the gamma table data is not set at the printer, it will receive a 4-character space character string "□□□□" (CR<0Dh> terminus, 4-byte padding).

[Sample Coding]

< Visual C >

```
char sum[256];
if(GetGammaTabelChecksum(PortNo, sum ) > 0 ){
        // successful
}
```

< VB.NET >

```
Dim sum As String = New String(“”, 256)
If GetGammaTabelChecksum (PortNo, sum ) > 0 Then
        ' successful
End If
```

Clear Data Table

[Format] BOOL SetDataTableClear(long lPortNum);

[Argument]
lPortNum : Port number

[Return]
Successful: TRUE
Failure: FALSE

[Explanation]
Clear the data of gamma table stored in the printer.

[Sample Coding]

< Visual C >

```
if(SetDataTableClear( PortNo )){
        // successful
}
```

< VB.NET >

```
If SetDataTableClear( PortNo ) Then
        ' successful
End If
```

Set Decurl Control

[Format] BOOL SetDecurlCtrl (long lPortNum, WORD wMode);

[Argument]
lPortNum : Port number
wMode : Decurl operation mode
0 : Do not perform the decurl operation
1 : Perform the decurl operation
2 : Automatic control (default)

[Return]
Successful: TRUE
Failure: FALSE

[Explanation]
Sets the paper decurl operation.
Decurl control settings are set before sending image data.
Decurl control settings are effective once for one image.
The printer will return to the default settings for each image.

[Sample Coding]

< Visual C >

```
if(SetDecurlCtrl(PortNo, 1 ) ){
        // successful
}
```

< VB.NET >

```
If SetDecurlCtrl(PortNo, 1 ) Then
        ' successful
End If
```

Current time notification

[Format] BOOL SetSysTime(long lPortNum, LPSTR lpSystemTime);

[Argument]
lPortNum : Port number
lpSystemTim : Current time (Format : yyyymmddHHMMSS)
range : 20000101000000 (Jan. 1, 2000 00:00:00) -
99991231235959 (Dec 31, 9999 23:59:59)

[Return]
Successful: TRUE
Failure: FALSE

[Explanation]
Notify the printer of the current time.
When the printer is turned on, this setting has been an initial value(Jan. 1, 2000 00:00:00).
If you set a value outside the range, it will be set with the minimum or maximum value of each set value.
This current time is used when the decurl control setting is “automatic control”.
Set it before executing SendImageData ().

Example of lpSystemTime input:
When the current time is 8:30:12 on April 26, 2019 => 20190426083012

[Sample Coding]

< Visual C >

```
if(SetSysTime( PortNo, "20190426083012" )){
        // successful
}
```

< VB.NET >

```
If SetSysTime( PortNo, "20190426083012" ) Then
        ' successful
End If
```

Get Supported Media Information(DP-DS620)

[Format] long GetSupportedMediaInfo (long lPortNum, LPSTR p);

[Argument]
lPortNum: Port number
p: Pointer to the receiving buffer

[Return]
Successful: The number of characters received in buffer p
Failure: -1

[Explanation] Get information on media supported by the printer.

■DP-DS620
The setting value is set in following 8-digit decimal ASCII numbers as the supported media information.

Setting value	Supported media			
	Premium Digital(PD) IAM Media	Luxury(LX) Media	Digital(SD) Media	Premium Digital(PD) Media
00000101	×	○	×	○
00000110	×	○	○	×
00000111	×	○	○	○
00001111	○	○	○	○

[Sample Coding]

< Visual C >
```
char rbuf[256];
if(GetSupportedMediaInfo(PortNo, (LPSTR)rbuf) > 0 ){
        // successful
}
```

< VB.NET >
```
Dim s As String = New String("", 255)
Dim i As Integer
i = GetSupportedMediaInfo(PortNo, s)
If i > 0 Then
        Text1.Text = VB.Left(s, i)
Else
        Text1.Text = "ERROR!"
End If
```

Setting crease line control

[Format] BOOL SetCreaseLine(long lPortNum, WORD wNumOfCreaseLine, WORD wNumOfPerPos, WORD wPitchOfPerPos, WORD wPosOfCreaseLine[]);

[Argument]

lPortNum:	Port number
wNumOfCreaseLine:	The number of crease lines to be arranged. setting value: 0(no crease line) - 2
wNumOfPerPos:	Number of crease lines per location. setting value: 1 - 9
wPitchOfPerPos:	The pitch of the crease lines to be placed in one place. setting value: 20 - 100 [Unit：0.1mm]
wPosOfCreaseLine[]:	An array that stores crease line placement locations. The array length is wNumOfCreaseLine. The data in the array represent the placement positions of the crease lines. setting value: 100 - 2950 [Unit：0.1mm]

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation] Places crease lines at specified locations. Valid only for cut sheets.
Please set before sending the first image.
If the following settings are made, the settings will be invalid.

- When the setting value is outside the setting range (less than 10 mm from both ends of the print).
- When the arrangement of crease lines is outside the range of the paper.
- When crease line placement is less than the minimum spacing (1mm).
- When the arrangement of crease lines overlaps.
- When the crease line placement locations have the following relationships.
  wPosOfCreaseLine[0] >= wPosOfCreaseLine[1]

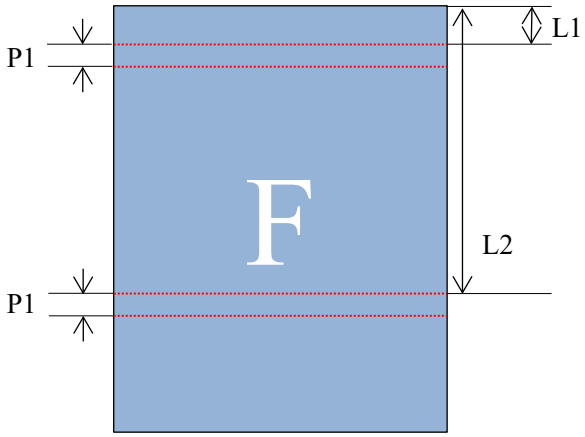
■Paper size and transport direction length of cut paper, and setting range

Paper size	Transport direction length [mm]	Minimum [mm]	Maximum [mm]
(8x4) (8x4)x2 (8x4)x3	101.6	10.0	91.6
(8x5)	127.0	10.0	117.0
(8x5)x2	127.0	10.0	113.3
(8x5.9)	150.0	10.0	140.0
(8x6) (8x6)x2	152.4	10.0	142.4
(8x6.5)	165.1	10.0	155.1
(8x7.1)	180.0	10.0	170.0
(8x8)	203.2	10.0	193.2
(8x10)	254.0	10.0	244.0
(8x10.35)	262.9	10.0	252.9
(8x10.50)	266.7	10.0	256.7
(8x10.75)	273.1	10.0	263.1
(8x11)	280.0	10.0	270.0
(8xA4)	297.0	10.0	287.0
(8x12)	304.8	10.0	294.8

[Attention] Specify this setting for each output.
The printer will return to its default settings (0) after printing.
The minimum recommended pitch for crease lines is 2mm. Settings less than 2mm should be thoroughly evaluated before deciding whether to apply them.

[Limitations] This feature does not support the (8x5.9)x2 size. Please do not set the crease line as it may cause a paper jam.
If you specify the (8x5)x2 size, you cannot set the crease line at a location greater than 113.3mm.

Crease lines when the surface image of double-sided printing is viewed from the top.



When the setting values are as follows, crease lines are arranged as shown in the left figure.
wNumOfCreaseLine:2
wNumOfPerPos:2
wPitchOfPerPos:P1(Positive integer)
wPosOfCreaseLine[0]:L1(Positive integer)
wPosOfCreaseLine[1]:L2(Positive integer)

Setting slitter control

[Format] BOOL SetSlitterCtrl(long lPortNum, WORD wPos);

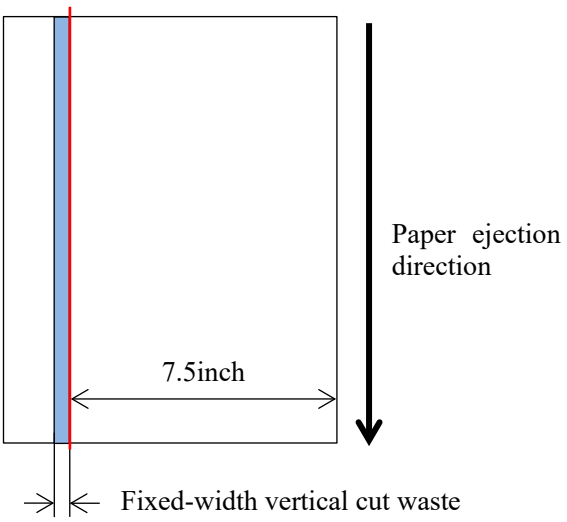
[Argument]
lPortNum: Port number
wPos: The number of crease lines to be arranged.
setting value: 0, 392 - 750 [Unit : 0.01inch]

[Return]
Successful: TRUE
Failure: FALSE

[Explanation]
Cut vertically at the specified position. Valid only for cut sheets.
Please set before sending the first image.
If the following settings are made, the settings will be invalid.
・When the setting value is outside the setting range.

[Attention]
Specify this setting for each output.
The printer will return to its default settings (0) after printing.

If the paper width is 8 inches and the setting value is 750 (7.5 inches), it will be cut vertically as shown below.



Creaser unit disabling settings

[Format] BOOL SetCreaserUnitDisable(long lPortNum, DWORD dwDisable);

[Argument]

lPortNum:	Port number
dwDisable:	Creaser unit disabling settings 0: Cancel creaser unit disable (default) 1: Creaser unit disabled

[Return]

Succesful:	TRUE
Failure:	FALSE

[Explanation]
Set to disable creaser unit operation.
Settings are stored in the printer. (It is valid even when you turn off the printer.)
If a creaser unit breaks down, it is possible to operate the printer without using the unit by disabling it with this function.
When recovery is achieved through repair or unit replacement, it is also possible to cancel the disablement using this function.

[Sample Coding]

```
< Visual C >
if( SetCreaserUnitDisable(PortNo, 1) == TRUE ){          // Creaser unit disabled
        // Successful
}

< VB.NET >
If SetCreaserUnitDisable(PortNo, 0) = True Then          ' Cancel creaser unit disable
        ' Successful
End If
```

Get the value of creaser unit disable setting

[Format] long GetCreaserUnitDisable(long lPortNum);

[Argument]

lPortNum:	Port number

[Return]

1 :	Creaser unit disabled
0 :	Cancel creaser unit disable
-1 :	Fail

[Explanation]
Gets the creaser unit disable setting.
The default value for this setting is " Cancel creaser unit disable (0)".

[Sample Coding]

```
< Visual C >
if( GetCreaserUnitDisable(PortNo) == 1 ){
        // Creaser unit disabled
}

< VB.NET >
If GetCreaserUnitDisable(PortNo) = 0 Then
        ' Cancel creaser unit disable
End If
```

Slitter unit disabling settings

[Format] BOOL SetSlitterUnitDisable(long lPortNum, DWORD dwDisable);

[Argument]
lPortNum: Port number
dwDisable: Slitter unit disabling settings
0: Cancel slitter unit disable (default)
1: Slitter unit disabled

[Return]
Succesful: TRUE
Failure: FALSE

[Explanation]
Set to disable slitter unit operation.
Settings are stored in the printer. (It is valid even when you turn off the printer.)
If a slitter unit breaks down, it is possible to operate the printer without using the unit by disabling it with this function.
When recovery is achieved through repair or unit replacement, it is also possible to cancel the disablement using this function.

[Sample Coding]

< Visual C >

```
if( SetSlitterUnitDisable(PortNo, 1) == TRUE ){          // Slitter unit disabled
        // Successful
}
```

< VB.NET >

```
If SetSlitterUnitDisable(PortNo, 0) = True Then          ' Cancel slitter unit disable
        ' Successful
End If
```

Get the value of slitter unit disable setting

[Format] long GetSlitterUnitDisable(long lPortNum);

[Argument] lPortNum: Port number

[Return]
1 : Slitter unit disabled
0 : Cancel slitter unit disable
-1 : Fail

[Explanation]
Gets the slitter unit disable setting.
The default value for this setting is " Cancel slitter unit disable (0)".

[Sample Coding]

< Visual C >

```
if( GetSlitterUnitDisable(PortNo) == 1 ){
        // Slitter unit disabled
}
```

< VB.NET >

```
If GetSlitterUnitDisable(PortNo) = 0 Then
        ' Cancel slitter unit disable
End If
```

Band unevenness prevention operation setting

[Format] BOOL SetPreventBandUnevenness(long lPortNum, DWORD dwEnable);

[Argument]

lPortNum:	Port number
dwDisable:	Band unevenness prevention operation setting 0: Disable (default) 1: Enable

[Return]

Succesful:	TRUE
Failure:	FALSE

[Explanation] Set the band unevenness prevention operation.
Depending on the media used, band-like unevenness may occur on the printed surface. In this case, use this function to enable the operation to prevent band unevenness.

[Attention] This setting will be remembered until the printer is turned off.
When the printer starts up, it is set to the default setting (0).

[Sample Coding]

< Visual C >

```
if( SetPreventBandUnevenness(PortNo, 1) == TRUE ){    // Band unevenness prevention operation enabled
        // Successful
}
```

< VB.NET >

```
If SetPreventBandUnevenness(PortNo, 0) = True Then     ' Band unevenness prevention operation disabled
        ' Successful
End If
```

Common Set Command

[Format] BOOL SetCommand(long lPortNum, LPSTR lpCmd, DWORD dwCmdLen);
BOOL CvSetCommand(long lPortNum, LPSTR lpCmd, DWORD dwCmdLen);

[Argument]

lPortNum:	Port number
lpCmd:	Pointer to the buffer where the command is stored
dwCmdLen:	The number of characters of the command

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation] Sends the command to the printer

Common Get Command

[Format]

```
long GetCommandEX( long lPortNum,
                   LPSTR lpCmd,
                   DWORD dwCmdLen,
                   LPSTR lpRetBuff,
                   DWORD dwRetBuffSize
                 );
long CvGetCommandEX( long lPortNum,
                     LPSTR lpCmd,
                     DWORD dwCmdLen,
                     LPSTR lpRetBuff,
                     DWORD dwRetBuffSize
                   );
```

[Argument]

lPortNum:	Port number
lpCmd:	Pointer to the buffer where the command is stored
dwCmdLen:	The number of characters of the command
lpRetBuff:	Pointer to the buffer to store receipt data
dwRetBuffSize:	Available size of receiving buffer

[Return]

Successful:	The number of bytes received (Receipt data by lpRetBuff)
Failure:	-1

[Explanation] After the command is sent to the printer, receipt data is stored in the buffer.

Get Minilab Tower Status

[Format] long GetMiniLabTowerStatus(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: Tower Error Status
Failure: STATUS_ERROR

[Explanation] This returns the minilab tower status.
Bit position of the status is defined in CspStat.h by macro.
Meanings of the symbols are as follows:

TGROUP_CXM0	[Shooter/Sorter] Group identification bit
TGROUP_CXM1	[Shooter/Sorter] Group identification bit
TGROUP_CXM2	[Shooter/Sorter] Group identification bit
TGROUP_CXM3	[Shooter/Sorter] Group identification bit
TGROUP_CXM4	[Shooter/Sorter] Group identification bit
/* tower normal */	
STATUS_CXM_TOWER_NORMAL	Tower Normal Operation
/* Sorter sensor */	
STATUS_CXM_SRT_SENSON	Sorter Unit Paper Max Position Sensor On
STATUS_CXM_SRT_MAXERR	Sorter Unit Paper Max Position Error
/* Multi Sorter */	
STATUS_CXM_SRT_SYSERR	Sorter Unit System Error
STATUS_CXM_SRT_MOTERR	Sorter Unit Motor Error
STATUS_CXM_SRT_TIMEOUT	Sorter Unit Working Timeout
/* Shooter Cover & Door Open Status */	
STATUS_CXM_SHT_COPENA	Shooter Cover A Open Error
STATUS_CXM_SHT_COPENB	Shooter Cover B Open Error
STATUS_CXM_SHT_COPENC	Shooter Cover C Open Error
STATUS_CXM_SHT_COPEND	Shooter Cover D Open Error
STATUS_CXM_SHT_DOPENA	Shooter Door A Open Error
STATUS_CXM_SHT_DOPENB	Shooter Door B Open Error
STATUS_CXM_SHT_CORNER_OPEN	Shooter Corner Unit Open
/* Back Print Unit Door Open Status */	
STATUS_CXM_BP_DOPENA	Back Print Unit Door A Open Error
STATUS_CXM_BP_DOPENB	Back Print Unit Door B Open Error
STATUS_CXM_BP_DOPENC	Back Print Unit Door C Open Error
STATUS_CXM_BP_DOPEND	Back Print Unit Door D Open Error
STATUS_CXM_BP_DOPENE	Back Print Unit Door E Open Error
/* Shooter Paper Jam Status */	
STATUS_CXM_SHT_PJAM_A	Shooter Paper Jam Error A
STATUS_CXM_SHT_PJAM_B	Shooter Paper Jam Error B
STATUS_CXM_SHT_PJAM_CORNER	Shooter Paper Jam Error Corner
/* Back Print Unit Paper Jam Status */	
STATUS_CXM_BP_PJAM_A	Back Print Unit Paper Jam Error A
STATUS_CXM_BP_PJAM_ABC	Back Print Unit Paper Jam Error A or B or C

/* Others */

STATUS_CXM_SHT_SYSERR	Shooter System Error
STATUS_CXM_RS422_NONC	Shooter RS422 Non connection Error
STATUS_CXM_BP_WDP	Back Print Unit Wire Dot Printer Error
STATUS_CXM_KIOSK_MODERR	KIOSK Mode Error (*note 1)
STATUS_CXM_BP_MODERR	Sorter Mode Error (*note 1)

Note) A switch on the tower itself allows you to switch between Kiosk mode and Sorter mode.
The Kiosk and Sorter mode errors are determined by the BP unit connection status, which depends on the operating mode selected.

Tower switch (operating mode)	Connection with the BP unit	Error
Kiosk mode	Yes	Kiosk mode error
Sorter mode	No	Sorter mode error

Minilab Print Data Group No., Serial No. Initial Setting

[Format] BOOL SetMiniLabGroupNoIni(long lPortNum, long lGrNo, long lSrNo, long lOption);

[Argument]

lPortNum:	Port Number	
lGrNo:	Group No. Initial value	1-10000 Normal operation 0x55AA Bulk transmission mode
lSrNo:	Serial No. Initial value	1-10000 Normal operation 0x55AA Bulk transmission mode
Option:	Operation options	0: No designation 1: Sorter on

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation]

This sets the initial value (pointer) for the tower controller group no. and serial no.
The sorter operates under the following conditions:

- When the operation option is set to 1 (Sorter on), and an image from a new group is delivered.
- When the quantity in one basket exceeds a specified amount.

In the Bulk Transmission Mode, the tower delivers prints as soon as they are ready and printed, regardless of the Group No./Serial No. order. In this case, the sorter operation does not switch depending on the Group No. / Serial No.

Note) This command is processed as a sequential execute command after printing is complete.
Be sure to send this command to the printer that printed the last image of the group.

Ex.)Transmission method for 3-printer set-up

① Gr.1 Sr1 image transmission -> printer 1
② Gr.1 Sr2 image transmission -> printer 2
③ [Initial value setting]SetMinilabGroupNoIni(Gr.2 Sr.1) -> printer 2
④ Gr.2 Sr1 image transmission -> printer 3

After the printing at ② is complete, the controller sets the GrNo./SrNo. initial value (pointer change) at ③.Then, it receives the new GrNo. at ④.

[Sample Coding]

< Visual C >

```
long lGrNo = 1;                    // Group number initialization
long lSrNo = 1;                    // Serial number initialization
char text[82];
char text2[41];
char len;

//  ①Group No 1 Serial No 1
SetMiniLabGroupNo(PortNo1, lGrNo);                          // Sends the group number
SetMiniLabSerealNo(PortNo1, lSrNo, SERIAL_NUM_SET);         // Sends the serial number
strcpy( text, "Group No.1 -\r" );                           // 1st line of data
strcpy( text2, "Serial No.1\r");                            // 2nd line of data
strcat( text, text2 );                                      // Combines 1st and 2nd lines of data
len = strlen(text);                                         // String length
SetMiniLabBackPrintData( PortNo1, (LPSTR)text, len);        // Sends back print data

//At this point, it sends the print data

//  ②Group No 1 Serial No 2(Finishes Group No 1)
lSrNo += 1;                                                 // Updates serial number
SetMiniLabGroupNo(PortNo2, lSrNo);                          // Sends the group number
SetMiniLabSerealNo(PortNo2, lSrNo, SERIAL_NUM_SET);         // Sends the serial number
strcpy( text, "Group No.1 –\r" );                           // 1st line of data
strcpy( text2, "Serial No.2\r");                            // 2nd line of data
strcat( text, text2 );                                      // Combines 1st and 2nd lines of data
len = strlen(text);                                         // String length
SetMiniLabBackPrintData( PortNo2, (LPSTR)text, len);        // Sends back print data

// At this point, it sends the print data
lGrNo += 1;                                                 // Updates group number
lSrNo = 1;                                                  // Initializes serial number

//  ③Group No 2 Serial No 1 Initialization
SetMinilabGroupNoIni(PortNo2, lGrNo, lSrNo, 1);             // Sorter Control

//  ④Group No 2 Serial No 1
SetMiniLabGroupNo(PortNo3, lGrNo);                          // Sends group number
SetMiniLabSerealNo(PortNo3, lSrNo, SERIAL_NUM_SET);         // Sends serial number
strcpy( text, "Group No.2 -\r" );                           // 1st line of data
strcpy( text2, "Serial No.1\r");                            // 2nd line of data
strcat( text, text2 );                                      // Combines 1st and 2nd lines of data
len = strlen(text);                                         // String length
SetMiniLabBackPrintData( PortNo3, (LPSTR)text, len);        // Sends back print data

// At this point, it sends the print data
```

< Visual Basic >

```
Dim lGrNo As Long
Dim lSrNo As Long
Dim bpText As String * 255
lGrNo = 1                                                    'Initializes group number
lSrNo = 1                                                    'Initializes serial number

'①Group No 1 Serial No 1
SetMiniLabGroupNo(PortNo, lGrNo)                             'Sends group number
SetMiniLabSerealNo(PortNo, lSrNo, SERIAL_NUM_SET)            'Sends serial number
bpText = "Group No.1 -" & Chr(&HD) & "Sereal No.1" & Chr(&HD)
SetMiniLabBackPrintData( PortNo, bpText, 24)                 'Sends back print data

' At this point, it sends the print data

'②Group No 1 Serial No 2
lSrNo = lSrNo + 1                                            'Updates serial number
SetMiniLabGroupNo(PortNo, lSrNo)                             'Sends group number
SetMiniLabSerealNo(PortNo, lSrNo, SERIAL_NUM_SET)            'Sends serial number
bpText = "Group No.1 -" & Chr$(&HD) & "Sereal No.2" & Chr$(&HD)
SetMiniLabBackPrintData( PortNo, bpText, 24)                 'Sends back print data

' At this point, it sends the print data

lGrNo = lGrNo + 1                                            'Updates group number
lSrNo = 1                                                    'Initializes serial number
'③Group No 2 Serial No 1 Initialization
SetMinilabGroupNoIni(PortNo2, lGrNo, lSrNo, 1)               'Sorter Control

'④Group No 2 Serial No 1
SetMiniLabGroupNo(PortNo, lGrNo)
SetMiniLabSerealNo(PortNo, lSrNo, SERIAL_NUM_SET)            'Sends serial number
bpText = "Group No.2 -" & Chr$(&HD) & "Sereal No.1" & Chr$(&HD)
SetMiniLabBackPrintData( PortNo, bpText, 24)

' At this point, it sends the print data
```

Setting the MiniLab Print Data Group Number

[Format] BOOL SetMiniLabGroupNo(long lPortNum, long lGrNo);

[Argument]
lPortNum: Port number
lGrNo: Print data group No.(1 - 10000)
No.0 is a special number.

[Return]
Successful: TRUE
Failure: FALSE

[Explanation]
This sets the group number for the print data.
CXM sorts the print material by group number.
No.0 is the number put out when the shooter is empty, regardless of the serial order.

[Sample Coding] Refer to the following section.

Setting the MiniLab Print Data Serial Number

[Format] BOOL SetMiniLabSerialNo(long lPortNum, long lSrNo, long lFinal);

[Argument]
lPortNum: Port number
lSrNo: Print data serial No(1 - 10000)
No.0 is a special number.
lFinal: Announces the end of the print data serial numbers

[Return]
Successful: TRUE
Failure: FALSE

[Explanation]
This sets the serial number for the print data.
1 serial number is designated for each image.
In the last data of the group, designate SERIAL_NUM_FINAL in lFinal.
Operation designation is macro-defined as CspStat.h.

SERIAL_NUM_SET Designates a normal serial number
SERIAL_NUM_FINAL The final serial number

When No.0 is set, the serial numbers in the tower are not increased.

Example) Regarding serial numbers in the tower

■ Normal order management

Serial numbers in the tower
Setting No.1 -> 2
Setting No.2 -> 3
Setting No.3 -> 4

■ When No.0 is used

Serial numbers in the tower
Setting No.1 -> 2
Setting No.2 -> 3
Setting No.0 -> remains 3, without being raised

[Sample Coding] Refer to the following section.

Setting MiniLab Back Print Data

[Format] BOOL SetMiniLabBackprintData(long lPortNum, LPSTR lpText, long lTextLen);

[Argument]

lPortNum:	Port number
lpText:	Back print data text string Add CR(0xD) at the end of the data
lTextlen:	Number of characters (1 - 82) Set the number of characters to include CR.

[Return]

Successful:	TRUE
Failure:	FALSE

[Explanation]
This sets the text data to be printed on the back side.
In setting 2 lines of data, show as [1st line text] + CR + [2nd line text] + CR.
Each line can have a maximum of 40 characters.

[Sample Coding]

< Visual C >

```
long lGrNo = 1;                    // Group number initialization
long lSrNo = 1;                    // Serial number initialization
char text[82];
char text2[41];
char len;

// Group No 1 Serial No 1
SetMiniLabGroupNo(PortNo, lGrNo);                          // Sends the group number
SetMiniLabSerealNo(PortNo, lSrNo, SERIAL_NUM_SET);         // Sends the serial number
strcpy( text, "Group No.1 - \r" );                         // 1st line of data
strcpy( text2, "Serial No.1\r");                           // 2nd line of data
strcat( text, text2 );                                     // Combines 1st and 2nd lines of data
len = strlen(text);                                        // String length
SetMiniLabBackPrintData( PortNo, (LPSTR)text, len);        // Sends back print data

//At this point, it sends the print data

// Group No 1 Serial No 2(Finishes Group No 1)
lSrNo += 1;                                                // Updates serial number
SetMiniLabGroupNo(PortNo, lSrNo);                          // Sends the group number
SetMiniLabSerealNo(PortNo, lSrNo, SERIAL_NUM_FINAL);       // Sends the final serial number
strcpy( text, "Group No.1 -\r" );                          // 1st line of data
strcpy( text2, "Serial No.2\r");                           // 2nd line of data
strcat( text, text2 );                                     // Combines 1st and 2nd lines of data
len = strlen(text);                                        // String length
SetMiniLabBackPrintData( PortNo, (LPSTR)text, len);        // Sends back print data

// At this point, it sends the print data
// Group No 2 Serial No 1
lGrNo += 1;                                                // Updates group number
lSrNo = 1;                                                 // Initializes serial number
SetMiniLabGroupNo(PortNo, lGrNo);                          // Sends group number
SetMiniLabSerealNo(PortNo, lSrNo, SERIAL_NUM_FINAL);       // Sends final serial number
strcpy( text, "Group No.2 -\r" );                          // 1st line of data
strcpy( text2, "Serial No.1\r");                           // 2nd line of data
strcat( text, text2 );                                     // Combines 1st and 2nd lines of data
len = strlen(text);                                        // String length
SetMiniLabBackPrintData( PortNo, (LPSTR)text, len);        // Sends back print data

// At this point, it sends the print data
```

< Visual Basic >

```
Dim lGrNo As Long
Dim lSrNo As Long
Dim bpText As String * 255
lGrNo = 1                                                    'Initializes group number
lSrNo = 1                                                    'Initializes serial number

'Group No 1 Serial No 1
SetMiniLabGroupNo(PortNo, lGrNo)                             'Sends group number
SetMiniLabSerealNo(PortNo, lSrNo, SERIAL_NUM_SET)            'Sends serial number
bpText = "Group No.1 -" & Chr(&HD) & "Sereal No.1" & Chr(&HD)
SetMiniLabBackPrintData( PortNo, bpText, 24)                 ' Sends back print data

' At this point, it sends the print data

'Group No 1 Serial No 2 (Finishes Group No 1))
lSrNo = lSrNo + 1                                            ' Updates serial number
SetMiniLabGroupNo(PortNo, lSrNo)                             'Sends group number
SetMiniLabSerealNo(PortNo, lSrNo, SERIAL_NUM_FINAL)          'Sends final serial number
bpText = "Group No.1 -" & Chr$(&HD) & "Sereal No.2" & Chr$(&HD)
SetMiniLabBackPrintData( PortNo, bpText, 24)                 'Sends back print data

' At this point, it sends the print data

'Group No 2 Serial No 1
lGrNo = lGrNo + 1                                            'Updates group number
lSrNo = 1                                                    'Initializes serial number
SetMiniLabGroupNo(PortNo, lGrNo)
SetMiniLabSerealNo(PortNo, lSrNo, SERIAL_NUM_FINAL)          'Sends final serial number
bpText = "Group No.2 -" & Chr$(&HD) & "Sereal No.1" & Chr$(&HD)
SetMiniLabBackPrintData( PortNo, bpText, 24)

' At this point, it sends the print data
```

Stop Printing Command

[Format] BOOL StopPrint(long lPortNum);

[Argument] lPortNum: port number

[Return] Successful: TRUE
Failure: FALSE

[Explanation] When a printer receives this command, if printing, it will complete printing, and if permitted, will deliver it to the shooter. If delivery is denied, the image will be cut into small pieces and destroyed.
If there is still data in the print buffer, the buffer is cleared.

[Sample Coding]

< Visual C >

```
StopPrint(PortNo);
```

< Visual Basic >

```
StopPrint(PortNo)
```

Designation of Paper Feed Destination

[Format] BOOL SetMiniLabOutputDest(long lPortNum, long lDest);

[Argument] lPortNum: Port number
lDest: Paper feed destination

[Return] Successful: TRUE
Failure: FALSE

[Explanation] This designates the paper feed destination when using the Mini-lab.
The paper feed destination is macro defined with CspStat.h.
The symbols have the following meanings.

OUTPUT_DEST_AUTO	(0)	Automatic setting by paper size (default)
OUTPUT_DEST_SORTER	(1)	Feed to sorter
OUTPUT_DEST_TRAY	(2)	Feed to front tray

[Sample Coding]

< Visual C >

```
SetMiniLabOutputDest(PortNo, OUTPUT_DEST_SORTER);
```

< Visual Basic >

```
SetMiniLabOutputDest(PortNo, OUTPUT_DEST_SORTER)
```

Set Sorter Paper MAX Position Detection Sensor ON/OFF

[Format] BOOL SetMiniLabSrtSensor(long lPortNum, long lSens);

[Argument]
lPortNum: Port number
lSens: Paper detection function switch

[Return]
Successful: TRUE
Failure: FALSE

[Explanation]
This sets the function for the Paper Maximum Position detection sensor.
Paper position detection sensor designation is macro defined as CspStat.h.
The symbols have the following meanings:

SORTER_MAX_OFF Ignores paper position detection during delivery
SORTER_MAX_ON Detects paper position.(Default)

[Sample Coding]
< Visual C >
```
SetMiniLabSrtSensor(PortNo, SORTER_MAX_ON);
```

< Visual Basic >
```
SetMiniLabSrtSensor(PortNo, SORTER_MAX_OFF)
```

Get Sorter Paper MAX Position Detection Sensor ON/OFF

[Format] long GetMiniLabSrtSensor(long lPortNum);

[Argument]
lPortNum: Port number

[Return]
Successful: Current paper detection function status
Failure: -1

[Explanation]
This gets the status for the Paper Maximum Position detection sensor.
Paper position detection sensor designation is macro defined as CspStat.h.
The symbols have the following meanings:

SORTER_MAX_OFF Ignores paper position detection during delivery
SORTER_MAX_ON Detects paper position.(Default)

[Sample Coding]
< Visual C >
```
long lSens;
lSens = GetMiniLabSrtSensor(PortNo, SORTER_MAX_ON);
```

< Visual Basic >
```
Dim lSens As Long
lSens = GetMiniLabSrtSensor(PortNo, SORTER_MAX_OFF)
```

Get Minilab Group No.

[Format] long GetMiniLabGroupNo(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: Group No(1-10000)
Failure: -1

[Explanation] This gets the current group number set at the tower controller.

Note) When the tower is in operation, the group numbers are constantly changing.
If attempting to get the same number as the application has, stop the tower before getting the group number.

[Sample Coding]

< Visual C >

```
long lGNo;
if((lGNo = GetMiniLabGroupNo(PortNo)) >= 0 ){
        // lGNo Group No.
}
```

< Visual Basic >

```
Dim lGNo As Long
lGNo = GetMiniLabGroupNo(PortNo)
```

Get Minilab Serial No.

[Format] long GetMiniLabSerialNo(long lPortNum);

[Argument] lPortNum: Port number

[Return] Successful: Serial No(1-10000)
Failure: -1

[Explanation] This gets the current serial number set at the tower controller.

Note) When the tower is in operation, the serial numbers are constantly changing.
If attempting to get the same number as the application has, stop the tower before getting the serial number.

[Sample Coding]

< Visual C >

```
long lSNo;
if((lSNo = GetMiniLabSerialNo(PortNo )) >= 0 ){
        // lSNo Serial No.
}
```

< Visual Basic >

```
Dim lSNo As Long
lSNo = GetMiniLabSerialNo(PortNo)
```

Get Printer Address

[Format] long GetMiniLabPrinterAdd(long lPortNum);

[Argument] lPortNum: port number

[Return] Successful: None Connect (0)
printer address (1 - 4)
Failure: -1

[Explanation] Printer gets the position set in the Tower.
No. 1 is the topmost.

[Sample Coding]

< Visual C >

```
long lAddr;
if((lAddr = GetMiniLabPrinterAdd(PortNo)) >= 0 ){
        // lAddr printer address
}
```

< Visual Basic >

```
Dim lAddr As Long
lAddr = GetMiniLabPrinterAdd(PortNo)
```

13.Appendix

(1) About DP-TC10 Single cut mode

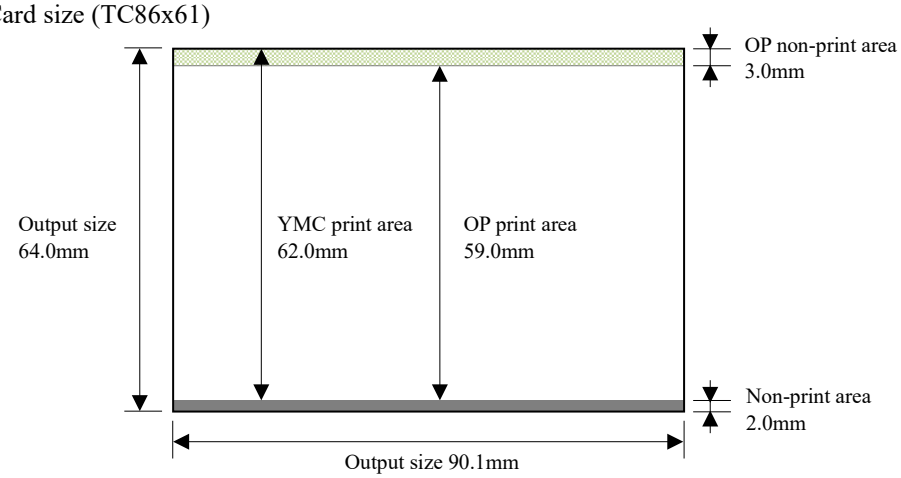
① Operation Overview

- In order to overlap the next printing, 3mm of the printing tip, not printing OP.
- Cut at the position of 2mm from the print tip.
- The printing end portion, so as to overlap 1mm to 2mm of last remaining print, to print up to 1mm from the terminal end paper.
- The printer receives the image data of the vertical 64mm, lower 2mm (300dpi: 24dot, 600dpi: 48dot) data is not printing.

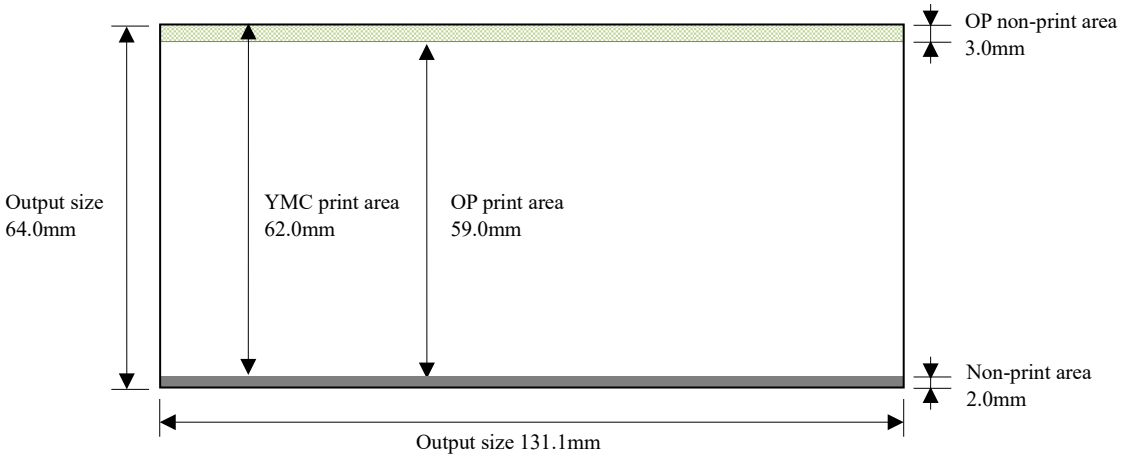
② Print area size

Paper Type	Image data size (W x H)	Image pixel size		Print size
		Resolution setting: 300x600dpi	Resolution setting: 300x600dpi	
TC86x61	90.1 x 64.0mm	1064 x 1512 pixel	1064 x 756 pixel	90.1 x 62mm
TC127x61	131.1 x 64.0mm	1548 x 1512 pixel	1548 x 756 pixel	131.1 x 62mm

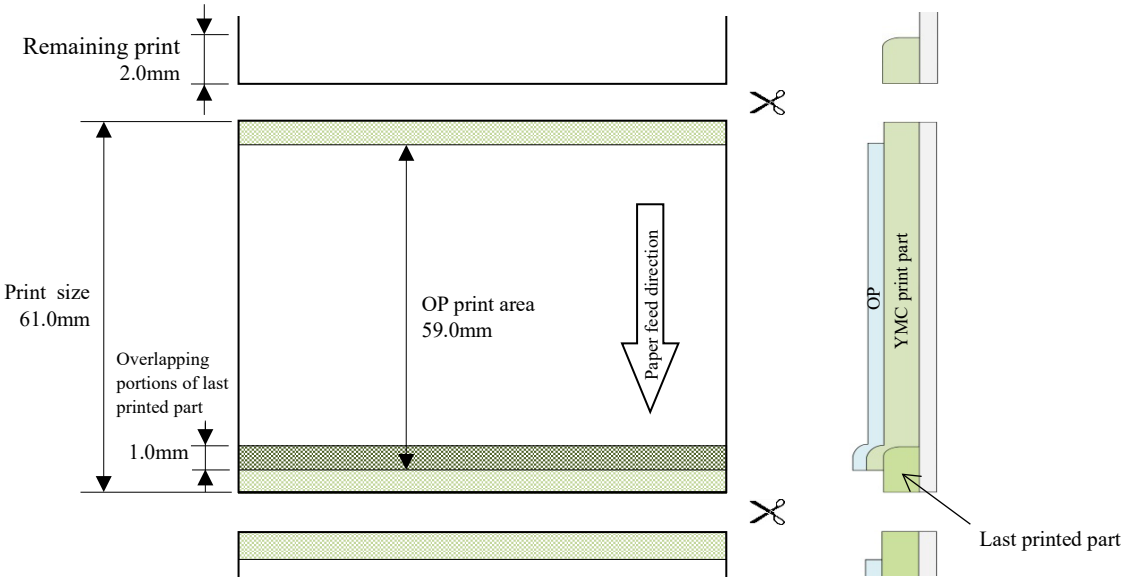
■ Card size (TC86x61)



■ L width size (TC127x61)



③ Print result



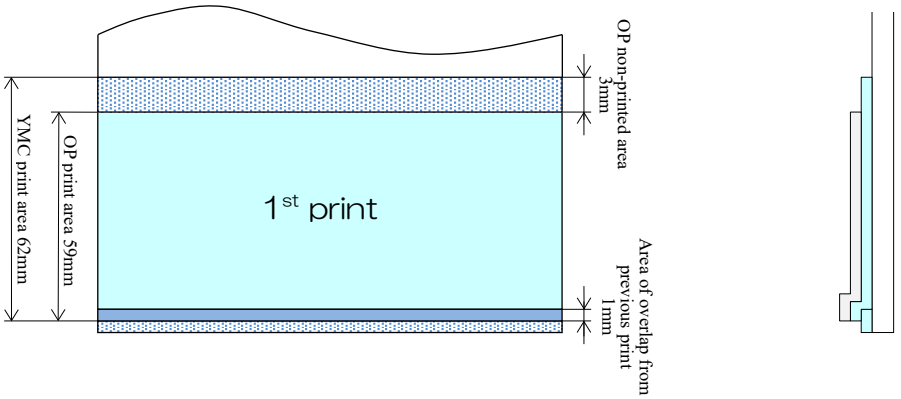
(2) About DP-TC10 L card size 2-image layout continuous printing

① Summary of Operations

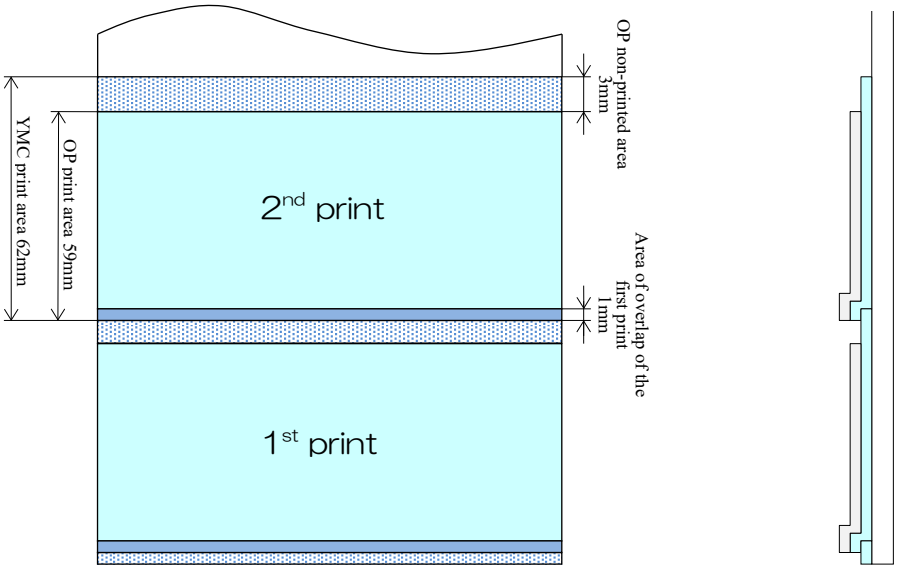
- With the command from the host, L card size 2-image layout continuous printing is carried out.
- After the L card size 2-image layout continuous print command is sent from the host, by sending 2 prints' data of L card size, the printer prints 2 images continuously.
- 2-image layout continuous printing can be done on L card size (127x61mm), and connot be done on card size (86x61mm).

② Print Area Size

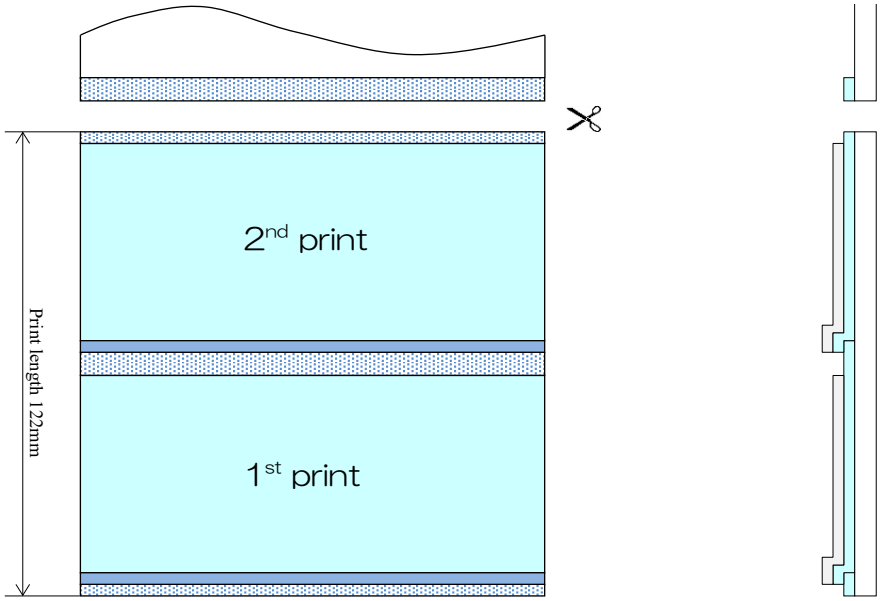
(I) 1st Print printing



(Ⅱ) 2nd Print printing



(Ⅲ) Cutting the prints



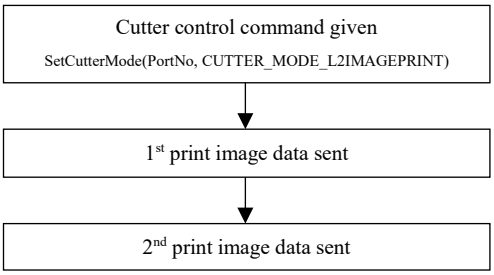
③ How to perform 2-image layout continuous printing

・Designation of the 2-image layout continuous printing can be done with the Direct DLL Cutter Control Command.

■ How to Designate

(Ⅰ) When performing 2-image layout continuous printing, before sending the data for the 1st print, use the Cutter Control Command to designate "CUTTER_MODE_L2IMAGEPRINT"(130)

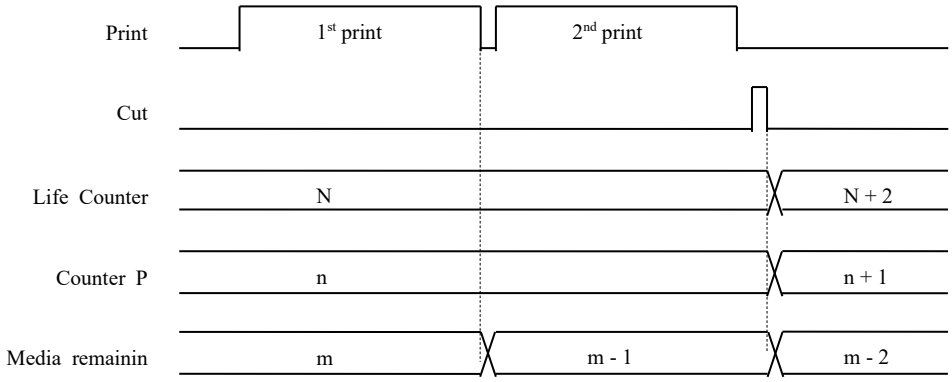
(Ⅱ) After the command is given, send the image data for 2 prints.



④ Life Counter operation

・When performing 2-image layout continuous printing, the life counter counts up +2 after printing and cutting the 2nd print.

・Counter P counts up +1 after the printing and cutting of the 2nd print has been completed.



⑤ Cautions

・When performing 2-image layout continuous printing, get the remaining media quantity from the printer, and check that the 2nd image can be printed. (Note that the actual media quantity can differ due to opening/shutting of the printer cover.)

・If Ribbon End occurs after printing the 1st print of 2-image layout continuous printing, the printer will cut the 1st print, and stop. (The printer status will be Ribbon End.)

・If the image data for the 2nd print of the 2-image layout continuous printing isn't sent, after approximately 20 seconds the print will be cut and the printer will stop. (The printer status will be Idle.)

(3) About Partial Matte Printing

① Outline of Partial Matte Printing.

・When the command designating the Partial Matte Mode is sent from the host, the printer will switch to Partial Matte Mode. After that, the partial matte pattern is sent, followed by the print image data, and partial matte printing is performed.

・In the Partial Matte Mode designation with the DP-TC10, the strength of the matte effect can be set in 5 levels. With the DP-DS620, you can choose between "Matte" and "Fine Matte".

・In the Partial Matte Mode, 1 image is printed, after which the printerwill revert to Normal Mode. When partial matte printing is being performed,it is necessary to send the command and partial matte pattern for eachprint.

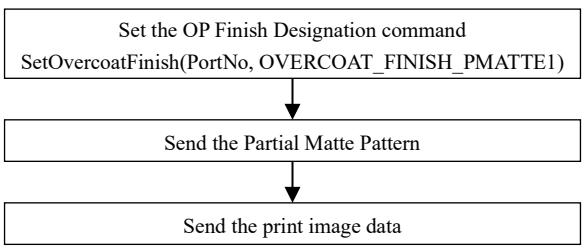
・When sending the partial matte pattern data, the pattern, created in black and white, is sent using the same process as is used when sending normal print data.

② How to Designate Partial Matte

(Ⅰ) When performing partial matte printing, first use the Overcoat Finish Control command "SetOvercoatFinish( )" to designate the setting from OVERCOAT_FINISH_PMATTE0 to 4 with the DP-TC10, or OVERCOAT_FINISH_PMATTE11 to 13 with the DP-DS620.

(Ⅱ) Then, send the partial matte pattern to the printer with "SendImageData( )", the same as for normal image data.

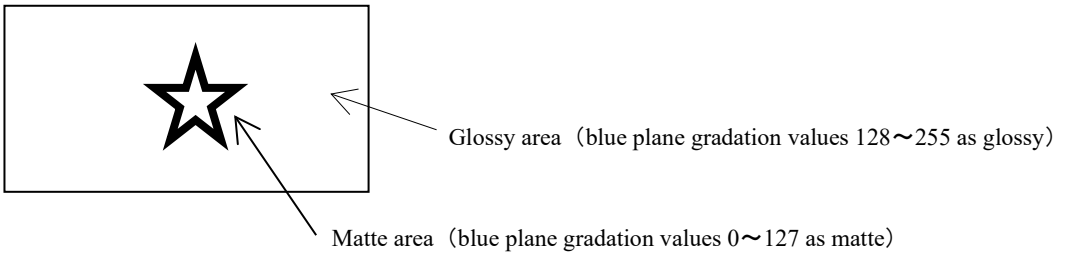
(Ⅲ) After the partial matte pattern, send the print image data.



③ About Partial Matte Patterns

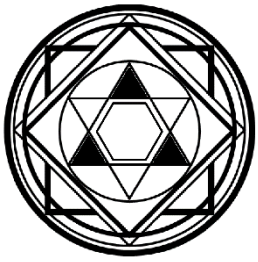
Create the partial matte pattern as a black and white image.

（Set the image format for the Direct DLL as RGB24bit color bitmap format.）



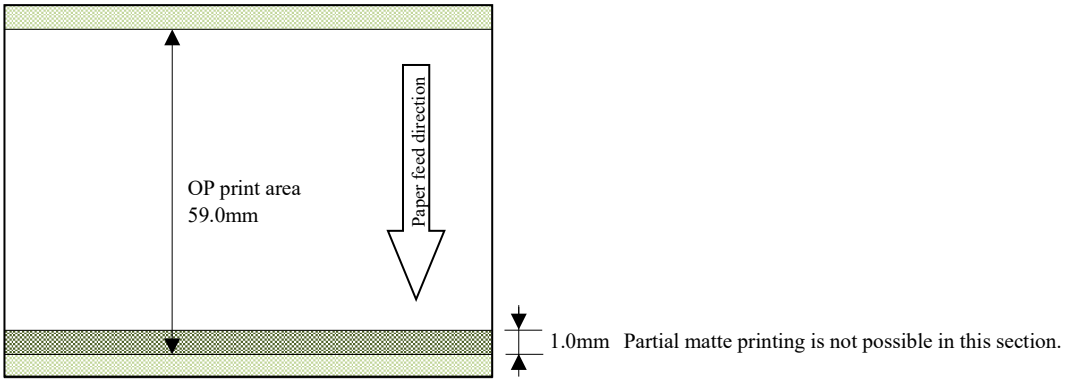
④ In-house confirmation pattern

With this Partial Matte mode, the print ratio for a design like the one below is expected to be 8～10%.



⑤ Limitations

When printing a partial matte pattern, partial matte printing is not possible at the end of the OP printing. (DP-TC10)



(4) About Ribbon End Check Operation

Tartget model: DP-DS620, DP-DS820, DP-DS820DX

The ribbons for printers until now had a black line at the head of each color panel, and ribbon end was determined after the ribbon was wound following OP printing, by whether the black line for the next Yellow panel was detected or not. The end of the ribbon was made such that the black line for the next Yellow panel was in a position that would not be detected by the ribbon sensor (it was designed with a lengthened pitch between the OP and Yellow).
Conversely, the ribbon for the target model's doesn't have a black line for detection at each color panel (it detects the panel color), and the panel pitch is shortened and the Yellow panel is in a position that can be detected by the ribbon sensor at the end of the ribbon. Therefore, for the target model's, the ribbon count is determined with the count recorded in the RF-ID tag. When the remaining count is 0, it determines ribbon end.
Normally, the actual remaining ribbon count and the ribbon count recorded in the RF-ID tag will match, but if trouble such as jamming occurs during printing, and the actual remaining ribbon count is 1 less than that recorded in the RF-ID tag, then even though it has used the last of the ribbon, it will detect the next Yellow panel and the RF-ID tag will show 1 remaining, so it won't show ribbon end.
In order to avoid this, only when the RF-ID tag shows the following remaining ribbon count, it will run the ribbon end check operation.

<Conditions for Ribbon End Check>

	RF-ID-tag Ribbon count (after OP printing)	
Media Size	Function: CvGetMediaCounter() / GetMediaCounter()	Function: GetMediaCounterH()
5x3.5 (L)	2 or 1	-
6x4 (PC)	2 or 1	-
5x7 (2L)	2 or 1	4 or 2 (5x3.5 conversion qty)
6x8 (A5)	2 or 1	4 or 2 (6x4 conversion qty)
6x9 (A5W) *1	2 or 1	4 or 2 (6x4.5 conversion qty)
8x10	2 or 1	4 or 2 (8x5 conversion qty)
8x12	2 or 1	4 or 2 (8x6 conversion qty)
A4	2 or 1	4 or 2 (A5 conversion qty)

*1 DP-DS620 firmware version 1.10 or later

When the RF-ID ribbon count after OP printing is as shown above, it will run the following ribbon end check operation.

<Ribbon End Check Operation>

After OP printing, it will wind the ribbon until the ribbon sensor detects the Yellow panel.
After the ribbon sensor detects the Yellow panel, it will continue to wind the ribbon for a predetermined length (about 20mm). Note that, ribbon winding at this time is wound at slowly, in order to prevent peeling of the adhesion part of a ribbon supply bobbin and a ribbon film.
After it has wound for the predetermined length, the ribbon will rewind, Yellow panel position will be set, and operation will continue.
If it cannot be wound the predetermined length (winding stops), it will be ribbon end, and operation will stop.

(5) Panorama Printing with White Border

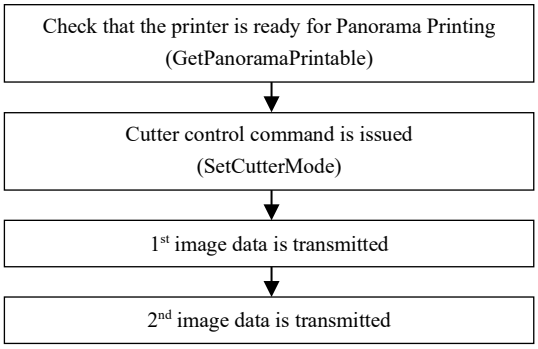
① Overview

- Panorama printing is performed by specifying a command from the host computer.
- After receiving the Cutter Control command (panorama printing operation), the printer does not cut the printout at the end of the printing but continues printing the data it received next. If the Panorama Printing Specification command is sent again before sending the data for the second image, the printer will not cut the media after printing the second image but will continue printing the data it received next.

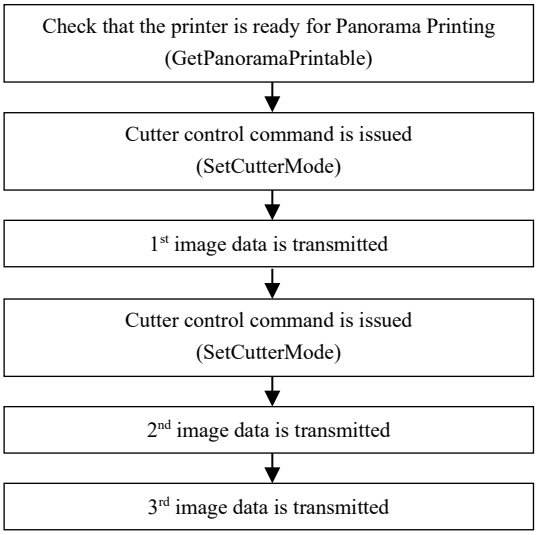
② Panorama Printing Operation Flow

1. When performing panorama printing, you can check whether the printer can perform panorama printing with the Panorama Printing Start Check function. (This can be skipped. When the head temperature is high, send the panorama print data without waiting for the status to change to show printing is possible.)
2. With panorama printing, specify panorama printing operation "CUTTER_MODE_PANORAMA" with the Cutter Control command before sending the image data.
3. After executing the command, send print data of 6x8, 8x10, 8x12, A4 size.

■ With 6x16, 8x20, 8x24, A4x2 printing

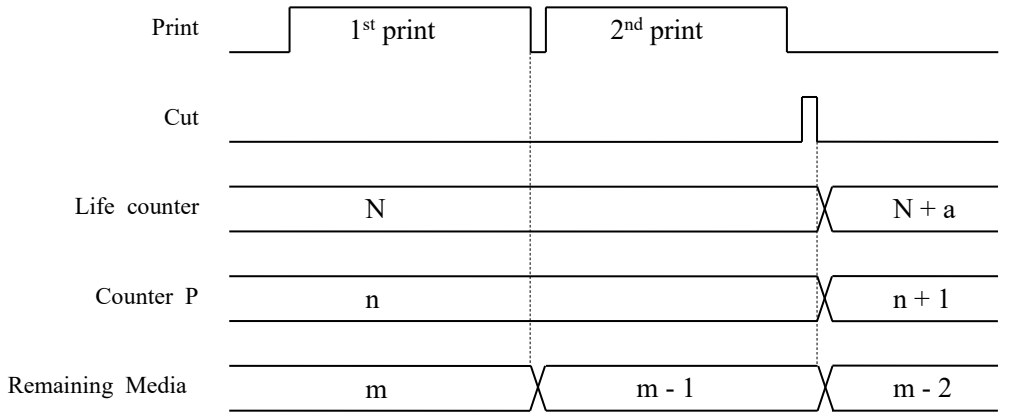


■ With 6x24, 8x30, 8x36, A4x3 printing



③ Life Counter operation

- For Panorama Printing, the counter increases after printing at the point when the print is cut.
  (For 6x16:+4, For 6x24:+6, For 8x20, 8x24, or A4x2:+2, For 8x30, 8x36, or A4x3:+3)
- The counter P counts up +1 after printing at the point when the print is cut.



DP-DS620: a=4
DP-DS820: a=2

④ Cautions

- When the head temperature is high, the printer will wait for the head to cool before starting printing.
- When performing panorama printing, get the remaining media quantity from the printer, and check that enough media remains to print the panorama before executing the command.
- If Ribbon End occurs during panorama printing, the printer will cut the printout and stop. The printer status will indicate Ribbon End.
- During panorama printing, the printer will wait until the next data transmission if it did not receive the data for the next printing. In such cases, note that marks may remain in the decurl part or where the media has been gripped.
- During panorama printing, if the next image data is not sent after approximately 60 seconds, the printout will be cut and the printer will stop. The printer status will indicate Idle.
- During panorama printing, the Print Retry Control command is disabled.
- During panorama printing, if the printer receives the Full Cutter Set-up command, the Non-scrap Cutter Operation command, or the 2inch Cut Operation command, the printout will be cut during the panorama printing operation and the printer will return to normal operation.
- During panorama printing, if the printer receives data for image size other than 6x8-size (DP-DS820: 8x10, 8x12, A4), the printout will be cut during the panorama printing operation and the printer will return to normal operation.
- Do not touch the paper output from the printer during panorama printing.
- Pay attention to the printer installation location since printout of up to approximately 36 inches can be output.